

Bringing Services Closer to People? Distance to Administrators and the Reach of the State*

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Abstract

As populations grow, extending the reach of the state to a larger and more dispersed population becomes increasingly challenging. Many developing countries have responded by creating new subnational administrative units, bringing local administrators physically closer to citizens. Leveraging an episode of district proliferation in Uganda, this paper studies whether geographical proximity to local government administrative centers improves access to public goods and services and changes citizens' engagement with government. To establish causality, I exploit a policy implementation rule that generated exogenous variation in households' proximity to new district headquarters. Households living closer to these centers have significantly better access to electricity, roads, health facilities, and water sources. Proximity also shifts engagement from contentious to participatory channels and reduces direct experiences of corruption. Despite these gains, proximity does not increase trust in government or improve democratic perceptions, including in politically aligned districts where engagement effects are stronger. Finally, each district split raises transfers to the pre-split district territory by roughly 14 percent, with most of the increase concentrated in administrative expenditures rather than service delivery budgets.

Keywords: Distance to Government, Public Service Delivery, Citizen Engagement, Decentralization, Administrative Unit Proliferation.

JEL codes: D72, D73, R12, O13, H77.

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1 Introduction

As populations grow, governments face a fundamental challenge: how to maintain the reach of the state over a larger number of citizens while extending it to communities where government presence remains limited. This is particularly relevant in rural areas of developing countries, where distance can make it harder for governments to learn about local needs, deliver public services, and interact with citizens. These pressures are especially salient in sub-Saharan Africa. The region’s population is projected to increase by two-thirds by 2050 (UN, 2024), and its rural population is still expanding (UN, 2018). Governments have responded in part by multiplying subnational administrative units. Since 1990, almost half of sub-Saharan African countries have increased their number of administrative units by at least 20 percent (Grossman & Lewis, 2014).¹ Rather than necessarily devolving new powers, these reforms reorganize the state spatially, creating new administrative centers closer to previously remote populations (Grossman & Lewis, 2014), often with the explicit objective of improving public service delivery (GoU, 1995).

The key margin changed by these reforms is the physical distance between citizens and local administrators. A long tradition emphasizes that state authority becomes harder to project over distance (Herbst, 2014), and empirical work shows that distance from political and administrative centers can weaken institutional penetration, political accountability, and public service provision (Michalopoulos & Papaioannou, 2014; Campante & Do, 2014; Asher et al., 2018). Administrative unit creation changes this relationship directly by establishing new centers of government, turning citizens’ distance to local administration into a policy margin. Theories of decentralization suggest that bringing government closer can improve information about heterogeneous local preferences and strengthen citizen engagement (Tiebout, 1956; Oates et al., 1972; Alesina & Spolaore, 1997; Faguet, 2023). Shorter distances may also facilitate interactions with government, increase policymakers’ responsiveness, and improve access to public services (Asher et al., 2018). Yet proximity need not translate into better government. Creating smaller units may eliminate economies of scale, intensify local elite capture (Oates et al., 1972), or generate administrative capacity constraints (Cohen, 2024). Moreover, in hybrid regimes, bringing government closer may extend central control rather than empower citizens (Martinez-Bravo et al., 2022). Whether deliberately reducing the distance between citizens and administrators strengthens the reach of the state therefore remains an open empirical question.

This paper studies whether geographical distance to local government administrative centers affects the provision of public goods and services in the context of administrative unit prolifer-

¹The pattern is not specific to Africa. Indonesia’s districts rose from 302 in 1999 to 514 in 2014 (Bazzi & Gudgeon, 2021), Brazil’s municipalities from 4,124 in 1989 to 5,507 in 1997 (Dahis & Szerman, 2024), India’s districts from 466 in 1991 to 640 in 2011 (Rajan & Malghan, 2022), and Vietnam’s provinces from 40 to 64 (Law, 2026). The United States went through the same process while its population was rural and growing: the number of counties rose from 292 in 1790 to 2,713 in 1900 (U.S. Bureau of the Census, 1994). Appendix H reports the series by country (Table A26).

ation and subdivision in Uganda. This country provides a relevant setting for several reasons. First, Uganda has gone through a large decentralization process, resulting in a rapid expansion of sub-national administrative units (e.g., the number of districts has grown from 38 in 1991 to 136 in 2020). Second, the national government established rules for the geographical location of new administrative headquarters, providing exogenous variation in distance to administrative centers. Third, the availability of geospatial data, combined with comprehensive data on households' access to and satisfaction with public goods provision, offers a unique opportunity to gain a broad overview of the effects of distance to administrators on public service delivery in a redistricting context.

To establish causality, I use an instrumental variable approach leveraging an implementation rule in the process of district creation, which introduced exogenous variation in distance to the new district headquarters. Specifically, governmental guidelines established that district headquarters should be located in the geographical middle of newly created districts (Green, 2010). Building on a strategy used by Campante & Do (2014) for the US, I use the distance from each household's enumeration area to its district's centroid, interacted with an indicator for districts created between 1991 and 2020, as an instrument for the household's distance to the district capital. To implement this strategy, I construct a granular spatial dataset that combines (i) rich household microdata on access to public services from the 2021 National Service Delivery Survey (NSDS) and the 2024 Afrobarometer, (ii) geospatial data and administrative records identifying newly created districts between 1991 and 2020, (iii) geocoded distances from each household's enumeration area to its district's headquarters and centroid, and (iv) administrative records of local-government budgets from the Ministry of Finance, Planning and Economic Development. I complement the main analysis with the 2015 NSDS and the 2017 Afrobarometer to examine how the effects of proximity evolved over time, as new district institutions became more established. Together, these data allow me to study how distance to the district's administrative center affects households' access to public goods and services, as well as their relationship with the state.

The findings show that geographical proximity to new district headquarters significantly improves access to public goods and services. Households closer to the district capital have better access to electricity and paved roads, shorter travel times to health facilities, and higher access to drinking water. The effects are strongest for centrally provided services, reflecting both direct infrastructure investments near new headquarters and the coordinating role of district administrators in channeling national programs to nearby communities. Using two rounds of a national service delivery survey, I also document that these effects strengthened between 2015 and 2021. While only agricultural inputs responded to proximity in 2015, by 2021 significant gains had emerged for health, water, and electricity access, consistent with the gradual consolidation of administrative capacity in newly created districts.

Various tests provide empirical support for the identification strategy. First, I examine whether

the identifying variation is independent of pre-existing local conditions. Headquarters locations within new districts are not predicted by pre-existing population density or night-time lights, and parishes later selected as headquarters show no differential trends in cropland or built-up area before their designation, providing empirical support showing that headquarters placement is not strategic. I also address non-random household exposure to the geometry generated by redistricting (Borusyak & Hull, 2023). Controlling for the expected instrument across feasible counterfactual redistricting configurations leaves the estimates stable or larger, while randomization inference over 1,000 counterfactual configurations provides further support for the main service delivery results. Second, I examine whether the instrument captures broader geographical centrality rather than proximity to the district administration itself. The first stage is unchanged when controlling for proximity to major urban centers, and instrumented proximity to district headquarters has no systematic effect on households' knowledge of or distance to lower-tier local governments (LCI, LCII, and LCIII), supporting the interpretation that the identifying variation is specific to the district level. The findings are also robust to placebo outcomes, alternative measures of proximity and the instrument, excluding households near headquarters towns, restricting the sample to newly created districts, weak-instrument-robust inference, and corrections for multiple hypothesis testing.

I leverage a unique dataset on service delivery to study two potential mechanisms behind these results: citizen engagement and accountability. On engagement, proximity increases state presence by raising citizens' knowledge of district level institutions and reducing reliance on traditional alternatives. As the state becomes more accessible, contentious engagement falls significantly: households closer to headquarters are less likely to join protests or contact the media. The evidence on participatory engagement is more suggestive, but the coefficients consistently point toward greater involvement in local decision making. Satisfaction with the Resident District Commissioner also increases significantly, suggesting that the decline in contentious engagement may partly reflect a shift toward formal participatory channels. On accountability, proximity reduces direct corruption experiences even as awareness of corruption remains high. This pattern is consistent with stronger monitoring. When citizens are closer to public officials, misconduct is easier to observe and may therefore become less common, even if awareness of corruption remains present.

Despite these improvements, proximity does not translate into higher trust in the government or more favorable democratic perceptions. Satisfaction with democracy, perceived freedom to vote and organize, rule of law evaluations, and trust in local and national institutions are all largely unaffected. Interestingly, while political trust is unchanged, proximity increases social capital. Households closer to headquarters are more likely to trust people of other religions and ethnicities and to engage in collective action to solve local problems, suggesting that the state getting closer builds community cohesion without generating political legitimacy.

Heterogeneity by political alignment between the elected district chairperson and the cen-

trally appointed Resident District Commissioner sheds further light on these findings. In aligned districts, proximity delivers larger gains for centrally provided infrastructure, roads in particular, while district-provided services show no differential response across the two groups. The shift in citizen engagement is also concentrated in aligned districts, where proximity lowers protest participation and collective civic action, raises involvement in council meetings and knowledge of state institutions, reduces bribery for medical care, and raises satisfaction with the RDC. Yet democratic perceptions remain unaffected even in this more favorable environment. In non-aligned districts, the gains for central infrastructure are smaller and no longer statistically significant, engagement does not respond, and proximity is associated with lower perceived freedom to vote and declining trust in local government and the ruling party, suggesting that administrative proximity can reveal frictions between elected and appointed leaders. Together, these results indicate that in hybrid regimes, bringing the state closer can improve services and accountability while leaving the underlying political relationship between citizens and the state unchanged, and could undermine it when local political incentives are misaligned.

Finally, I quantify the fiscal cost of this proliferation strategy using the approved local-government budget estimates published by the Ministry of Finance, Planning and Economic Development for fiscal years 2012/13 to 2023/24 ([Ministry of Finance, Planning and Economic Development, 2024](#)). I aggregate intergovernmental transfers to fixed territories defined by the 1991 district borders and estimate a staggered difference in differences model showing that real transfers rise by roughly 14 percent once a territory is first split. This increase is concentrated in duplicated administrative payroll rather than in conditional service delivery grants. The composition of these additional resources suggests that the service gains are not mechanically driven by larger delivery budgets. At the same time, it reveals the limit of the strategy. Each split expands the administrative apparatus, so bringing the state closer becomes increasingly costly with every additional district.

This paper contributes to three strands of literature. First, it adds to the emerging literature on administrative unit proliferation. Seminal contributions in this area study why central governments create new subnational units during decentralization processes, emphasizing motives such as patronage ([Green, 2010](#)), gerrymandering ([Awortwi & Helmsing, 2014](#)), parliamentary and legislative control ([Resnick, 2017](#); [Green, 2010](#)), and recentralization ([Lewis, 2018](#)). The literature highlights that episodes of administrative unit proliferation reflect a confluence of national executives' objectives and local elites' and citizens' demands ([Grossman & Lewis, 2014](#); [Pierskalla et al., 2019](#)). A smaller set of studies examines the consequences of proliferation for service delivery and finds mixed effects. Cross-country panel evidence shows that more subnational units are on average associated with higher service provision, with decreasing returns as the number of units grows ([Grossman et al., 2017](#)). Country-level difference-in-differences analyses report that administrative splits increased service delivery in Brazil ([Dahis & Szerman, 2024](#)) and service delivery inputs in Uganda ([Cohen, 2024](#)), but had adverse effects

in Burkina Faso (Billing, 2019). This paper contributes by bringing causal evidence from a natural experiment on the effect of geographical distance to administrative centers, within a context of rapid district proliferation, on access to public goods and services. I further go beyond existing work by isolating the spatial margin directly changed by administrative unit creation, the distance between households and their new headquarters, and by identifying two channels through which proximity operates: citizen engagement with local government and accountability, in a setting where redistricting is widespread and policy-relevant for many African countries.

Second, the paper relates to the literature on spatial inequality and local development. This work has emphasized the role of proximity to urban centers and markets in shaping labor outcomes (Fafchamps & Shilpi, 2005; Fafchamps & Wahba, 2006), specialization (Fafchamps & Shilpi, 2005), and economic growth (Jacoby, 2000; Michaels et al., 2012; Storeygard, 2016). A related literature on the spatial reach of the state emphasizes that authority becomes harder to project over distance (Herbst, 2014). Empirical work shows that the influence of national institutions weakens with distance from the capital (Michalopoulos & Papaioannou, 2014), and that populations living farther from their state capital face weaker political accountability and more corruption (Campante & Do, 2014). Much of this literature, however, takes the location of political and administrative centers, and therefore citizens' distance to them, as given, and focuses primarily on national or state capitals. Much less is known about proximity to local administrative centers, where many public services are coordinated. A notable exception is Asher et al. (2018), who document that greater distance to administrative headquarters reduces access to public goods and worsens economic outcomes in India. I contribute to this literature by providing causal evidence on the effects of proximity to newly created local government centers. This allows me to move beyond documenting spatial gradients and study the causal effects of administrative proximity in a setting where governments actively create new centers of administration. I also quantify the fiscal cost of this strategy, showing that district creation requires substantial additional government resources, largely through higher administrative expenditures. Together, the results provide evidence on both the benefits and fiscal costs of extending the spatial reach of the state into initially remote rural areas.

Lastly, this paper contributes to the literature on decentralization and the provision of public goods and services, with a particular focus on hybrid regimes. Existing work emphasizes that greater local autonomy can help tailor policies to heterogeneous local preferences and improve service delivery (Oates et al., 1972; Alesina & Spolaore, 1997; Gadenne & Singhal, 2014; Faguet, 2004; Galiani et al., 2008; Kis-Katos & Sjahrir, 2017; Canavire-Bacarreza et al., 2020). More recent research shows that, in non-democratic or hybrid settings, decentralization reforms can be strategically designed to extend central control and weaken local accountability rather than empower citizens (Martinez-Bravo et al., 2022), and that citizens in such contexts respond less to government performance in terms of trust in the government (Besley et al., 2026). This paper contributes by showing that decentralization-induced increases in access to

public goods and services do not necessarily translate into higher trust in the government or increased democratic perceptions in hybrid regimes.

The remainder of the paper is organized as follows. Section 2 describes the institutional background. Section 3 summarizes the conceptual framework guiding the empirical analysis, with a formal model presented in the Appendix A. Section 4 details the data construction and displays descriptive statistics for the main variables. Section 5 outlines the empirical framework. Section 6 presents the main results, while sections 7 – 11 detail the mechanisms, effects on democratic perceptions, heterogeneities, dynamic effects, and fiscal costs, respectively. Section 12 concludes.

2 Institutional Background

The decentralization process in Uganda began in 1987 with the establishment of a five-tier administrative structure consisting of villages (LCI), parishes (LCII), sub-counties (LCIII), counties (LCIV), and districts (LCV), with districts being the largest administrative unit. While each level has distinct operational responsibilities, local government councils, which serve as the highest political authority within their jurisdictions, operate specifically at the district and sub-county levels. These councils typically consist of a chairperson, vice-chairperson, and five councilors, though additional members may be included, such as representatives of women, youth, and people with disabilities. They are responsible for planning and implementing public service-related projects within their jurisdictions. At the district level, two key figures share administrative responsibilities: the Local Council V (LCV) chairperson, who is directly elected by local voters and leads the district council, and the Resident District Commissioner (RDC), who is appointed by the President and represents the central government within the district. While the LCV is accountable to local constituents, the RDC is accountable upward to the national government, and the degree of political alignment between these two figures can shape how effectively district resources are coordinated and services delivered.

The constitutional framework for decentralization and devolution of powers in Uganda was established in the 1995 Ugandan Constitution and the 1997 Local Governments Act. Notably, the 1995 constitution granted the central government the authority to create sub-national units. Article 179 states that: “Parliament may: (a) alter the boundaries of districts; and (b) create new districts [...] Any measure for the alteration of the boundaries of the creation of districts or administrative units shall be based on the necessity for effective administration and the need to bring services closer to the people” (GoU, 1995). This marked a turning point in the decentralization policy, as, since 1997, district creation increased substantially from 39 districts in 1994 to 112 in 2014, reaching a total of 136 districts in 2020.²

²The literature does not have a consensus on the underlying motives for splitting a district. While Green (2010) argues that there is no concrete evidence that districts have been created in deprived areas, Grossman & Lewis (2014) suggests that districts with politically marginalized minorities were more likely to be split.

The 1997 Local Governments Act define the division of service delivery responsibilities between district governments and the central state (GoU, 1997). Districts are mandated to provide primary education, primary health care through health centers and district hospitals, rural water supply for domestic use, agricultural extension services, district and feeder roads, and community development services (Ahmad et al., 2009). The central government is responsible for the provision of services with national reach such as the national and trunk-road network, electricity transmission and rural electrification, regional and national referral hospitals, and large-scale water infrastructure (Mushemeza, 2019). This institutional structure motivates the categorization of outcomes by the level of government primarily responsible for delivery, distinguishing between services provided by district governments and those under central mandate.

The legal procedure for creating new local government units involves several stages. First, local residents, through their council representative, present a request to the district council. If approved, a resolution to create a new district is submitted to the Ministry of Local Government. After approval, the proposal is referred to the Cabinet for its evaluation. If the Cabinet supports the request, it is then submitted to Parliament for final approval (Awortwi & Helmsing, 2014). Upon successful completion of the process, the district issuing the request is split, allowing the splitter district to establish a new administrative system. Once a district is approved, the new administrative unit is authorized to recruit over 200 technical and administrative staff, a process that has been linked to patronage networks (Green, 2010) and electoral incentives, including vote-buying (Awortwi & Helmsing, 2014). Importantly, the governmental guidelines establish that new districts should locate their headquarters in the geographical middle of the newly created district (Green, 2010).³

According to the Ugandan Constitution, the main objectives of creating new administrative units are political devolution and improving public service delivery. Political devolution aims to empower leaders at lower administrative levels, enabling them to manage local affairs more effectively (Mushemeza, 2019). This process is legitimized by local elections held every four years, which have witnessed significant participation in recent periods. Simultaneously, the creation of new administrative units seeks to enhance public service delivery through a bottom-up approach, where local governments provide opportunities for local participation, leading to better utilization of available resources (Mushemeza, 2019).

3 Conceptual Framework

As described in the previous section, the creation of new districts in Uganda introduces two administrative figures with different accountability relationships operating simultaneously at the district level: the elected LCV chairperson, who is accountable to local voters, and the centrally appointed RDC, who represents the national government. The geographical place-

³Fieldwork interviews with district officials across twelve districts confirm that the centrality norm is a consistent practice. Officials in every district visited stated that the council deliberately chose a central location to guarantee equal access for local leaders and citizens attending meetings at the headquarters.

ment of the new district headquarters determines the physical distance between households and these administrators and the administrative services they are in charge of. This section summarizes the conceptual framework that guides the empirical analysis, describing how proximity to the administrative center and political alignment between the LCV and RDC jointly shape service delivery, citizen engagement, and accountability. A formal model, building on [Martinez-Bravo et al. \(2022\)](#), developing these predictions is presented in Appendix A, with proofs in Appendix B.

Provision of public goods and services. Administrative effort translates into household-level access to public goods and services. Administrators choose an overall level of investment for the district, but the household-level return to these investments depends on distance to the headquarters. Infrastructure radiates outward from the administrative center: electricity grids, paved roads, health facilities, and water sources are more accessible near the headquarters. While the marginal welfare gains from public goods may be largest in remote, underserved areas, the spatial decay of delivery means these gains are the hardest to realize. This argument motivates creating new headquarters closer to previously remote areas.

The degree of political alignment between the LCV and RDC affects the total effort directed toward locally valued services. When both administrators share political affiliation, their incentives are aligned: the elected LCV channels local preferences into service delivery decisions while the appointed RDC coordinates the allocation of centrally financed programs, and both pull in the same direction. When they are not aligned, the RDC's career concerns and political loyalty rewards may be larger than local welfare, reducing the effort the RDC devotes to services valued locally. Importantly, distance and alignment interact. Proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed. This complementarity predicts that the effects of proximity on service delivery should be stronger in politically aligned districts.

Citizen engagement. The quality of public service delivery depends not only on administrative effort but also on the information that administrators receive about local needs. A central insight of the decentralization literature is that citizens have an informational advantage regarding local conditions and the performance of local officials ([Björkman & Svensson, 2009](#)). How effectively citizens can exercise this advantage depends on how they choose to engage with the state, which is shaped by the costs they face.

Citizens can engage through participatory channels, such as attending district council meetings, meeting the RDC or the LCV, or using formal feedback mechanisms, which transmit detailed information about local priorities directly to decision-makers but require traveling to the headquarters. Alternatively, they can engage through contentious channels, like joining protests, contacting the media, or organizing collective pressure campaigns, which are noisier but can be pursued locally at costs that do not depend on distance. This cost structure implies that proximity reduces the relative cost of participatory engagement, shifting citizens from

contentious to formal channels. The pattern is one of substitution: proximity does not suppress citizen voice but rather redirects it toward channels that are more informative for administrators. Political alignment further shapes this choice. When the RDC internalizes local welfare, the state is more responsive to citizen input, increasing the returns to participatory engagement. In non-aligned districts, the lower responsiveness of the state reduces these returns, and citizens may not shift away from contentious channels even when the headquarters is nearby. Together, these mechanisms predict that proximity should reduce contentious engagement and increase participatory engagement, and that this shift should be more pronounced in aligned districts.

Accountability. A further channel through which proximity affects service delivery is the monitoring of local officials. Citizens who live closer to the headquarters interact more frequently with district institutions and are better positioned to observe and report misconduct, raising the expected cost of corruption for administrators. Two forces reinforce each other in deterring corrupt behavior: external monitoring by citizens, which increases with proximity, and internal alignment with local welfare, which increases with political congruence between the LCV and RDC. In aligned districts, the RDC internalizes the welfare cost of resource diversion, making corruption less attractive even when citizen monitoring is imperfect. These mechanisms predict that proximity should reduce corruption experiences and that the reduction should be stronger in politically aligned districts.

Taken together, the framework identifies three reinforcing channels (provision, engagement, and accountability) through which proximity to the administrative center improves public service delivery. All three channels are amplified by political alignment between the elected and appointed district leaders. The main results in Section 6 test the distance predictions, while Sections 7 examine the engagement and accountability channels. The alignment predictions are tested in the heterogeneity analysis of Section 9, which splits the sample by political congruence between the LCV chairperson and the RDC.

4 Data

To study the effects of the geographical distance between citizens and public administrators in the Ugandan redistricting context, I combine various sources of geolocalized secondary data, as well as earth observation data. This section describes the data sources and construction as well as the main variables of interest.

4.1 Sources and Dataset Construction

4.1.1 Household Survey Data

The main data source is the 2021 National Service Delivery Survey (NSDS), conducted by the Uganda Bureau of Statistics on behalf of the Ministry of Public Services to assess the efficiency and effectiveness of service delivery from the perspectives of both providers and recipients

(UBOS, 2022). The survey covers 10,246 households across 1,088 enumeration areas in all 136 districts as of 2020 and is, by design, representative of each district. Of these households, 9,313 form the sample for analysis after excluding those with insufficient geolocalization data and households belonging to Kampala and its metropolitan area Wakiso.⁴ The survey provides rich household and community-level information on education, health, water, electricity, roads, and agricultural inputs, covering services provided by both district and central governments. I complement the analysis using the 2015 wave of the NSDS to examine how the effects of proximity evolve over time. Detailed variable definitions and index construction are described in Appendix C.

Measures of citizen engagement, corruption experiences, and democratic perceptions come from the tenth round of the Afrobarometer survey, conducted in Uganda in 2024. The Afrobarometer is a nationally representative survey covering 2,400 respondents in 105 districts and more than 270 enumeration areas, providing detailed information on citizens' interactions with government institutions, experiences of bribery and petty corruption, satisfaction with democracy, perceptions of the rule of law, and trust in local and national institutions. While the NSDS offers broader household coverage and richer service delivery measures, the Afrobarometer provides attitudinal outcomes that are not available in the NSDS, making the two surveys complementary.

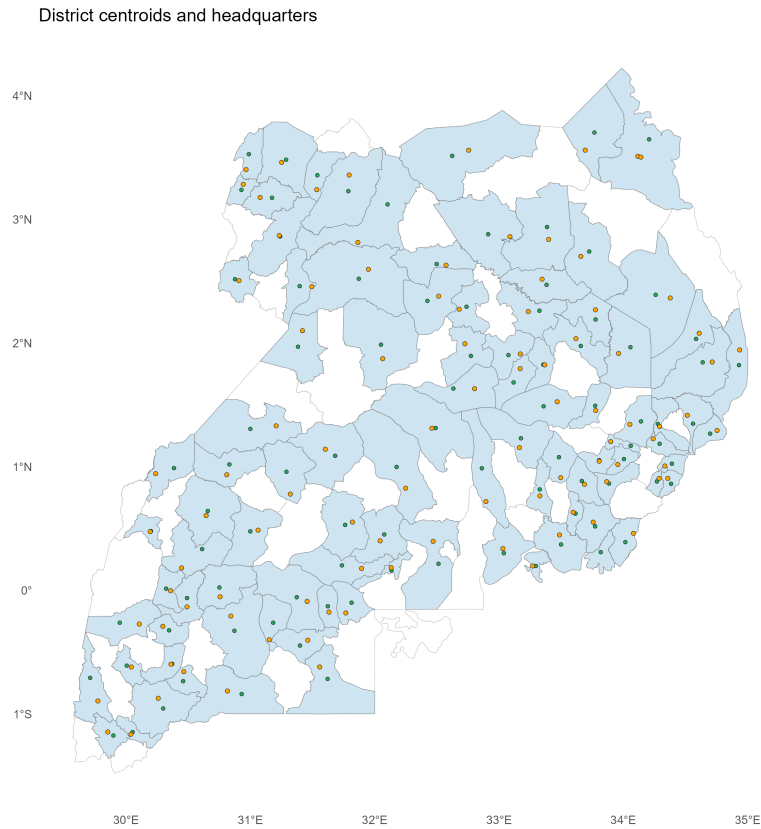
4.1.2 Distance Variables

To assess the impact of a household's proximity to the district's administrative center on public service delivery, I leverage a government policy mandating district headquarters to be located near the district's geographical center. Within these guidelines, however, the specific placement of headquarters may also be influenced by factors such as road access, proximity to other constituencies, and the availability of infrastructure. These factors could introduce bias when analyzing the effects of proximity on service provision. To address this, I compute the straight-line distance from each household's enumeration area to the district's centroid, which serves as an instrument for household proximity to the district headquarters. Figure 1 illustrates the spatial relationship between headquarters and district centroids for newly created districts.

Two key proximity variables are constructed from the household questionnaire. *SR Proximity Household – District Headquarters* is defined as the negative log of the NSDS self-reported distance (in kilometers) from the household to the district capital. *Proximity Household –*

⁴These households are excluded as the capital city service delivery is governed by a separate institutional framework under the Kampala Capital City Authority. Additionally, as noted by Cohen (2024), Kampala is an extreme outlier across a variety of service access indicators. In addition, for eleven enumeration areas, the survey's administrative district label does not match the district in which the enumeration area's coordinates are located. For nine of these cases, the coordinates fall within a neighboring district that was carved from the same 1991 mother district, so I use the district implied by the spatial coordinates. The remaining two enumeration areas are dropped because their coordinates lie implausibly far (more than 100 kilometers) from the district reported in the survey.

FIGURE 1: SPATIAL DISTRIBUTION OF NEW DISTRICTS' HEADQUARTERS



Notes: The map displays new districts (shaded) and old districts (both mother and never split districts) for the period 1991-2020. Green dots mark district centroids and orange dots district headquarters.

District Centroid is the negative log of the straight-line distance between the household's enumeration area geocoordinates and the district's centroid. Relying on proximity calculations using log distance throughout ensures that higher values of all proximity variables correspond to households located closer to the reference point, so that a positive coefficient in the regression analysis is consistent with the expected beneficial effects of administrative proximity.

4.1.3 Evolution of Districts

I combine geospatial data with national district registries to document geographical changes associated with decentralization and district splits between 1991 and 2020. The geospatial data, obtained from UBOS, includes the geographical borders of districts for different years, enabling the identification of district splits and distinguishing between mother and splitter districts. The geographical data is complemented with administrative records on the creation of districts for years in which district's borders are not available. This dataset enables the construction of the variable $NewDistrict_{1991-2020}$, an indicator variable that takes the value of one for newly created districts between 1991 and 2020.

4.1.4 Remote Sensing Data

I combine household and community survey data with three remote sensing products to capture dynamic development patterns across land use, economic activity, and population over time. First, I use land classification data from the MODIS Land Cover Type product, which provides annual observations at a 500-meter resolution from 2001 to 2023.⁵ From these data I construct parish-level variables capturing the share of land under cultivated cropland and cropland mixed with natural vegetation.⁶ Second, to proxy for local economic activity and urbanization, I use the harmonized DMSP-OLS/VIIRS nighttime lights for 2020. I aggregate mean radiance to the parish level. Third, I measure local population using the Gridded Population of the World (GPW v4.11) from CIESIN/SEDAC, which provides population count and density estimates at a 30-arc-second (1 km) resolution for 2020. These parish-level datasets are used as evidence of non-strategic placement of district headquarters.

4.1.5 Qualitative Fieldwork

The empirical analysis is further supported by fieldwork that was conducted in Uganda from July to August 2024. The fieldwork covered twelve districts spanning both old (created in or before 2010) and newly created (after 2010) administrative units. In each district, I conducted semi structured interviews with district-level officials, including Chief Administrative Officers, district planners, finance officers, and Resident District Commissioners (RDCs), as well as with Local Council 1 (LC1) leaders and households in towns located both near to and far from district headquarters. This design comparing officials and citizens across old and new districts and close and far towns was intended to capture qualitative variation along the dimensions exploited in the empirical analysis. The evidence gathered informs the institutional background (Section 2), motivates several robustness checks (Appendix D), and informs the analysis of the engagement, accountability, and political alignment channels.

4.2 Descriptive Statistics

Table 1 contains summary statistics, detailing information on distance variables, the main service delivery-related outcomes, and mechanisms.

In the NSDS, households report living, on average, 16 kilometers from their district headquarters. Using the enumeration area's geocoordinates, the average straight-line distance to the

⁵The MODIS product classifies land cover using the IGBP scheme into 17 categories. I focus on cultivated cropland (>60% cultivated) and cropland mixed with natural vegetation (40–60% cultivation), as these classes together account for the majority of rural land cover in Uganda and provide the most informative measure of agricultural intensification and local economic activity around newly established administrative centers. Changes in built-up area are studied separately using the JRC Global Human Settlement Layer built-up surface product (GHS-BUILT-S), available at 100-meter resolution in five-year epochs from 1990 to 2025, whose long span provides genuine pre-periods for the headquarters cohorts.

⁶Parishes are the lowest administrative unit with stable boundary data.

district headquarters is approximately 12 kilometers, while the corresponding distance to the district centroid is 14.6 kilometers. Under the 1991 district boundaries, these same households would have been, on average, 25.5 kilometers from their district headquarters, approximately 60 percent farther. Nearly half of the households in the sample belong to districts created between 1991 and 2020, reflecting the substantial redistricting that took place over this period.

Turning to the main outcomes, access to services is limited. While 71 percent of households with primary school age children report school attendance, and 86 percent of households that vaccinated their children in the past year did so at a government facility, infrastructure access is considerably lower. Only 16 percent of households have access to improved roads and 47 percent have electricity. Access to improved water sources during the dry season reaches 83 percent, with households traveling 0.8 kilometers on average to their water source. The outcome indices used in the main analysis aggregate these and other related measures. They are centered at zero by construction, with standard deviations between 0.41 and 0.78, reflecting meaningful variation in service access across households.⁷ Among the mechanism variables, 25 percent of households report involvement in LCV meetings and 10 percent report having met the RDC, while direct experiences of corruption are relatively low: 3 and 7 percent of households report being victims of nepotism and absenteeism respectively. Satisfaction with the RDC is reported by 33 percent of households, and 38 percent of households know their magistrate court, suggesting moderate but heterogeneous awareness of district-level institutions.

In the Afrobarometer sample, 36 percent of respondents report engaging in collective civic action, while protest participation is considerably lower at 4 percent. Bribery for medical care is reported by 41 percent of respondents who sought health services, pointing to substantial corruption in healthcare access. Trust in local government stands at around 50 percent, while trust in the ruling party is somewhat higher at 59 percent. Finally, 74 percent of households in the sample are located in politically aligned districts, where the LCV chairperson and RDC share the same political affiliation.

5 Empirical Framework

Measuring the effects of geographical proximity to government administrative centers is challenging because the location of these centers is typically endogenous. Administrative centers may be located in areas with higher population density, greater economic activity, better road access, or higher existing service provision. At the same time, household remoteness may be correlated with unobserved household and local characteristics that also affect the outcomes of interest. As a result, distance to administrative centers may capture factors other than administrative proximity itself.

To circumvent these challenges, this paper leverages quasi-random variation in households' proximity to district headquarters generated by Uganda's redistricting process and the rules

⁷Appendix Table A1 lists the components included in each index.

TABLE 1: Summary Statistics

	N	Mean	SD	Min	Max
<i>Panel A: Proximity Variables</i>					
Distance HH – District Headquarters (km)	8640	16.03	14.59	0.10	90.00
Distance HH – District Headquarters 1991 (km)	8640	25.51	22.45	0.04	105.02
Distance EA – District Headquarters (km)	9315	12.31	10.45	0.05	84.90
Distance EA – District centroid (km)	9315	14.58	8.30	0.28	57.07
New district (1991–2020)	9503	0.50	0.50	0.00	1.00
<i>Panel B: Main Outcomes (NSDS 2021)</i>					
School attendance (prim. age)	4726	0.71	0.41	0.00	1.00
Gov. health facility (immunization)	2361	0.86	0.34	0.00	1.00
Improved water source (dry s.)	8738	0.83	0.37	0.00	1.00
Water distance (dry s., km)	7474	0.79	0.92	0.00	5.00
Improved roads	8731	0.16	0.37	0.00	1.00
HH has electricity	8752	0.47	0.50	0.00	1.00
Education index (district)	7811	0.00	0.73	-2.53	0.92
Health index (district)	8400	-0.01	0.49	-3.37	2.21
Water index (district)	9472	0.00	0.47	-2.29	1.19
Roads index (district)	9471	0.00	0.41	-1.42	2.26
Agricultural inputs index (district)	8752	0.03	0.63	-0.47	2.40
Education index (central government)	6009	0.00	0.73	-2.26	0.83
Health index (central government)	9315	0.03	0.78	-0.89	2.43
Water index (central government)	9436	-0.01	0.53	-7.89	0.62
Roads index (central government)	9471	-0.06	0.73	-2.52	1.08
Electricity index (central government)	9482	-0.03	0.61	-2.26	1.92
<i>Panel C: Mechanisms (NSDS 2021)</i>					
Involved in LCV meetings	8752	0.25	0.43	0.00	1.00
Household meets with RDC	8752	0.10	0.30	0.00	1.00
Victim of nepotism	8752	0.03	0.18	0.00	1.00
Victim of absenteeism	8752	0.07	0.26	0.00	1.00
Satisfied with RDC performance	8752	0.33	0.47	0.00	1.00
Family uses community services	8728	0.06	0.23	0.00	1.00
Knows magistrate court	8729	0.38	0.48	0.00	1.00
<i>Panel D: Mechanisms (Afrobarometer Round 10)</i>					
Civic and collective action	2182	0.36	0.48	0.00	1.00
Engagement in protest	2184	0.04	0.19	0.00	1.00
Bribe for medical care	1604	0.41	0.49	0.00	1.00
Trust in local government	2132	0.51	0.50	0.00	1.00
Trust in ruling party	2184	0.59	0.49	0.00	1.00
<i>Panel E: Local leader's political alignment (LCV and RDC)</i>					
Aligned in 2016 - 2021	9524	0.74	0.44	0.00	1.00

Notes: Summary statistics for the main variables used in the analysis. Panels A–C use NSDS 2021 data. Panel D uses Afrobarometer Round 10 data.

governing the placement of new administrative centers. According to the governmental guidelines, the headquarters of a newly created district should be located near its geographical center (Green, 2010). As a result, the geographical shape of the new district helps determine where its headquarters is located and, consequently, how close households are to it. Following the strategy used by Campante & Do (2014) for the United States, I use households' proximity to the district centroid as a source of exogenous variation in proximity to the headquarters. Specifically, I interact proximity to the district centroid with an indicator for districts created between 1991 and 2020, which isolates the variation generated by the placement rule in newly created districts while accounting for differences in pre-split remoteness. Formally, the instrumental variable first-stage and second-stage are given by:

$$\begin{aligned} ProximityHH_HQ_{i,d} = & \gamma_0 + \gamma_{CN}(ProximityHH_Centroid \times NewDistrict_{1991-2020})_{i,d} \\ & + \gamma_C ProximityHH_Centroid_i + \gamma_N NewDistrict_{1991-2020,d} \\ & + \mathbf{X}'_{i,d} \gamma_X + v_{i,d} \end{aligned} \quad (1)$$

$$\begin{aligned} Y_{i,d} = & \alpha + \beta \widehat{ProximityHH_HQ}_{i,d} + \beta_C ProximityHH_Centroid_i \\ & + \beta_N NewDistrict_{1991-2020,d} + \mathbf{X}'_{i,d} \beta_X + \varepsilon_{i,d}, \end{aligned} \quad (2)$$

where $ProximityHH_Centroid_i$ is the log proximity of household i 's enumeration area to the district centroid, $NewDistrict_{1991-2020,d}$ indicates whether district d was created between 1991 and 2020, their interaction is the excluded instrument, $ProximityHH_HQ_{i,d}$ is the household's self-reported proximity to the district headquarters, and $\mathbf{X}_{i,d}$ is a vector of controls.⁸ Finally, $v_{i,d}$ is the error term clustered at the parish level.

The second-stage estimates in Equation 2 capture the causal effects of households' proximity to the district headquarters on public service delivery outcomes, $Y_{i,d}$, including education, health, water, electricity, roads, and agricultural inputs. Throughout the analysis, the key variable of interest is *proximity* to the district headquarters, so that the coefficient of interest, β , has a straightforward interpretation: a positive estimate indicates that households located closer to the district headquarters experience better public service delivery outcomes, consistent with the government's stated objective of bringing services closer to citizens through district creation.

The conceptual framework in Section 3 distinguishes between two administrators with different accountability structures operating at the district level. The elected LCV chairperson is responsible for coordinating district-mandated services, while the appointed RDC is responsible

⁸Controls include district ever split between 1991-2020, Lake Victoria district, districts' area in Km² as of 1991, household's distance to district's capital as of 1991 (in log.), presence of lakes other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. Lake controls account for districts where water bodies occupy a large share of the territory, which affects the location and interpretation of the geographical centroid.

for channeling centrally financed programs. To map this distinction to the empirical analysis, I construct outcome indices separately for services under district mandate (primary education, district health facilities, rural water supply, district and feeder roads, and agricultural extension and inputs) and services under central government mandate (secondary education, national and referral hospitals, large-scale water infrastructure, the national roads network, and electricity) (GoU, 1997, Second Schedule, Part 2). This distinction allows me to examine whether the effects of administrative proximity differ across levels of government and whether these effects vary with political alignment between the elected and appointed district leaders.

Identifying assumptions. For the instrumental variables estimates to be interpreted causally, three main assumptions must hold: relevance, independence, and the exclusion restriction.

The relevance assumption holds as long as policymakers comply with the governmental guidelines. If this is the case, households' proximity to a district's centroid should increase the likelihood of being closer to the district's headquarters. This assumption is directly testable and is examined in Section 5.1.

The independence assumption rests on the arbitrariness of the new district's centroid. As the location of a district's centroid depends exclusively on the geographical shape of the new administrative unit, it is unlikely to be related to unobservable characteristics that determine public service delivery outcomes, conditional on the set of controls. In other words, new districts' centroids are located as good as randomly, since they only depend on the respective district's shape.

For the exclusion restriction to hold, the interaction between a household's distance to the district's centroid and the indicator for districts created between 1991 and 2020, conditional on the set of controls, should affect public service delivery outcomes only through the household's distance to the district headquarters. This appears plausible, as a household's distance to the new district's centroid is unlikely to affect any relevant outcome other than through access to administration and related services (Campante & Do, 2014) once its distance to the pre-split district headquarters is controlled for. Restricting the identifying variation to newly created districts addresses concerns associated with pre-existing remoteness (i.e., households that were remote both before and after belonging to a new district).

A set of controls helps strengthen the plausibility of the exclusion restriction. All specifications control for whether a district was split between 1991 and 2020. As the size of a district could affect both the decision to split it and the distance to its centroid, specifications control for district area at the beginning of the period (as of 1991). Controlling for households' distance to district capitals as of 1991 captures potential confounders arising from differential proximity to pre-existing urban centers. Finally, controls for the presence of water bodies, elevation, and macro-regions capture potential confounders coming from geographical factors.

Threats to validity. The main threat is that district boundaries are drawn so that the centroid, and hence the headquarters, falls near a favored group, in which case the centroid would no longer be an arbitrary geometric point. Three institutional features make this unlikely. District creation follows a multi-stage legal procedure in which the initial petition depends on the cooperation of local council representatives, limiting both Parliament’s ability to select constituencies and any single actor’s ability to shape the outcome (Section 2). New district borders are built from county borders drawn during colonial times, which constrains the feasible shapes. And qualitative and anecdotal evidence indicates that headquarters placement is jointly constrained by the centrality norm and the availability of public land, with new offices often built from scratch (Green, 2010). Section 5.2 complements these institutional arguments with empirical evidence on strategic placement and addresses a separate concern arising from non-random household exposure to the geometry generated by redistricting. Section 5.3 tests the exclusion restriction directly.

5.1 Relevance: First Stage

The spatial correlation between district centroids and headquarters suggests that policymakers comply with the placement rule (Figure 1). Table 2 confirms this relationship. In districts created between 1991 and 2020, a 10 percent increase in proximity to the district centroid is associated with a 4.29 percent increase in proximity to the district headquarters. The first-stage F-statistics are around 28 in the NSDS and 19 in the Afrobarometer.

TABLE 2: FIRST-STAGE: PROXIMITY HOUSEHOLD - DISTRICT’S HEADQUARTERS

	Log SR Proximity Household - District Headquarters	
	NSDS 2021	Afrobarometer R10
Log Proximity Household - District’s centroid $\times$ New District 91-20	0.429*** (0.080)	0.474*** (0.107)
Log Proximity Household - District’s centroid	-0.051 (0.063)	0.231*** (0.064)
New district between 1991-2020	1.696*** (0.220)	2.099*** (0.322)
Observations	8,417	2,184
Number of Parishes	967	271
IV F-stat Kleibergen-Paap rk Wald	28.75	19.45
Controls	Yes	Yes

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Regression of self-reported household’s distance to the district’s capital on household’s distance to the district’s centroid, an indicator for new districts between 1991-2020, and the interaction between the former two. Both distance variables are in logarithms. Controls include: district ever split between 1991-2020, lake Victoria district, districts’ area in Km^2 as of 1991, household’s distance to district’s capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), four major regions. The instrument for Household’s distance to the district’s capital is the interaction of distance from the household to the district’s centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Figure 2 displays the identifying variation behind these estimates. After residualizing on the full set of controls, it plots binned means of self-reported proximity to the headquarters against proximity to the centroid, separately for new and old districts. The gradient is strong and positive where the placement rule binds and absent in old districts. Appendix Figure A1

presents the corresponding reduced-form relationships for the main outcome indices.

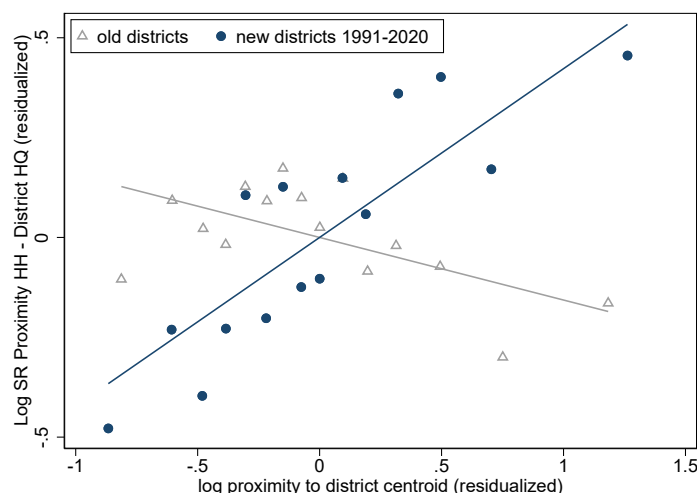


FIGURE 2: First stage binned scatter of residualized self-reported proximity to the district headquarters against residualized proximity to the district centroid for both households in new districts (1991–2020) and old districts. Both variables are residualized on the main specification’s control set (survey-weighted). Each dot is a ventile of the residualized centroid proximity.

5.2 Independence: Headquarters Placement and Household Exposure

I first provide two pieces of empirical evidence addressing the possibility of strategic headquarters placement. I then consider a separate concern. Even if headquarters placement is not strategic, households may be systematically exposed to different instrument values because of their location within the pre-split district. I address this possibility using a design-based recentering approach that accounts for expected exposure across feasible counterfactual redistrictings.

5.2.1 Headquarters Placement

If boundaries were drawn to favor centrally located groups, new headquarters would be more likely to be placed in areas that were already more populated or developed. I test this by analyzing whether pre-existing sub-county population density in 2010⁹ and night lights in 2013¹⁰ predict headquarters placement among the 24 districts created between 2014 and 2020. The results show that neither measure predicts which sub-county becomes the headquarters (Appendix Table A2). Figure 3 illustrates this pattern for Kabarole, where the new headquarters was placed near the geographic center of the new district rather than in the pre-existing population and economic center.

⁹Data obtained from the Gridded Population of the World Database constructed by NASA.

¹⁰The Annual global VIIRS nighttime lights come from the Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines.

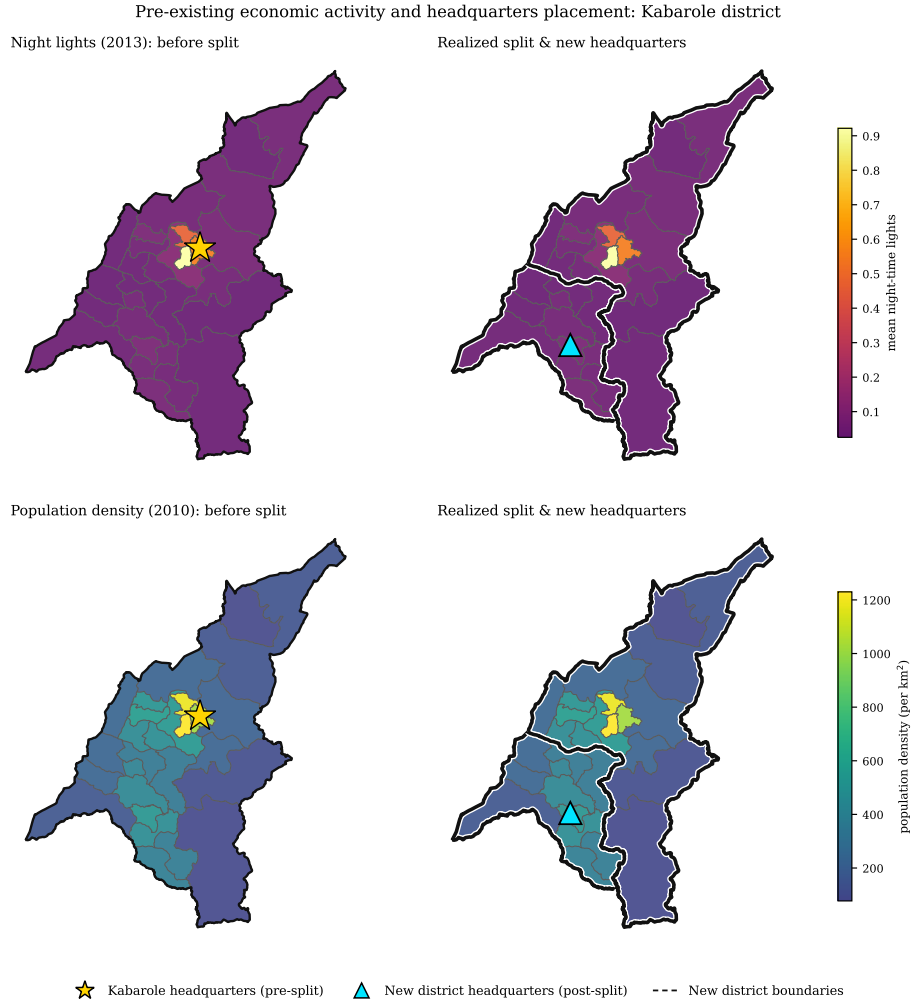


FIGURE 3: Pre-existing economic activity and headquarters placement. The panels show the sub-counties of the pre-split mother district of Kabarole, shaded by night-time lights (2013, top) and population density (2010, bottom). Left panels mark the pre-split district headquarters (star); right panels overlay the realized new-district boundaries and the new headquarters (triangle). The new headquarters is located away from the brightest and most densely populated sub-counties, consistent with the placement rule and with the null estimates in Appendix Table A2.

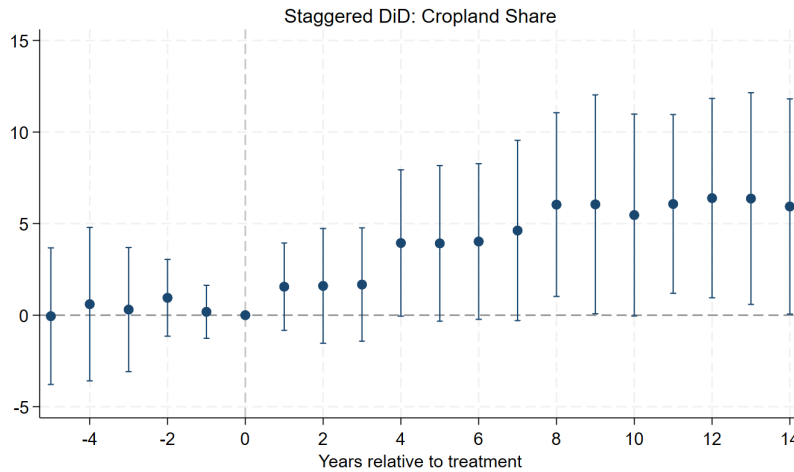
5.2.2 Pre-trends in Land Use

I complement the headquarters' placement results by analyzing whether parishes that become district headquarters were already developing differently before their designation. Using MODIS land classification data (2001–2023, 500m resolution) and the staggered difference-in-differences estimator of [de Chaisemartin & D'Haultfœuille \(2020\)](#) at the parish level, Figure 4 shows no differential pre-trends in cultivated cropland between future headquarters parishes and not-yet-treated parishes. I find a similar pattern for built-up area using the GHS-BUILT-S product.¹¹

¹¹The control group for the cropland analysis is based on 79 not-yet-treated parishes which became district headquarters between 2001 and 2023. Appendix Figure A3 presents the corresponding event study for built-up area using the GHS-BUILT-S product, available from 1990 to 2025.

In both cases, the outcomes begin to rise only after headquarters creation, consistent with local development following headquarters establishment rather than preceding it. Section 10.2 returns to these post-designation dynamics.

FIGURE 4: EVENT STUDY: DISTRICT HEADQUARTER CREATION ON PARISH CROPLAND SHARE



Notes: Event study graph of the estimated coefficients of the [de Chaisemartin & D’Haultfœuille \(2020\)](#) staggered difference in difference estimator of the parish cropland share on headquarters creation.

5.2.3 Non-random Exposure and Design-based Inference

Even if headquarters placement is plausibly exogenous, households may be systematically exposed to the geometry of redistricting. Following [Borusyak & Hull \(2023\)](#), I enumerate all feasible contiguous re-partitions of each splitting 1991 mother district into its realized number of 2020 districts, using predetermined 1991 county borders as building blocks, and compute each household’s expected instrument μ_i across these counterfactual redistrictings. Three results, presented in Appendix F, support the design. Controlling for this expected exposure leaves the main estimates stable or slightly larger, and weak-instrument-robust Anderson–Rubin confidence intervals continue to exclude zero for the main service outcomes. Randomization inference across counterfactual redistrictings also places the realized first stage and reduced-form estimates in the tails of their respective distributions.

5.3 Exclusion restriction: Specificity

A remaining concern is that the instrument captures broader geographical centrality or proximity to government institutions rather than proximity to the district administration itself. I examine whether instrumented proximity to the district headquarters affects households’ knowledge of or distance to lower-tier local government institutions (LC1, LC2, and LC3), which operate below the district level and should not be directly affected by the creation of a new

district headquarters. Appendix Table A5 shows no systematic effects on these outcomes. Eleven of the twelve coefficients are statistically indistinguishable from zero at the 5 percent level, and the only exception does not survive correction for multiple hypothesis testing. This supports the interpretation that the identifying variation is specific to the district level rather than reflecting a general reduction in distance to government institutions.

Appendix D reports additional identification and robustness exercises. These show that the first stage is unchanged when controlling for proximity to major cities and that proximity to the district headquarters does not predict placebo outcomes unrelated to administrative proximity. I also consider specifications excluding households around headquarters towns, GIS-based measures of distance, a comparison of OLS and 2SLS estimates, alternative constructions of the instrument, and estimates restricted to newly created districts. The conclusions are unchanged across these exercises.

6 Results

6.1 Main Results - Effects on Public Service Delivery

The National Service Delivery Survey offers a rich set of variables on public goods and services, allowing the analysis of a broad range of outcomes. As predicted by the conceptual framework (Proposition 1), proximity to district headquarters should improve access to public goods and services.

The outcomes cover education, health, water, and roads, each constructed separately for district and central government provision, while electricity is provided only by the central government and agricultural inputs only by districts. I follow Anderson (2008) and construct summary indices to account for multiple hypothesis testing. To create each index, I standardize the component variables to have a mean of 0 and a standard deviation of 1, and compute the index as the average of the standardized components available for each household. The components entering each index are listed in Appendix Table A1.

Table 3 presents the causal effects of household proximity to the district headquarters on these summary indices. Panel A reports the estimates for services under the responsibility of district governments, while Panel B reports those for services under the responsibility of the central government, where the RDC plays a coordinating role at the district level. The table also reports Anderson’s sharpened False Discovery Rate (FDR) q-values to account for multiple hypothesis testing across outcome groups.

The results show that proximity to new district headquarters improves access to several key public services, with effects that are stronger and more precisely estimated for centrally provided services. Among district-provided services (Table 3, Panel A), households living closer to the district headquarters have significantly better health access (Column 2), and

TABLE 3: Effects on Services and Proximity to District Headquarters (2SLS)

	School (1)	Health (2)	Water (3)	Roads (4)	Agricultural inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
Log Proximity to District HQ	0.282 (0.214)	0.387*** (0.121)	0.279* (0.157)	0.089 (0.110)	-0.085 (0.145)
p-val	[0.187]	[0.001]	[0.076]	[0.416]	[0.556]
q-val	{0.234}	{0.008}	{0.179}	{0.452}	{0.452}
Observations	7137	7595	8417	8417	8417
Number of clusters	958	951	967	967	967
Mean of DV	-0.003	-0.016	-0.004	0.008	0.030
IV F-stat (KP rk Wald)	22.243	33.349	28.745	28.745	28.745
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
Log Proximity to District HQ	0.151 (0.192)	0.233 (0.201)	0.372** (0.146)	0.706*** (0.261)	0.428*** (0.137)
p-val	[0.432]	[0.248]	[0.011]	[0.007]	[0.002]
q-val	{0.209}	{0.142}	{0.015}	{0.014}	{0.009}
Observations	5553	8417	8417	8417	8417
Number of clusters	951	967	967	967	967
Mean of DV	-0.001	0.013	-0.017	-0.070	-0.043
IV F-stat (KP rk Wald)	22.621	28.745	28.745	28.745	28.745

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

water (Column 3). The indices for education and roads point in the expected direction but are not statistically significant. The agricultural inputs index shows a small negative and insignificant coefficient, which likely reflects the fact that input provision operates primarily at the sub-county level, making it less sensitive to district-level administrative proximity.

The effects are most consistent and precisely estimated for centrally provided services (Table 3, Panel B). Households closer to district headquarters have significantly better access to water (Column 3), roads (Column 4), and electricity (Column 5). These significant results are robust

to multiple hypothesis testing corrections, with FDR q-values below 0.10. Disaggregated results for the individual components underlying each index are presented in Appendix Tables A15–A20. The individual tables show that the improvements detected in the indexes correspond to increases in several indicators and are not driven by specific components. In addition, Appendix Table A14 contains results for service availability from the Afrobarometer adding evidence to the higher service access in areas proximate to district headquarters.

Complementing the service delivery findings, Table 4 shows that households living closer to new district headquarters report higher satisfaction with local administrators. While all three types of leaders (the elected district politician (Local Council V), the centrally appointed Resident District Commissioner (RDC), and civil servants) show positive coefficients, only satisfaction with the RDC is statistically significant. This result is consistent with the stronger effects found for centrally provided services. Since the RDC is the main link between the district and the central government, s/he plays a central role in coordinating resource allocation and program implementation from above. Households living closer to the headquarters may therefore not only benefit more from these services, but also be more likely to recognize the RDC’s coordinating role. In terms of magnitude, a 10 percent increase in proximity raises the likelihood of being satisfied with the RDC’s performance by 1.2 percentage points, equivalent to 3.8 percent of the sample mean.

TABLE 4: Satisfaction with Local Leaders and Proximity to District Headquarters (2SLS)

	LCV performance (1)	RDC satisfaction (2)	Civil servants performance (3)
<i>Panel A: NSDS 2021</i>			
Log Proximity to District HQ	0.084 (0.076)	0.122** (0.059)	0.103 (0.085)
Observations	8417	8417	7451
Number of clusters	967	967	966
Mean of DV	0.428	0.329	0.547
IV F-stat (KP rk Wald)	28.745	28.745	30.047

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of satisfaction with local leaders’ variables on self-reported household’s distance to the district’s capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts’ area in Km² as of 1991, household’s enumeration area distance to district’s capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household’s distance to the district’s capital is the interaction of distance from the household’s enumeration area to the district’s centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

Taken together, the main results describe a consistent pattern. Reducing the distance between

households and their district administrative center improves access to public goods and services and raises satisfaction with appointed local leaders, who are responsible for coordinating centrally financed programs. The following section assesses the robustness of these findings before turning to the mechanisms: citizen engagement with the government and accountability.

7 Mechanisms

To investigate the underlying mechanisms behind the main findings, this study focuses on two potential channels through which the distance to new district headquarters may impact the provision of public goods and services. First, the level of citizen engagement with local governments could influence policymakers' awareness of the most pressing issues within the community. Second, individuals' perceptions of corruption may reflect problems associated with corruption in public service delivery, potentially leading to increased demands for accountability.

7.1 Citizens' Engagement with the Local Government

One of the main objectives behind the creation of new administrative units was to empower local populations and enhance their participation in decision-making. As discussed in the conceptual framework, reducing the geographic gap between citizens and government lowers the cost of formal engagement with district institutions, shifting citizens from contentious to participatory channels (Propositions 4–5). By minimizing information and communication frictions between administrators and citizens, closer proximity to administrative centers may foster greater awareness of state institutions, shift citizens away from reliance on traditional alternatives, and ultimately translate into more effective engagement with formal government. This section studies whether proximity to new district headquarters influences these dimensions of citizen-state relations.

Before turning to direct measures of citizen engagement, I first analyze whether households living closer to district headquarters are more aware of district-level institutions and whether proximity reshapes the balance between formal state institutions and traditional ones. Table 5, Panel A shows that proximity to headquarters significantly increases citizens' knowledge of important state institutions. Households living closer are more likely to know about magistrate courts, the police, the Office of the Auditor General, and the judiciary system, reflecting that the creation of new headquarters is strengthening state presence and legitimacy in previously remote areas. The institutional knowledge index in Column (6), which aggregates the five indicators, confirms that this is a broad-based effect rather than the result of any single institution (the coefficient of 0.36 is significant at the 1 percent level). Additionally, households closer to headquarters are more likely to have a family member employed as a civil servant, suggesting a broader deepening of state-society linkages (Panel B, Column 1).

Panel B further reveals that this increased awareness of formal institutions is accompanied by

TABLE 5: Institutional Knowledge and State-Traditional Substitution (2SLS)

	Knows magistrate court (1)	Knows police (2)	Knows land office (3)	Knows OAG (4)	Knows judiciary (5)	Knowledge index (6)
<i>Panel A: NSDS 2021</i>						
Log Proximity to District HQ	0.273*** (0.090)	0.088* (0.053)	0.096 (0.063)	0.088** (0.039)	0.162** (0.080)	0.361*** (0.123)
Observations	8413	8415	8413	8417	8417	8415
Number of clusters	967	967	967	967	967	967
Mean of DV	0.375	0.930	0.156	0.092	0.384	0.001
IV F-stat (KP rk Wald)	28.592	28.639	28.592	28.745	28.745	28.639
	Family member civil servant (1)	Conflict resolved by government (2)	Conflict resolved by trad. leader (3)	Contacted MP (4)	Contacted trad. leader (5)	State reliance index ^a (6)
<i>Panel B: NSDS 2021 and Afrobarometer R10</i>						
Log Proximity to District HQ	0.077** (0.031)	0.096 (0.144)	-0.080 (0.110)	0.076 (0.071)	-0.181* (0.106)	0.240 (0.176)
Observations	8414	2131	2131	2178	2146	2184
Number of clusters	967	271	271	271	271	271
Mean of DV	0.057	0.556	0.211	0.168	0.277	-0.001
IV F-stat (KP rk Wald)	28.655	19.705	19.705	19.462	19.819	19.446

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of state/traditional institutions variables on self-reported household's distance to the district's capital. Column (6) of Panel A is a summary index of Columns (1)–(5). ^a Column (6) of Panel B is a state-reliance index of Columns (2)–(5), with the two traditional-institution indicators reversed so that higher values indicate greater reliance on the state; Column (1) is excluded because it comes from the NSDS. Indices are constructed to have mean 0 and standard deviation 1. All regressions control for district ever split between 1991–2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991–2020. Standard errors clustered at the Parish level are in parenthesis.

a substitution away from traditional ones. The NSDS measures whether any household member is employed as a civil servant, while the Afrobarometer captures the institutions citizens turn to for conflict resolution and representation. Among respondents who experienced a dispute in the past year, I distinguish whether they first sought resolution from the police, a court, or a government official, rather than from a traditional or religious leader, elders, or family. I also measure whether respondents contacted a Member of Parliament or a traditional leader at least once in the past year. Proximity increases the probability that a household includes a civil servant, while the coefficients on state-based dispute resolution and contacting a Member of Parliament are positive but imprecisely estimated. More importantly, respondents living closer to the headquarters are significantly less likely to have contacted a traditional leader, pointing to a shift in the institutions citizens use when seeking for conflict resolution or representation.¹² Together, these results indicate that administrative proximity makes the state more visible and accessible, gradually displacing traditional institutions.

Table 6 presents indicators of citizen participation with district-level governmental actors, differentiating between participatory and contentious forms of engagement. The results show that increased proximity reduces contentious engagement (Panel B). A 10 percent increase

¹²The state-reliance index in Column (6) combines the four Afrobarometer indicators, with the two traditional-institution items entering with reversed sign, so that higher values denote greater reliance on formal state channels; it points in the same direction (0.24) but is imprecisely estimated ($p = 0.17$).

in proximity to district headquarters significantly reduces the likelihood of participating in protests and contacting the media by 0.7 and 1.2 percentage points, respectively (equivalent to 19 and 13 percent of their respective sample means). The contentious engagement index confirms that this is not driven by a single indicator, with a negative coefficient of -0.38 , significant at the 5 percent level. Together with the results on higher state presence above, these results are consistent with citizens having less need to engage in protests to raise their worries. Adding evidence to this interpretation, all coefficients on participatory engagement in Panel A are positive, indicating that proximity increases formal engagement with district officials, even if neither the individual coefficients nor the participatory engagement index (0.17 , $p = 0.13$) are statistically significant.¹³

TABLE 6: Citizen Engagement and Proximity to District Headquarters (2SLS)

	Involved in LCV meetings (1)	Met the RDC (2)	Attended RDC baraza (3)	Participatory index (4)
<i>Panel A: NSDS 2021</i>				
Log Proximity to District HQ	0.086 (0.066)	0.045 (0.043)	0.033 (0.027)	0.169 (0.112)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.255	0.102	0.052	0.020
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745
	Collective civic action (1)	Protest (2)	Contacted media (3)	Contentious index (4)
<i>Panel B: Afrobarometer R10</i>				
Log Proximity to District HQ	-0.176 (0.112)	-0.071** (0.034)	-0.116** (0.058)	-0.381** (0.160)
Observations	2182	2184	2184	2184
Number of clusters	271	271	271	271
Mean of DV	0.360	0.037	0.093	-0.000
IV F-stat (KP rk Wald)	19.413	19.446	19.446	19.446

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of citizens' involvement with the government variables on self-reported household's distance to the district's capital. Column (4) of each panel is a summary index of Columns (1)–(3), constructed to have mean 0 and standard deviation 1. All regressions control for district ever split between 1991–2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991–2020. Standard errors clustered at the Parish level are in parenthesis.

Additional evidence suggests that this shift toward formal engagement is translating into public investments that match local demands. The NSDS community questionnaire asks each

¹³Importantly, Appendix Table A5 confirms that proximity to headquarters does not systematically affect engagement with LC1 leaders (no coefficient is significant at the 5 percent level), reinforcing that the effects documented here are specific to institutions belonging to the districts.

community to rank the projects it considers most important and, for each ranked project, whether it was implemented in the village or parish in the past three years. Appendix Table A21 uses this information at the community level. The results show that communities closer to district headquarters are significantly more likely to have received the classroom, health facility, and road projects they prioritized, and a 10 percent increase in proximity raises the share of top-three priorities implemented by about 3 percentage points, roughly 10 percent of the sample mean. The coefficients for water and electricity are positive but lack statistical significance, and there is no effect for new schools. This pattern suggests that greater interaction with district officials helps channel local preferences into tangible projects.

Taken together, these results support the engagement channel described in the conceptual framework (Proposition 4) by showing that increasing proximity between citizens and the district administrative center increases state presence and institutional knowledge, shifts citizens away from traditional institutions toward formal ones, reduces the need for contentious political action, and increases participatory engagement.

7.2 Corruption Perceptions

A further potential mechanism through which proximity affects public service delivery is accountability. The conceptual framework predicts that proximity strengthens citizen monitoring, deterring corruption (Proposition 6). Table 7 presents two sets of measures: Panel A presents results on households' corruption awareness and their direct experience of victimization, while Panel B contains bribery experiences across different service contexts.

TABLE 7: Corruption and Proximity to District Headquarters (2SLS)

	Aware of nepotism (1)	Aware of embezzlement (2)	Aware of absenteeism (3)	Awareness index ^a (4)	Victim of nepotism (5)	Victim of embezzlement (6)	Victim of absenteeism (7)	Victimisation index ^b (8)
<i>Panel A: NSDS 2021</i>								
Log Proximity to District HQ	0.084 (0.082)	0.044 (0.077)	-0.011 (0.081)	0.078 (0.139)	-0.031* (0.019)	-0.029 (0.025)	-0.086* (0.045)	-0.219** (0.098)
Observations	8412	8412	8413	8413	8417	8417	8417	8417
Number of clusters	967	967	967	967	967	967	967	967
Mean of DV	0.494	0.647	0.487	0.000	0.034	0.038	0.075	0.006
IV F-stat (KP rk Wald)	28.613	28.613	28.656	28.656	28.745	28.745	28.745	28.745
	Bribe for medical care (1)	Bribe for police help (2)	Bribe to avoid police (3)	Bribe for ID document (4)	Bribery index (5)			
<i>Panel B: Afrobarometer R10</i>								
Log Proximity to District HQ	-0.287** (0.118)	-0.274 (0.183)	-0.151 (0.166)	0.092 (0.151)	-0.458** (0.211)			
Observations	1604	345	904	396	1886			
Number of clusters	271	196	257	208	271			
Mean of DV	0.409	0.771	0.523	0.487	-0.047			
IV F-stat (KP rk Wald)	20.664	19.947	12.130	17.949	18.161			

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of corruption variables on self-reported household's distance to the district's capital. Columns (4)^a and (8)^b of Panel A are summary indices of Columns (1)–(3) and (5)–(7), and Column (5) of Panel B of Columns (1)–(4), where each bribery indicator is defined only for respondents who sought the service. Indices are constructed to have mean 0 and standard deviation 1; higher values indicate more corruption. All regressions control for district ever split between 1991–2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991–2020. Standard errors clustered at the Parish level are in parenthesis.

The results reveal that proximity to district headquarters is associated with slightly higher awareness of nepotism, embezzlement, and absenteeism, though none of these coefficients is statistically significant. This suggests that households living closer to administrative centers are not necessarily exposed to more corruption. Importantly, when examining direct victimization, the results show that households closer to headquarters are significantly less likely to have personally experienced nepotism and absenteeism, with a 10 percent increase in proximity reducing the probability of experiencing nepotism or absenteeism by approximately 0.3 and 0.8 percentage points respectively, equivalent to 9 and 11 percent of their sample means. The victimization index, which aggregates the three experiences, is negative and significant at the 5 percent level (-0.22). Consistent with these findings, Panel B reveals that proximity significantly reduces the likelihood of paying a bribe to access health services by approximately 3 percentage points (about 7 percent of the sample mean). The bribery index is also negative and significant at the 5 percent level, reflecting the general pattern in columns (1-4) of Panel B.

These results suggest that administrative proximity strengthens accountability by reducing direct corruption experiences even as awareness of corruption remains high or increases slightly. The combination of high awareness, together with lower victimization, is consistent with a monitoring mechanism. Households closer to district headquarters engage more frequently with local officials and are better able to observe and report misconduct, which in turn may deter corrupt behavior.

8 Democratic perceptions

The evidence so far suggests that bringing the administration closer changes what the state delivers and how citizens interact with it. This section examines whether these changes also affect political attitudes, moving from what the state does to how citizens perceive it. The theoretical case for decentralization has been based on a sequence running from proximity to performance to legitimacy: a government that is closer is easier to observe and hold accountable, performs better as a result, and thereby earns citizens' trust and support for the political system (Oates et al., 1972; Bardhan, 2002; Faguet, 2023). The first links of this chain have empirical support. Decentralization can improve delivery and responsiveness (e.g. (Faguet, 2004)), as the previous sections confirm for Uganda. However, the final link is often assumed rather than established. Credible evidence is scarce because political attitudes and state presence are jointly determined. Governments may expand where support is high or where contestation is politically costly, and communities that trust the state may both demand and receive more of it. As a result, observed correlations between government performance and political trust have limited causal content. Moreover, in hybrid regimes, there are reasons to expect this link to fail. Decentralization can be designed to extend central control rather than empower citizens (Martinez-Bravo et al., 2022), and improvements in government performance not necessarily translate into political trust where democratic institutions are weak (Besley et al., 2026). The setting of this paper helps address this gap. The proximity induced by redistricting provides an exogenous shock

to citizens' exposure to the state, one that improved services, engagement, and accountability. This section studies whether these improvements translate into democratic attitudes, using the same specification as before.

Table 8 details the relationship between proximity to district headquarters and a broad set of democratic perception indicators (Panel A), rule of law perceptions (Panel B), and trust in government (Panel C). The last column of each panel reports a summary index of the outcomes in that panel. The results show that across all fourteen outcomes (including satisfaction with democracy, perceived freedom to vote and organize, evaluations of judicial independence, official accountability, and trust in local government, parliament, the president, the ruling party, the opposition, and the courts) the estimated coefficients are small and statistically insignificant. The three summary indices confirm that this is not a matter of individual outcomes being too noisy. The coefficients on the democratic perceptions, rule of law, and trust indices are small and all have p -values above 0.75. This pattern is robust across specifications and holds for both local and national institutions, suggesting that the gains in service delivery and accountability documented above do not translate into measurable shifts in how citizens evaluate the democratic system or the state more broadly.¹⁴

Interestingly, while trust in government remains unchanged, Appendix Table A22 reveals that proximity does affect other dimensions of trust and collective capacity. Households closer to new district headquarters are significantly more likely to trust people of other religions and other ethnicities, and are more likely to engage in collective action to solve local problems. These findings suggest that administrative proximity generates an environment that fosters social capital even if political trust remains unaffected.

¹⁴Appendix Table A23 shows that the same null pattern holds in the 2017 Afrobarometer round. The six trust in government coefficients are small and statistically insignificant, suggesting that the null is not driven by political trust responding more slowly than service delivery, as it holds both before and after the service effects documented in Section 10 materialized.

TABLE 8: Democratic Perceptions and Proximity to District Headquarters (2SLS)

	Country is a democracy (1)	Satisfied with democracy (2)	Free to vote (3)	Free to join organisations (4)	Democracy index (5)		
<i>Panel A: Democratic perceptions</i>							
Log Proximity to District HQ	-0.012 (0.053)	0.072 (0.085)	-0.015 (0.044)	-0.064 (0.060)	-0.031 (0.112)		
Observations	2079	2108	2159	2151	2179		
Number of clusters	271	271	271	271	271		
Mean of DV	0.882	0.593	0.934	0.847	0.001		
IV F-stat (KP rk Wald)	19.352	19.744	20.486	21.665	20.323		
	President ignores law (1)	Unequal under law (2)	Judges influenced (3)	Officials go unpunished (4)	Rule of law index (5)		
<i>Panel B: Rule of law perceptions</i>							
Log Proximity to District HQ	-0.009 (0.098)	0.020 (0.101)	-0.100 (0.108)	-0.029 (0.111)	-0.057 (0.180)		
Observations	1859	2112	1980	2075	2161		
Number of clusters	271	271	271	271	271		
Mean of DV	0.366	0.640	0.601	0.607	-0.011		
IV F-stat (KP rk Wald)	22.695	20.535	19.684	20.123	19.437		
	Trust local government (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)	Trust index (7)
<i>Panel C: Trust in government</i>							
Log Proximity to District HQ	0.029 (0.079)	-0.066 (0.084)	0.043 (0.099)	0.044 (0.084)	0.017 (0.086)	0.024 (0.083)	0.028 (0.111)
Observations	2132	2116	2184	2184	2184	2087	2184
Number of clusters	271	271	271	271	271	271	271
Mean of DV	0.507	0.695	0.503	0.589	0.315	0.526	-0.005
IV F-stat (KP rk Wald)	20.692	19.058	19.446	19.446	19.446	21.146	19.446

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of democratic perception variables on self-reported household's distance to the district's capital. The last column of each panel is a summary index of the outcomes in that panel, constructed to have mean 0 and standard deviation 1; higher values indicate more favourable democratic perceptions (Panel A), more negative rule-of-law perceptions (Panel B), and more trust in government (Panel C). All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

9 Heterogeneous Effects by Local Political Alignment

The conceptual framework predicts that political alignment between the elected LCV and the appointed RDC amplifies the effects of proximity on service delivery (Proposition 3), participatory engagement (Proposition 5), and accountability (Proposition 7). This section tests these predictions. As described in Section 2, the LCV is accountable to local voters, while the RDC is accountable to the central government. When these two figures share a political affiliation, their incentives are more likely to be complementary, easing coordination

over resource allocation and program implementation. When they are not aligned, frictions may emerge as each administrator responds to different principals with potentially different priorities. I classify a district as aligned when the LCV chairperson and the RDC belonged to the same party following the 2016 elections.¹⁵ I then examine whether the effects of proximity to district headquarters vary with political alignment between appointed and elected local leaders.

Tables 9 to 14 report the 2SLS estimates for the main outcomes and mechanisms analyzed above, including the effect of proximity in non-aligned districts, the interaction between proximity and alignment, and the sum of these two coefficients (which gives the effect in aligned districts). For outcomes for which the framework predicts the sign of the interaction term, the tables also report the predicted sign and the corresponding one-sided p -value (significance stars correspond to two-sided tests). In addition, the tables report the Kleibergen–Paap F-statistic for each group’s first stage separately. It ranges between 12 and 17 in the NSDS, but falls to between 3 and 9 in the aligned Afrobarometer sample, with the lowest values for the bribery items, whose samples are restricted to respondents who sought the corresponding service. The Afrobarometer panels therefore also report Anderson–Rubin 95 percent confidence intervals, which remain valid under weak instruments. Appendix Tables A11 and A12 report these confidence intervals for every coefficient in the six tables.

9.1 Service Delivery

As alignment enters the framework through the appointed RDC, the complementarity predicted by Proposition 3 should be strongest for services in which the RDC plays a larger coordinating role. In the Ugandan setting, this applies primarily to services financed by the central government (Section 2), since the RDC represents the central government at the district level and helps coordinate the implementation of centrally funded programs. Within this group, the effects of alignment should be particularly pronounced for infrastructure such as roads and electricity, where the spatial allocation of investments is relatively discretionary. By contrast, district-provided services operate mainly through the elected council’s own administrative structure and should therefore respond to proximity in both aligned and non-aligned districts.

Table 9 provides evidence consistent with these predictions. For district-provided services (Panel A), proximity improves health access in both aligned and non-aligned districts, while none of the interaction terms is statistically significant. Among central government services (Panel B), the interaction is largest for roads, where the effect of proximity is 0.81 standard deviations larger in aligned districts (significant at the 10 percent level). The interaction is also positive for electricity (0.17), although it is not statistically significant. These are the two services for which the pooled effects of proximity are strongest in Table 3. By contrast, the interactions are close to zero for central schooling, health, and water. Agricultural inputs are the

¹⁵In the sample, 101 of the 134 districts are classified as aligned.

only district-provided service with a sizeable interaction (0.38), although this pattern is driven by a negative effect of proximity in non-aligned districts rather than by a larger positive effect in aligned ones. Overall, the evidence for Proposition 3 is concentrated in central infrastructure, where the RDC's coordinating role and discretion over implementation are likely to matter the most.

TABLE 9: Effects on Services and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	School (1)	Health (2)	Water (3)	Roads (4)	Agricultural inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
Proximity, non-aligned	-0.018 (0.275)	0.383** (0.157)	0.308 (0.203)	0.007 (0.150)	-0.277* (0.165)
Proximity $\times$ Aligned	0.432 (0.412)	-0.058 (0.229)	-0.121 (0.298)	0.094 (0.217)	0.382 (0.255)
Proximity, aligned (sum)	0.413 (0.307)	0.324* (0.167)	0.187 (0.218)	0.101 (0.157)	0.105 (0.195)
Predicted sign of interaction	(+)	(+)	(+)	(+)	(+)
One-sided p -value	0.147	0.600	0.658	0.333	0.067
Observations	7,137	7,595	8,417	8,417	8,417
Number of clusters	958	951	967	967	967
Mean of DV	-0.003	-0.016	-0.004	0.008	0.030
KP F (aligned / non-aligned)	9.9 / 16.7	17.0 / 11.8	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
Proximity, non-aligned	-0.026 (0.244)	0.070 (0.344)	0.337 (0.213)	0.202 (0.274)	0.254 (0.167)
Proximity $\times$ Aligned	-0.000 (0.393)	0.087 (0.436)	-0.096 (0.282)	0.811* (0.472)	0.170 (0.255)
Proximity, aligned (sum)	-0.026 (0.309)	0.157 (0.268)	0.241 (0.184)	1.013*** (0.384)	0.424** (0.193)
Predicted sign of interaction	(+)	(+)	(+)	(+)	(+)
One-sided p -value	0.500	0.421	0.634	0.043	0.253
Observations	5,553	8,417	8,417	8,417	8,417
Number of clusters	951	967	967	967	967
Mean of DV	-0.001	0.013	-0.017	-0.070	-0.043
KP F (aligned / non-aligned)	9.4 / 9.7	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, two-sided tests. Each column is a 2SLS regression of the outcome on log proximity to the district headquarters and its interaction with an indicator for politically aligned districts. The rows report the effect of proximity in non-aligned districts, the interaction, and their sum, the effect in aligned districts. The predicted sign is the sign of the interaction implied by Proposition 3, and the one-sided p -value tests the interaction against that sign. Standard errors clustered at the parish level are in parentheses. Observations and the mean of the dependent variable refer to the pooled sample; the indices have mean 0 and standard deviation 1 in the full sample. The Kleibergen–Paap rk Wald F-statistic is reported for the aligned and the non-aligned first stage separately. Anderson–Rubin 95 percent confidence sets for every coefficient in this table are reported in Appendix Table A11. Controls and weights as in Table 3; the excluded instruments are the one used there and its interaction with the alignment indicator.

9.2 Citizen Engagement and State Presence

The engagement results provide strong support for the predictions of the theoretical framework. Table 10, Panel B, shows that the reduction in contentious engagement documented in Section 7.1 is concentrated in aligned districts. In these districts, a 10 percent increase in proximity lowers protest participation by 1.6 pp and collective civic action by 5.0 pp (equivalent to 44 and 14 percent of their respective means), while the coefficients in non-aligned districts are small and positive. The corresponding interaction terms are -0.19 for protest and -0.56 for collective action, both significant at the 5 percent level. For media contact, the interaction is -0.17 and is significant at the 10 percent level in the one-sided test. These results are also reflected in the summary index, which shows that the decrease in contentious engagement is driven by aligned districts. Importantly, these patterns are not driven by the weak first stage in the aligned Afrobarometer sample. The Anderson–Rubin confidence intervals for the aligned coefficients exclude zero for all three outcomes, while the non-aligned confidence intervals are narrow and contain zero.

The evidence on participatory engagement (Panel A) points towards stronger effects of proximity for aligned districts. Involvement in LCV meetings rises with proximity only in aligned districts (0.18, significant at the 10 percent level), with the interaction term reaching 0.19 and having a one-sided p -value of 0.07. The interactions for meeting the RDC and attending an RDC baraza also have the predicted sign, although they are not statistically significant. The participatory engagement index shows the same pattern. It rises with proximity only in aligned districts (0.32, significant at the 10 percent level), with an interaction of 0.32 and a one-sided p -value of 0.07, while it is close to zero and not statistically significant in non-aligned districts.

Table 11 shows that the stronger presence of the state near district headquarters is also concentrated in aligned districts. Knowledge of the magistrate court and the police rises with proximity in aligned districts (0.39 and 0.12, respectively) but not in non-aligned districts, and the corresponding interaction terms are positive and significant at the 5 percent level. The exception is knowledge of the Auditor General, which increases with proximity only in non-aligned districts, although the difference across groups is not statistically significant. Importantly, the summary index in column 6 shows that the increase in knowledge is differentially more concentrated in aligned districts. The results in Panel B also point toward stronger substitution away from traditional leaders in aligned districts. Proximity reduces contact with traditional leaders by 0.28 in aligned districts, compared with -0.01 in non-aligned districts, with an interaction of -0.26 (with a one-sided p -value of 0.107). At the same time, the share of households with a family member in the civil service rises with proximity only in aligned districts (0.09, $p < 0.05$). The state-reliance index, which combines the four Afrobarometer indicators with the traditional-institution items reversed, is larger in aligned districts (0.36 against 0.11) but neither coefficient nor their difference is statistically significant.

Taken together, the patterns in Tables 10 and 11 are consistent with the threshold mechanism

TABLE 10: Citizen Engagement and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	Involved in LCV meetings (1)	Met the RDC (2)	Attended RDC baraza (3)	Participatory index (4)
<i>Panel A: NSDS 2021 – Participatory engagement</i>				
Proximity, non-aligned	-0.011 (0.083)	0.010 (0.043)	0.000 (0.040)	0.004 (0.126)
Proximity $\times$ Aligned	0.190 (0.128)	0.062 (0.079)	0.065 (0.058)	0.317 (0.213)
Proximity, aligned (sum)	0.179* (0.097)	0.072 (0.066)	0.065 (0.042)	0.321* (0.172)
Predicted sign of interaction	(+)	(+)	(+)	(+)
One-sided p -value	0.068	0.218	0.134	0.068
Observations	8,417	8,417	8,417	8,417
Number of clusters	967	967	967	967
Mean of DV	0.255	0.102	0.052	0.020
KP F (aligned / non-aligned)	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5
	Collective civic action (1)	Protest (2)	Contacted media (3)	Contentious index (4)
<i>Panel B: Afrobarometer R10 – Contentious engagement</i>				
Proximity, non-aligned	0.058 (0.059)	0.024 (0.026)	-0.025 (0.049)	0.054 (0.116)
AR 95% set	[-0.11, 0.21]	[-0.03, 0.11]	[-0.17, 0.09]	[-0.24, 0.36]
Proximity $\times$ Aligned	-0.555** (0.256)	-0.187** (0.080)	-0.170 (0.119)	-0.913** (0.400)
Proximity, aligned (sum)	-0.498** (0.249)	-0.164** (0.076)	-0.195* (0.108)	-0.860** (0.383)
AR 95% set	[-3.60, -0.19]	[-1.06, -0.07]	[-1.50, -0.06]	[-5.98, -0.43]
Predicted sign of interaction	(-)	(-)	(-)	(-)
One-sided p -value	0.015	0.010	0.075	0.011
Observations	2,182	2,184	2,184	2,184
Number of clusters	271	271	271	271
Mean of DV	0.360	0.037	0.093	-0.000
KP F (aligned / non-aligned)	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 9. The predicted sign is the sign of the interaction implied by Proposition 5, and the one-sided p -value tests the interaction against that sign. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A11.

captured by $\hat{d}(\alpha)$. When the state is more responsive to formal input, citizens living closer to the headquarters are more likely to recognize state institutions and shift away from contentious engagement toward participatory channels. When responsiveness is lower, proximity has little

TABLE 11: Institutional Knowledge and State-Traditional Substitution (2SLS) – Aligned versus Non-Aligned Districts

	Knows magistrate court (1)	Knows police (2)	Knows land office (3)	Knows OAG (4)	Knows judiciary (5)	Knowledge index (6)
<i>Panel A: NSDS 2021 – Knowledge of state institutions</i>						
Proximity, non-aligned	0.010 (0.120)	-0.054 (0.041)	0.032 (0.062)	0.125** (0.061)	-0.039 (0.078)	0.049 (0.123)
Proximity × Aligned	0.375** (0.172)	0.173** (0.088)	0.086 (0.106)	-0.072 (0.079)	0.188 (0.129)	0.364* (0.203)
Proximity, aligned (sum)	0.385*** (0.123)	0.120 (0.077)	0.119 (0.086)	0.052 (0.050)	0.149 (0.102)	0.413** (0.161)
Predicted sign of interaction	(+)	(+)	(+)	(+)	(+)	(+)
One-sided <i>p</i> -value	0.015	0.024	0.208	0.821	0.072	0.036
Observations	8,413	8,415	8,413	8,417	8,417	8,417
Number of clusters	967	967	967	967	967	967
Mean of DV	0.375	0.930	0.156	0.092	0.384	0.001
KP F (aligned / non-aligned)	14.3 / 12.4	14.3 / 12.5	14.3 / 12.4	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5
	Family member civil servant (1)	Conflict resolved by government (2)	Conflict resolved by trad. leader (3)	Contacted MP (4)	Contacted trad. leader (5)	State reliance index (cols 2-5) (6)
<i>Panel B: NSDS 2021 and Afrobarometer R10 – State-traditional substitution</i>						
Proximity, non-aligned	0.044 (0.059)	0.152* (0.085)	-0.011 (0.050)	0.043 (0.073)	-0.011 (0.073)	0.113 (0.085)
AR 95% set	[-0.08, 0.21]	[-0.01, 0.46]	[-0.14, 0.12]	[-0.19, 0.19]	[-0.18, 0.21]	[-0.12, 0.33]
Proximity × Aligned	0.041 (0.071)	-0.145 (0.240)	-0.147 (0.201)	0.125 (0.144)	-0.264 (0.212)	0.249 (0.331)
Proximity, aligned (sum)	0.086** (0.040)	0.007 (0.224)	-0.158 (0.195)	0.168 (0.125)	-0.275 (0.199)	0.362 (0.320)
AR 95% set	[0.01, 0.21]	[-0.64, 1.29]	[-1.78, 0.22]	[-0.07, 1.16]	[-2.38, 0.04]	[-0.19, 3.29]
Predicted sign of interaction	(+)	(+)	(-)	(+)	(-)	(+)
One-sided <i>p</i> -value	0.280	0.727	0.232	0.193	0.107	0.226
Observations	8,414	2,131	2,131	2,178	2,146	2,184
Number of clusters	967	271	271	271	271	271
Mean of DV	0.057	0.556	0.211	0.168	0.277	-0.001
KP F (aligned / non-aligned)	14.4 / 12.4	6.3 / 20.7	6.3 / 20.7	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8

Notes: **p*<0.1, ***p*<0.05, ****p*<0.01, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 9. The predicted sign is the sign of the interaction implied by Proposition 5, and the one-sided *p*-value tests the interaction against that sign. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A11.

effect on how citizens engage with the state.

9.3 Accountability

The framework links proximity to corruption through two channels. Citizens living closer to the headquarters monitor administrators more intensely, while in aligned districts the RDC internalises the welfare cost of diversion, extending the range of distances over which corruption is deterred (Proposition 7). Table 12 analyzes these channels, distinguishing awareness of corruption and direct experiences of it. As above, Panel A reports whether households have heard of specific corrupt practices, which proxies for the monitoring intensity $\mu(d_i)$ in the model, alongside whether they have been victims of these practices in the past year. Panel B complements these measures with bribery experiences from the Afrobarometer.

The awareness results provide evidence of stronger monitoring close to responsive district headquarters. In aligned districts, households closer to the headquarters are more likely to have heard of nepotism (0.22, significant at the 10 percent level), embezzlement (0.17), and absenteeism (0.06). In non-aligned districts, the pattern reverses: all three coefficients are negative and the index falls by 0.32 (significant at the 10 percent level). The interaction terms are positive for every item and are significant at the 1 percent level for nepotism and at the 5 percent level for embezzlement and for the summary index. Proximity to a responsive headquarters therefore seems to make citizens better informed about misconduct, while proximity to a non-aligned administration does not generate the same increase in awareness. The positive interaction is nevertheless what one would expect if citizens acquire information about misconduct through the participatory channels that Proposition 5 predicts to be more active in aligned districts.¹⁶

Greater awareness goes together with fewer direct experiences of corruption, in line with Proposition 6, although this reduction is not concentrated in aligned districts. The victimisation index falls with proximity in both groups (-0.18 in aligned and -0.28 in non-aligned districts, the latter significant at the 10 percent level), leaving the interaction close to zero.

The bribery results in Panel B provide stronger support for Proposition 7. The bribery index falls by 0.82 in aligned districts (significant at the 10 percent level) and is unchanged in non-aligned districts, with an interaction of -0.82 and a one-sided p -value of 0.06. The effect is concentrated in bribes for medical care, which is the outcome with the largest sample. The aligned first stage is weak for these outcomes (Kleibergen–Paap F between 3 and 7), so the Anderson–Rubin confidence intervals are wide. They nevertheless exclude zero for the medical-care bribe and for the index, while the corresponding confidence intervals for non-aligned districts are narrow and contain zero. Bribes for police assistance fall with proximity in both groups, while the remaining two items are imprecisely estimated in smaller samples.¹⁷

¹⁶The framework predicts higher awareness closer to the headquarters, since monitoring intensity $\mu(d_i)$ depends on distance alone, but it does not generate a prediction for the interaction, which is why the awareness columns carry no predicted sign in the table.

¹⁷The aligned bribery results should be taken with care as the first-stage F -statistics is low, and the Anderson–Rubin confidence interval for the index is unbounded below. The evidence supports the sign of the effect and its

Taken together, the panels are consistent with the monitoring mechanism of the framework rather than with worse governance near the headquarters. In aligned districts, citizens living closer to the headquarters are more aware of corruption, report fewer direct experiences of it, and pay fewer bribes. In non-aligned districts, proximity lowers both awareness and reported victimisation. This pattern is consistent with weaker exposure to (or observation of) an administration that is less responsive to local citizens.

TABLE 12: Corruption and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	Aware of nepotism (1)	Aware of embezzlement (2)	Aware of absenteeism (3)	Awareness index (4)	Victim of nepotism (5)	Victim of embezzlement (6)	Victim of absenteeism (7)	Victimisation index (8)
<i>Panel A: NSDS 2021</i>								
Proximity, non-aligned	-0.225** (0.104)	-0.138 (0.113)	-0.106 (0.094)	-0.321* (0.188)	0.002 (0.022)	-0.080* (0.043)	-0.108 (0.083)	-0.277* (0.159)
Proximity × Aligned	0.447*** (0.157)	0.308** (0.155)	0.169 (0.146)	0.630** (0.268)	-0.050 (0.037)	0.070 (0.056)	0.047 (0.099)	0.093 (0.206)
Proximity, aligned (sum)	0.221* (0.117)	0.170 (0.106)	0.063 (0.112)	0.308 (0.190)	-0.047 (0.030)	-0.011 (0.036)	-0.061 (0.055)	-0.184 (0.132)
Predicted sign of interaction	n.p.	n.p.	n.p.	n.p.	(−)	(−)	(−)	(−)
One-sided <i>p</i> -value					0.092	0.893	0.683	0.674
Observations	8,412	8,412	8,413	8,413	8,417	8,417	8,417	8,417
Number of clusters	967	967	967	967	967	967	967	967
Mean of DV	0.494	0.647	0.487	0.000	0.034	0.038	0.075	0.006
KP F (aligned / non-aligned)	14.3 / 12.5	14.3 / 12.5	14.3 / 12.5	14.3 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5	14.4 / 12.5
	Bribe for medical care (1)	Bribe for police help (2)	Bribe to avoid police (3)	Bribe for ID document (4)	Bribery index (5)			
<i>Panel B: Afrobarometer R10</i>								
Proximity, non-aligned	-0.019 (0.066)	-0.299** (0.121)	0.034 (0.112)	-0.034 (0.141)	-0.008 (0.112)			
AR 95% set	[−0.18, 0.16]	[−0.81, −0.09]	[−0.25, 0.42]	[−0.47, 0.32]	[−0.28, 0.29]			
Proximity × Aligned	-0.464* (0.256)	-0.100 (0.404)	-0.149 (0.329)	0.317 (0.281)	-0.816 (0.512)			
Proximity, aligned (sum)	-0.483* (0.248)	-0.398 (0.385)	-0.115 (0.309)	0.284 (0.243)	-0.823* (0.500)			
AR 95% set	[−3.42, −0.15]	[−3.78, 0.26]	(−∞, ∞)	[−0.42, 2.13]	(−∞, −0.20]			
Predicted sign of interaction	(−)	(−)	(−)	(−)	(−)			
One-sided <i>p</i> -value	0.035	0.403	0.326	0.871	0.056			
Observations	1,604	345	904	396	1,886			
Number of clusters	271	196	257	208	271			
Mean of DV	0.409	0.771	0.523	0.487	-0.047			
KP F (aligned / non-aligned)	6.5 / 26.1	6.8 / 16.8	2.9 / 16.0	6.6 / 12.9	4.8 / 24.2			

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 9. The predicted sign is the sign of the interaction implied by Proposition 7, and the one-sided *p*-value tests the interaction against that sign. n.p. marks outcomes for which the framework makes no prediction. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A11.

9.4 Satisfaction with Local Leaders

Table 13 shows that the higher satisfaction with the RDC documented in Section 6 is concentrated in aligned districts. In these districts, a 10 percent increase in proximity raises the probability of reporting satisfaction with the RDC by 2.1 pp, while it has a negative non-statistically significant coefficient in non-aligned districts. The interaction term is 0.26 and is significant at the 5 percent level. Ratings of the elected LCV chairperson display the opposite pattern. Proximity increases satisfaction with the LCV only in non-aligned districts, although concentration in aligned districts, but not its magnitude.

the interaction term is not statistically significant. The framework does not generate a specific prediction for this outcome, but the contrast is informative. In districts where the two leaders are not politically aligned, households closer to the headquarters appear to attribute improvements more to the elected LCV than to the appointed RDC, while in aligned districts households closer show higher satisfaction with the RDC work.

TABLE 13: Satisfaction with Local Leaders and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	LCV performance (1)	RDC satisfaction (2)	Civil servants performance (3)
<i>NSDS 2021</i>			
Proximity, non-aligned	0.237* (0.130)	-0.056 (0.084)	0.062 (0.107)
Proximity $\times$ Aligned	-0.182 (0.164)	0.264** (0.120)	0.046 (0.161)
Proximity, aligned (sum)	0.055 (0.101)	0.208** (0.086)	0.108 (0.120)
Observations	8,417	8,417	7,451
Number of clusters	967	967	966
Mean of DV	0.428	0.329	0.547
KP F (aligned / non-aligned)	14.4 / 12.5	14.4 / 12.5	14.7 / 12.6

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 9. The framework makes no prediction for these outcomes, so no predicted sign is reported. Anderson–Rubin 95 percent confidence sets for every coefficient in this table are reported in Appendix Table A11.

9.5 Democratic Perceptions

While the framework does not generate a prediction for democratic perceptions, the heterogeneity by political alignment is informative. In aligned districts, proximity leaves assessments of democracy, perceived political freedoms, rule-of-law evaluations, and trust in political institutions largely unchanged (Table 14). In contrast, in non-aligned districts proximity lowers perceived freedom to vote (-0.11 , significant at the 1 percent level), increases the perception that the president ignores the law (0.15 , $p < 0.05$), and reduces trust in both local government and the ruling party (both -0.13 , $p < 0.10$). The interaction terms for these four outcomes are significant at the 5 or 10 percent level, while the corresponding effects in aligned districts are close to zero. These results for non-aligned districts are supported by strong first stages, with F-statistics between 17 and 21. The Anderson–Rubin confidence intervals also exclude zero for freedom to vote, the perception that the president ignores the law, and trust in local government, while the set for trust in the ruling party, marginally includes zero. Taken together, these patterns suggest that proximity does not build political legitimacy even where administrative coordination is stronger and may instead erode it in non-aligned districts. One possible interpretation is that citizens living closer to the headquarters are more exposed to

frictions between the elected and appointed administrators.

TABLE 14: Democratic Perceptions and Proximity to District HQ (2SLS) – Aligned versus Non-Aligned Districts

	Country is a democracy (1)	Satisfied with democracy (2)	Free to vote (3)	Free to join organisations (4)	Democracy index (5)		
<i>Panel A: Democratic perceptions</i>							
Proximity, non-aligned	-0.002 (0.038)	0.031 (0.063)	-0.113*** (0.032)	-0.033 (0.050)	-0.109 (0.078)		
AR 95% set	[−0.11, 0.09]	[−0.15, 0.18]	[−0.21, −0.04]	[−0.19, 0.08]	[−0.35, 0.06]		
Proximity × Aligned	-0.013 (0.097)	0.108 (0.173)	0.160** (0.082)	-0.067 (0.114)	0.138 (0.213)		
Proximity, aligned (sum)	-0.015 (0.089)	0.138 (0.161)	0.047 (0.075)	-0.099 (0.102)	0.029 (0.198)		
AR 95% set	[−0.39, 0.30]	[−0.15, 1.43]	[−0.09, 0.54]	[−0.55, 0.15]	[−0.45, 1.09]		
Observations	2,079	2,108	2,159	2,151	2,179		
Number of clusters	271	271	271	271	271		
Mean of DV	0.882	0.593	0.934	0.847	0.001		
KP F (aligned / non-aligned)	6.4 / 21.1	6.4 / 21.5	7.1 / 18.9	7.8 / 19.6	6.9 / 20.1		
	President ignores law (1)	Unequal under law (2)	Judges influenced (3)	Officials go unpunished (4)	Rule of law index (5)		
<i>Panel B: Rule of law perceptions</i>							
Proximity, non-aligned	0.151** (0.067)	-0.113 (0.089)	-0.014 (0.086)	-0.014 (0.088)	-0.011 (0.131)		
AR 95% set	[0.02, 0.41]	[−0.43, 0.06]	[−0.30, 0.16]	[−0.31, 0.17]	[−0.44, 0.26]		
Proximity × Aligned	-0.327* (0.182)	0.222 (0.175)	-0.194 (0.219)	-0.072 (0.204)	-0.145 (0.331)		
Proximity, aligned (sum)	-0.176 (0.169)	0.108 (0.150)	-0.208 (0.201)	-0.087 (0.184)	-0.156 (0.303)		
AR 95% set	[−1.00, 0.13]	[−0.34, 0.70]	[−1.95, 0.11]	[−1.18, 0.34]	[−2.21, 0.47]		
Observations	1,859	2,112	1,980	2,075	2,161		
Number of clusters	271	271	271	271	271		
Mean of DV	0.366	0.640	0.601	0.607	-0.011		
KP F (aligned / non-aligned)	8.8 / 16.7	7.4 / 18.5	6.7 / 17.7	6.8 / 19.5	6.6 / 19.0		
	Trust local government (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)	Trust index (7)
<i>Panel C: Trust in government</i>							
Proximity, non-aligned	-0.129* (0.077)	-0.083 (0.067)	0.012 (0.079)	-0.126* (0.074)	0.079 (0.059)	0.091 (0.056)	-0.048 (0.092)
AR 95% set	[−0.45, −0.01]	[−0.28, 0.06]	[−0.21, 0.20]	[−0.40, 0.01]	[−0.04, 0.28]	[−0.10, 0.20]	[−0.37, 0.13]
Proximity × Aligned	0.249* (0.145)	0.034 (0.160)	0.011 (0.173)	0.273* (0.156)	-0.077 (0.158)	-0.059 (0.154)	0.142 (0.204)
Proximity, aligned (sum)	0.121 (0.123)	-0.050 (0.146)	0.023 (0.155)	0.146 (0.137)	0.002 (0.146)	0.032 (0.144)	0.094 (0.181)
AR 95% set	[−0.18, 0.78]	[−0.64, 0.65]	[−0.62, 0.60]	[−0.15, 1.11]	[−0.75, 0.44]	[−0.37, 0.70]	[−0.43, 1.07]
Observations	2,132	2,116	2,184	2,184	2,184	2,087	2,184
Number of clusters	271	271	271	271	271	271	271
Mean of DV	0.507	0.695	0.503	0.589	0.315	0.526	-0.005
KP F (aligned / non-aligned)	7.0 / 20.8	6.0 / 20.0	6.4 / 19.8	6.4 / 19.8	6.4 / 19.8	7.0 / 20.4	6.4 / 19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Specification, rows, standard errors, pooled statistics, first-stage F-statistics, controls and instruments as in Table 9. The framework makes no prediction for these outcomes, so no predicted sign is reported. In the Afrobarometer panels, where the aligned first stage is weak, the rows labelled AR 95% set report Anderson–Rubin confidence sets for the two subsample coefficients (none exists for the interaction); sets for all coefficients are in Appendix Table A11.

9.6 Summary

Overall, the alignment results guided by the theoretical framework help explain both the main findings and the limits of decentralization in hybrid regimes. When the incentives of the elected and appointed district leaders are aligned, bringing the state closer improves service

delivery, increases the visibility of state institutions, shifts citizen engagement toward more formal channels, and reduces direct experiences of corruption. However, these gains do not translate into stronger democratic support. When the two leaders are not aligned, the benefits of proximity are more limited, engagement responds little, and proximity is associated with lower political trust and less favorable democratic perceptions along some dimensions. Taken together, these results suggest that reducing the physical distance between citizens and government can improve how the state functions locally, but that the benefits of decentralization depend on the political environment in which local institutions operate.

10 Dynamics

So far the results analyzed documented the effects of distance to headquarters on public service delivery from a static perspective. However, it is likely that these effects vary depending on the stage of development of the administrative centers. This section investigates the potential dynamic effects of distance on public service delivery using data from the NSDS 2015 and the NSDS 2021, and then uses remote-sensing data to document how land use in the headquarters parishes themselves evolves after designation.

Given the cross-sectional nature of the NSDS, the different rounds of data collection do not capture information from the same households. Nevertheless, comparing cross-sectional coefficients of the impacts of distance to headquarters on public service delivery within a 5-year time span in the same districts can provide valuable insights of the evolution of these effects over time. To do so, I focus on the evolution of the importance of the distance to administrative headquarters for households belonging to districts created between 1991 and 2014 (one year before the earliest available dataset). To guide the analysis, I take the NSDS 2015 as the benchmark and analyze how the magnitude and significance of the distance to administrative headquarters change in both samples 2015 and 2021.

Since the NSDS 2015 does not include household geo-coordinates nor consistent parish names, I approximate a household's location using the household's sub-county centroid as its geocoordinate. Thus, I calculate the distance to the district's centroid by using the distance between the household's sub-county centroid and the district's centroid. Although this measure may contain some measurement error, its primary purpose is to capture the exogenous variation in distance introduced by the creation of a new district.

10.1 Dynamics - Main results

Table 15 presents results for the major outcome indices in districts created between 1991 and 2014, using data from the NSDS 2015 (Panel A) and the NSDS 2021 (Panel B). Comparing the two panels reveals an informative evolution in the effects of proximity over time.

In 2015, proximity to district headquarters had limited effects on most service domains.

TABLE 15: DYNAMICS IMPACT OF HOUSEHOLD'S DISTANCE TO HEADQUARTERS ON INDICES OF MAJOR OUTCOMES

	Education district (1)	Education central gov (2)	Health district (3)	Water district (4)	Water central gov (5)	Electricity central gov (6)	Roads district (7)	Agr inputs district (8)
<i>Panel A. NSDS 2015 - Period 1991-2014</i>								
Log SR Proximity HH - District HQ	0.155	0.063	0.014	0.067	-0.115	-0.172	0.103	1.274***
SE	(0.143)	(0.122)	(0.157)	(0.197)	(0.200)	(0.154)	(0.197)	(0.391)
p-value	[0.280]	[0.605]	[0.929]	[0.734]	[0.565]	[0.264]	[0.603]	[0.001]
sharpened q-value	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{0.010}
Mean of Dependent Variable	-0.017	-0.006	0.104	0.004	0.010	-0.019	0.005	0.026
Observations	5,271	4,048	8,918	8,934	8,928	8,940	8,940	8,940
Number of Parishes	946	938	953	953	953	953	953	953
IV F-stat Kleibergen-Paap rk Wald	16.620	18.056	17.799	17.512	17.598	17.505	17.505	17.505
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Panel B. NSDS 2021 - Period 1991-2014</i>								
Log SR Proximity HH - District HQ	0.262	0.173	0.397***	0.295**	0.377***	0.346***	0.085	-0.093
SE	(0.193)	(0.171)	(0.107)	(0.131)	(0.129)	(0.116)	(0.097)	(0.129)
p-value	[0.174]	[0.313]	[0.000]	[0.025]	[0.004]	[0.003]	[0.382]	[0.468]
sharpened q-value	{0.162}	{0.264}	{0.002}	{0.033}	{0.009}	{0.009}	{0.280}	{0.306}
Mean of Dependent Variable	-0.016	0.004	-0.035	-0.093	-0.036	-0.060	-0.018	0.087
Observations	6,555	5,148	6,959	7,752	7,752	7,752	7,752	7,752
Number of Parishes	885	879	878	894	894	894	894	894
IV F-stat Kleibergen-Paap rk Wald	27.382	28.469	40.570	36.997	36.997	36.997	36.997	36.997
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. All regressions control for district ever split between 1991-2014, lake Victoria district, districts' area in Km² as of 1991, household's parish centroid distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2014. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values are in curly brackets.

The only statistically significant result is a large positive effect on agricultural inputs, robust to multiple hypothesis testing corrections. All remaining indices in education, health, water, electricity, and roads are small and insignificant. By 2021, this pattern changes considerably. Proximity is no longer affecting the agricultural inputs index, while it now significantly improves access to health services, water at both district and central government levels, and electricity.

These dynamics suggest that the relevance of administrative proximity for public service delivery strengthened considerably between 2015 and 2021, and that the domains in which it matters shifted over time. Two complementary forces could explain this pattern. First, the gradual consolidation of administrative capacity in newly created districts: the rapid pace of district creation between 2005 and 2010 imposed significant burdens on government capacity, generating delays in the effective establishment of district headquarters (Green, 2015; Cohen, 2024), so that many districts surveyed in 2015 may still have been in the early stages of administrative development. Second, the differential timeline of public investments: a new administration can quickly influence access to fast-turnaround services such as the distribution of agricultural inputs through existing extension services structures, but infrastructure investments require several years from planning and procurement through to completion. The pattern in Table 15 is consistent with this. While in 2015 the only significant effect was on agricultural inputs, an outcome that responds quickly to administrative capacity, by 2021 infrastructure investments near new headquarters materialized, producing the significant effects on health,

water, and electricity access documented in the main results. While the evidence is suggestive, the combination of consolidating administrative capacity and the timeline of infrastructure provision goes in line with the main results and with existing research of the administrative constraints facing newly created districts in Uganda (Cohen, 2024).

To complement the NSDS results, I use Afrobarometer to provide further evidence on this consolidation pattern. The Afrobarometer records service availability at the enumeration area level, based on direct observation by enumerators rather than household self reports, and uses identical questions in the 2017 and 2024 rounds (Appendix Table A24). The results show that in 2017, proximity to district headquarters has no significant effect on five of the six availability indicators, with only a marginal effect on piped water. By 2024, the effects on piped water, paved roads, and police stations are large and statistically significant. The evidence coming from the Afrobarometer mirrors the evolution observed in the NSDS, reinforcing the interpretation that the returns to administrative proximity increased as new district institutions became more established.

10.2 Land Use Dynamics After Headquarters Establishment

The consolidation pattern documented above should also leave a physical footprint in the headquarters parishes themselves. Figure 4 shows that the share of parish land devoted to cultivated cropland, defined as pixels with more than 60 percent cultivation, increases continuously after a parish acquires headquarters status, and the effect persists up to 14 years after creation. This expansion is accompanied by a decline in the share of cropland mixed with natural vegetation, a land class that includes pixels with 40 to 60 percent cultivation (Appendix Figure A4), consistent with cultivation intensifying on land that was already partly farmed rather than with new land being cleared. Built-up area moves in the same direction. Appendix Figure A3, based on the GHS-BUILT-S product, shows that the built-up share of headquarters parishes rises steadily and monotonically after creation, by roughly 1.1 percentage points over twenty years, on the order of 40 percent of the baseline built-up share (joint nullity $p = 0.003$), with no pre-existing upward trend.

Taken together, these patterns indicate that headquarters designation spurs local agricultural activity and urbanization, and that this footprint builds up gradually over more than a decade rather than appearing at once. This is the same timing that the survey-based results above suggest: the returns to administrative proximity are limited in the years immediately after creation and strengthen as district institutions, and the towns that host them, become established.

11 The Fiscal Cost of District Creation

The former sections establish that proximity to new district headquarters improves access to public goods and services, increases citizen participation, and strengthens accountability. While these benefits are important, it is also necessary to understand the costs associated

with district creation. This section provides evidence from government budgets to examine the fiscal implications of creating new districts. Central transfers to districts take two main forms. Conditional grants are earmarked for specific services such as education, health, water, and roads, and are allocated through formulas that scale with population and service needs. By contrast, the unconditional block grant finances the district administration itself, including the salaries of the LCV chairperson, the chief administrative officer, and headquarters staff, and is allocated to every district regardless of its size. Creating a new district therefore entails a fixed administrative cost, since each district requires its own administrative apparatus financed largely through the unconditional grant. District creation therefore generates a growing fiscal burden that constrains how far the strategy of bringing services closer to citizens can be efficiently extended.

To estimate the cost of district creation, I use a panel of approved local-government transfers from the Ministry of Finance, Planning and Economic Development (MoFPED), covering fiscal years 2012/13 to 2023/24. These data allow me to construct a detailed decomposition of each district's annual transfer grants. All amounts are expressed in real fiscal year 2017/18 shillings, deflating nominal budgets with the Uganda consumer price index.¹⁸ The unit of observation is the district as a budget vote, the budget-holding unit through which MoFPED channels transfers to each local government. Municipal and city councils are excluded: they are separate urban local-government votes with their own transfers and administrative payroll, nested within districts or, since 2020, carved out of them, so including them would mix a different tier of government into the comparison and count part of a territory twice. Kampala, governed by the Kampala Capital City Authority under a separate framework, and Wakiso, the metropolitan outlier, are excluded as in the household analysis. For each of the 24 districts created during the sample period (calendar years 2015–2020), I build an event-time fiscal accounting around the split. Specifically, I compare the mother district's average budget in the two fiscal years before the split with its average budget in the two years after the split. I then add the child district's average budget during its first two years of operation to obtain the net change in total spending across the territory. The wage component of the unconditional block grant (which finances the LCV chairperson, chief administrative officer, district headquarters staff, and other political officeholders) provides the cleanest measure of duplicated administrative payroll costs.

Table 16 presents the dynamics of fiscal cost reallocation. Panel A shows an inverse relationship between the total transfers received by a district and the relative importance of unconditional grants. The unconditional block grant represents a substantially larger share of central transfers in districts with smaller budgets. This pattern appears in both the wage and non-wage components. By contrast, the share allocated to conditional grants, which are tied to service delivery formulas, is largely unrelated to district fiscal size. Similarly, Panel B shows that recently created districts devote roughly two percentage points more of their budgets

¹⁸World Development Indicators series FP.CPI.TOTL; each fiscal year is assigned the average of the two overlapping calendar years.

to unconditional grants than established districts, with most of the difference concentrated in the wage component, while the share allocated to conditional grants remains essentially unchanged. Because conditional grants scale with service needs, whereas unconditional grants finance the administrative apparatus required to operate as a district, this contrast suggests that the larger administrative share observed in small and newly created districts reflects a fixed cost of district administration rather than differences in service provision.

TABLE 16: Administrative overhead share and recently-created districts

	Wage bill (1)	Uncond. (admin) (2)	Uncond. wage (3)	Uncond. non-wage (4)	Cond. (services) (5)
<i>Panel A. Cross-sectional scale (log of total transfers)</i>					
ln(transfers_full)	0.0168 (0.0129)	-0.0539*** (0.0053)	-0.0419*** (0.0042)	-0.0120*** (0.0015)	0.0064 (0.0103)
Observations	1,468	1,468	1,468	1,468	1,468
R-squared	0.387	0.584	0.485	0.650	0.516
<i>Panel B. Recently-created district premium (created within 3 FYs)</i>					
Created within last 3 FYs	-0.0108 (0.0162)	0.0202*** (0.0060)	0.0156*** (0.0052)	0.0046*** (0.0015)	-0.0050 (0.0176)
Observations	1,468	1,468	1,468	1,468	1,468
R-squared	0.383	0.255	0.140	0.536	0.515

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each column is a separate regression whose dependent variable is the share of the district's total transfers going to that grant category; unconditional grants (Columns 2–4) finance the fixed administrative apparatus, while conditional grants (Column 5) are tied to service-delivery formulas. Panel A regresses each share on the log of the district's total transfers (its fiscal size); Panel B, on an indicator for districts created within the last 3 fiscal years. Column 5 shares the same denominator as the overhead measures and serves as a placebo: its flat relationship with size rules out a mechanical correlation between the overhead share and total transfers. All columns include financial year fixed effects. Sample: district-FY observations FY2012/13–FY2023/24, excluding Kampala, Wakiso, and municipal/city councils. Standard errors clustered at the district level are in parenthesis.

Finally, it is important to address whether changes of budget are a zero-sum exercise. Appendix Table A25 therefore traces the fiscal accounting across the 24 splits in real terms and benchmarks each split against the contemporaneous budget growth of never-split 1991 territories. Relative to this counterfactual, total spending in the territory increases by 5.37 billion shillings on average, equivalent to 19 percent of the mother district's pre-split budget (12 percent at the median), with a positive excess in 19 of the 24 splits.

To complement this analysis, I aggregate the panel to fixed territories defined by the 1991 district borders. This means that transfers to a mother district and to any districts later carved out of it are added within the same geographic unit, netting out reallocations between old and new districts. I then estimate a staggered event study of log real territory transfers using the estimator of [de Chaisemartin & D'Haultfœuille \(2020\)](#). A territory is classified as treated in the first fiscal year in which a newly created district carved out of it receives its own budget vote. Of the 37 territories, 18 experience such a split during the sample period, while the remaining 19 either are never split or were already subdivided before the panel begins, and therefore serve as controls. Figure 5 shows that real transfers to treated territories increase by approximately 14 percent after the first split (corresponding to an average total effect of 0.135 log points, with a standard error of 0.042). The effects grow from 9 to 17 percent over the following four years, with no evidence of differential pre trends (with a joint placebo p value of 0.73). The composition of this increase mirrors the allocation mechanism documented in Table 16. The wage component of the unconditional grant, which captures the duplicated administrative

payroll, rises by 26 percent, compared with only 7 percent for conditional service delivery grants (Appendix Figure A7).¹⁹ Consistently, Table 17 shows that, holding the territory fixed, the share of transfers devoted to the administrative payroll increases with the number of districts into which the territory is divided.

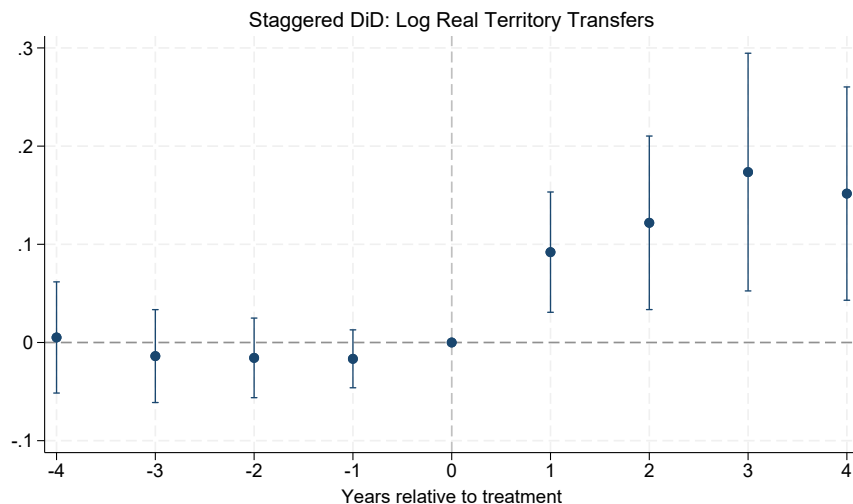


FIGURE 5: Event study: log real territory transfers around the first district split. Balanced panel of 37 1991-mother territories over fiscal years 2012/13–2023/24 (real FY2017/18 shillings); 18 territories treated at the first fiscal year in which a newly created district receives its own budget vote, and 19 control territories with no split during the sample window (their subdivision, where any, was completed before FY2012/13). Estimates from the heterogeneity-robust estimator of [de Chaisemartin & D’Haultfœuille \(2020\)](#) with four pre- and four post-periods; the joint test of the placebo estimates has a p-value of 0.73. Kampala, Wakiso, and municipal/city councils excluded.

TABLE 17: Administrative Overhead and Territory Fragmentation

	UCG-wage share (1)	Unconditional share (2)
Log number of districts in territory	0.021*** (0.004)	0.020*** (0.007)
Territory FE	Yes	Yes
FY FE	Yes	Yes
Observations	444	444
Territories	37	37

Notes: *p<0.1, **p<0.05, ***p<0.01. Balanced panel of 37 1991-mother territories over fiscal years 2012/13–2023/24, aggregating the approved transfers of all constituent districts. The dependent variables are the shares of the territory’s total approved transfers allocated to the wage component of the unconditional block grant (Column 1) and to the unconditional block grant as a whole (Column 2). The regressor is the logarithm of the number of districts in the territory in each fiscal year, where a district is counted from the first year in which it holds its own budget vote, the budget-holding unit in the MoFPED transfer data. Kampala, Wakiso, and municipal/city councils are excluded throughout. Standard errors clustered at the territory level are in parenthesis.

These results show that there is a limit to the strategy of bringing government closer to citizens through district creation. Each new district requires its own administrative apparatus,

¹⁹The joint placebo test for the unconditional wage component is significant, with a p value of 0.02.

making further subdivision increasingly costly. Importantly, these fiscal costs are different from the mechanism identified in the main analysis. The estimated benefits of proximity are not driven by higher service delivery budgets in newly created districts, since the additional resources are concentrated in administrative transfers rather than conditional grants. The evidence therefore suggests that district creation can improve service access, engagement, and accountability, but does so through a fiscally costly expansion of the state apparatus.

12 Conclusions

This paper provides empirical evidence on the impact of proximity to administrative centers on public service delivery and state-citizen relations in the context of Uganda's redistricting process. By leveraging a governmental rule governing the placement of new district headquarters through an instrumental variable approach, this study identifies the causal effects of households' proximity to new administrative centers using granular georeferenced survey data combined with detailed geographical information on districts.

The findings show that proximity to administrative centers significantly improves access to public goods and services. Households closer to new district headquarters have better access to electricity and high-quality roads, shorter travel times to health facilities, and improved access to water sources. The stronger effects for centrally provided services reflect not only direct infrastructure investments but also the coordinating role of district administrators in channeling and implementing national programs for nearby communities. At the same time, proximity increases engagement through formal institutions, reduces direct corruption experiences, and raises satisfaction with appointed local leaders, consistent with stronger monitoring and accountability when citizens and officials are within closer reach. Comparing survey rounds, these effects strengthen as new administrations consolidate, becoming more pronounced by 2021 than in the years immediately following district creation.

Despite these improvements, proximity to new district headquarters does not translate into higher trust in government or more favorable democratic perceptions. Satisfaction with democracy, perceived political freedoms, rule of law, and trust in both local and national institutions remain largely unchanged.

Heterogeneity by political alignment between the elected district chairperson and the centrally appointed RDC sheds further light on this finding. In aligned districts, proximity is associated with larger gains for centrally provided infrastructure, a shift in engagement from protest toward formal participation, lower bribery for medical care, and greater satisfaction with the RDC. However, democratic perceptions remain unaffected even in this more favorable environment. In non-aligned districts, the gains for central infrastructure are smaller and no longer statistically significant, engagement does not respond, and closer distance to headquarters is associated with lower perceived freedom to vote and declining trust in local government and the ruling party, suggesting that proximity can reveal frictions between elected and appointed

leaders. Together, these results suggest that in hybrid regimes, bringing the state closer to citizens can simultaneously improve services and accountability while leaving the underlying political relationship between citizens and the state unchanged.

These benefits come at a fiscal cost. Each new district requires its own administrative apparatus, and the intergovernmental transfer system finances a fixed administrative core regardless of district size. As a result, district creation increases total public spending across the original territory by around 14 percent. This additional spending is concentrated in administrative overhead, especially the duplicated wage bill of headquarters staff and political officeholders, rather than in service delivery grants. The service improvements are therefore not driven by proportionally larger service budgets. Instead, they appear to reflect a more effective deployment of resources when administration is closer. At the same time, administrative duplication implies diminishing returns to further subdivision, since each additional district absorbs a growing share of transfers into the fixed costs of administration.

One of the main limitations of the instrumental variable approach is that the estimates provide a Local Average Treatment Effect, focusing on compliers — in this context, households whose proximity to the new district's centroid influenced their likelihood of living closer to the new district's capital. While these estimates offer valuable insights into the effects of proximity in newly created districts that followed governmental placement rules, they do not speak to the effects of distance to headquarters in non-split or mother districts.

The findings have several policy implications. Reducing the geographical distance between citizens and administrators can be an effective way to improve service delivery, particularly when accompanied by institutional arrangements that facilitate meaningful participation in local decision-making. Proximity may also strengthen accountability and reduce inefficiencies in public spending. However, the alignment results caution that the governance environment matters: the benefits of administrative proximity are larger and more consistent when elected and appointed local leaders share political incentives. The fiscal evidence adds a further caution: because each new district carries a fixed administrative cost, improving proximity through subdivision faces diminishing returns, and its benefits must be weighed against a growing administrative burden. Finally, while redistricting can be an effective tool for bridging the gap between government and citizens, the evidence that democratic perceptions remain unresponsive suggests that administrative decentralization alone is unlikely to foster deeper democratic legitimacy in hybrid regimes.

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Appendix

A Conceptual Framework

This section develops a reduced-form model of public good provision with two local administrators who differ in their accountability relationships and citizens who choose how to engage with the state. The model formalizes how geographical distance to the administrative center and political alignment between local leaders shape service delivery, citizen engagement, and accountability (proxied by corruption).

A.1 Setup

Consider a district with two administrators: an elected Local Council V (LCV) chairperson, indexed by L , who is directly accountable to local voters, and a centrally appointed Resident District Commissioner (RDC), indexed by R , who represents the national government within the district.²⁰ A household i is located at distance $d_i \geq 0$ from the district headquarters.

Public good production and spatial delivery. Each administrator simultaneously chooses effort $e_j \geq 0$, $j \in \{L, R\}$. Effort translates into higher public goods and services for the district. However, the benefits of this effort are decreasing with distance to the headquarters. The public good available to household i at distance d_i from the headquarters is

$$g_i = (e_L + \alpha e_R) \phi(d_i), \quad (3)$$

where $\alpha \in (0, 1]$ is a *political alignment* parameter governing the share of RDC effort that translates into local public goods, and $\phi(d_i)$ is a *spatial attenuation function* satisfying $\phi(0) = 1$, $\phi'(d_i) < 0$, and $\phi''(d_i) \leq 0$.

The spatial attenuation function $\phi(d_i)$ captures the fact that infrastructure coverage, service quality, and administrative reach decline with distance from the district headquarters. This reflects the empirical patterns that electricity grids and national roads are less available farther from administrative centers, health workers tend to concentrate near headquarters, and oversight of service delivery weakens at the periphery (Asher et al., 2018). Importantly, the attenuation does not reflect a deliberate choice by administrators to favor nearby households. Instead, it reflects the spatial structure of service delivery: administrators choose effort once for the district, and the household-level realization of that effort depends on where the household is located relative to the headquarters.

²⁰The distinction between elected and appointed local officials builds on Besley & Coate (2003), who study the trade-offs between these two selection mechanisms. A related dual-principal structure is detailed in the working paper version of Martinez-Bravo et al. (2022) with village chiefs in rural China being simultaneously accountable to villagers and to the autocratic central government.

The alignment parameter α captures the weight the RDC places on local welfare relative to a fixed outside payoff from central loyalty. When the LCV chairperson and the RDC share political affiliation (α close to 1), local welfare dominates the RDC's payoff, and the RDC exerts high effort on locally valued services. When they are not aligned (α is small), the fixed outside payoff (which captures career concerns, patronage rents, or political loyalty rewards) dominates the RDC's payoff, and effort on local services falls accordingly.²¹

Administrator preferences. Administrators choose effort to maximize their respective payoffs. The LCV, being accountable to local voters, maximizes a payoff that is increasing in aggregate household welfare and decreasing in effort cost:

$$U_L = (e_L + \alpha e_R) \bar{\phi} - \frac{1}{2} e_L^2, \quad (4)$$

where $\bar{\phi} \equiv \int \phi(d) dF(d)$ denotes the average spatial attenuation across the district population. The LCV cares about the welfare of the representative constituent, whose access to public goods depends on $\bar{\phi}$. The RDC internalizes local welfare with weight α and receives a fixed outside benefit $(1 - \alpha)\bar{u}$ from serving the central government's agenda, capturing career concerns, patronage rents, or political loyalty rewards:

$$U_R = \alpha(e_L + \alpha e_R) \bar{\phi} + (1 - \alpha) \bar{u} - \frac{1}{2} e_R^2. \quad (5)$$

A.2 Equilibrium Provision

Administrators choose effort simultaneously in a Nash equilibrium. The first-order conditions are:

$$\text{LCV: } e_L^* = \bar{\phi}, \quad (6)$$

$$\text{RDC: } e_R^* = \alpha^2 \bar{\phi}. \quad (7)$$

The LCV's optimal effort equals $\bar{\phi}$, the average spatial return to effort in the district. The RDC's effort scales with $\alpha^2 \bar{\phi}$. Substituting into (3), the equilibrium public good available to household i is

$$g_i^* = (1 + \alpha^3) \bar{\phi} \phi(d_i). \quad (8)$$

The equilibrium expression separates two components. The term $\bar{\phi}$ captures the district-level supply of public goods: it determines how much effort administrators exert and reflects the average accessibility of the population. The term $\phi(d_i)$ captures the household-level realization:

²¹This specification follows [Martinez-Bravo et al. \(2022\)](#) in modeling local officials under dual accountability, where preference congruence between vertically and locally accountable principals shapes governance outcomes, abstracting from the within-agent reallocation of effort across competing tasks between the central and the local government

it determines what share of the district's effort effectively reaches household i . Together, they imply that a household benefits from both living in a compact district (high $\bar{\phi}$) and from being located close to the headquarters (high $\phi(d_i)$). In the context of redistricting, the creation of a new district with a centrally located headquarters raises $\bar{\phi}$ (the average distance falls for the splitter district population) and, for many households, also increases $\phi(d_i)$ (the household is now closer to its headquarters). These two forces reinforce each other.

Proposition 1 (Provision decreasing in distance). *Equilibrium public good provision for household i is strictly decreasing in its distance to the headquarters:*

$$\frac{\partial g_i^*}{\partial d_i} = (1 + \alpha^3) \bar{\phi} \phi'(d_i) < 0.$$

Households located closer to the district headquarters have better access to public goods and services. The result follows directly from the spatial attenuation of service delivery: infrastructure and administrative capacity radiate outward from the administrative center, and households farther away receive less of the public good regardless of the effort level chosen by administrators. This prediction is consistent with the government's stated objective of bringing services closer to citizens through district creation.

Proposition 2 (Alignment premium). *Equilibrium provision is strictly increasing in political alignment:*

$$\frac{\partial g_i^*}{\partial \alpha} = 3\alpha^2 \bar{\phi} \phi(d_i) > 0.$$

When the RDC and LCV share political incentives, a greater share of RDC effort is channeled toward locally valued services, raising total provision. The convexity in α implies that the marginal return to alignment is largest when alignment is already high, reflecting the compounding nature of incentive congruence between the two administrators.

Proposition 3 (Complementarity between proximity and alignment). *The marginal benefit of proximity is strictly increasing in alignment:*

$$\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) = -3\alpha^2 \bar{\phi} \phi'(d_i) > 0.$$

This cross-derivative establishes that distance and alignment are complements in the production of local public goods. The intuition is that proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed. In non-aligned districts, even nearby households benefit less from proximity because RDC effort is partly diverted to central objectives. This predicts that the effects of proximity on service delivery should be stronger in politically aligned districts.

A.3 Citizen Engagement

The improvements in service delivery documented above depend not only on administrative effort but also on the quality of information that administrators receive about local needs. This section models how proximity shapes citizens' decisions to engage with the state and, consequently, the information available to administrators.

Citizen preferences. A citizen located at distance d_i from the headquarters derives utility from the public good g_i^* and may choose to engage with the state. Engagement takes one of two forms: *participatory* (attending district council meetings, meeting the RDC, using formal complaint and feedback channels) or *contentious* (joining protests, contacting the media, organizing collective pressure campaigns). The citizen's payoff is:

$$V_i = g_i^* + \max \left\{ \underbrace{b_P(\alpha) - c_P(d_i)}_{\text{participatory}}, \underbrace{b_C - \bar{c}_C}_{\text{contentious}}, 0 \right\}, \quad (9)$$

where $b_P(\alpha)$ is the benefit of participatory engagement, $c_P(d_i)$ is its cost, b_C is the benefit of contentious engagement, and $\bar{c}_C$ is the cost of contentious action. The citizen chooses whichever form of engagement yields the highest net benefit, or does not engage at all.

Two features of the cost structure are important. First, the cost of participatory engagement is increasing in distance ($c'_P(d_i) > 0$): attending a district council meeting, visiting the RDC's office, or filing a formal complaint all require traveling to the headquarters, with time and transport costs that rise with d_i . Second, the cost of contentious engagement $\bar{c}_C$ is independent of distance to the headquarters (e.g. protests can be organized locally).

The benefit structure includes the role of alignment. Participatory engagement is assumed to be more effective at influencing policy outcomes ($b_P(\alpha) > b_C$ for sufficiently high α) because it transmits detailed information about local priorities directly to decision-makers, while contentious action is a noisier signal. Additionally, $b_P(\alpha)$ is increasing in alignment: when the RDC internalizes local welfare, the state is more responsive to citizen input, making participatory engagement more rewarding. In non-aligned districts, citizens who engage through formal channels may find their input disregarded by an RDC pursuing central directives, reducing the returns to participation.

Engagement equilibrium. Given these assumptions, there exists a distance threshold $\hat{d}(\alpha)$, defined by $b_P(\alpha) - c_P(\hat{d}) = b_C - \bar{c}_C$, such that citizens at $d_i < \hat{d}(\alpha)$ choose participatory engagement, while citizens at $d_i > \hat{d}(\alpha)$ choose contentious engagement or do not engage at all. Since $b'_P(\alpha) > 0$, the threshold $\hat{d}(\alpha)$ is increasing in alignment: in aligned districts, participatory engagement dominates over a wider range of distances.

Proposition 4 (Proximity shifts engagement from contentious to participatory). *A reduction in*

distance d_i increases participatory engagement and decreases contentious engagement:

$$\frac{\partial \Pr(\text{participatory})}{\partial d_i} < 0, \quad \frac{\partial \Pr(\text{contentious})}{\partial d_i} > 0.$$

The mechanism is that proximity reduces the cost of formal engagement with district institutions. When the headquarters is within reach, citizens can attend council meetings and communicate their preferences directly to local administrators, making contentious action unnecessary. When the headquarters is distant, the cost of formal engagement rises and citizens resort to protests and media contact as lower-cost alternatives for voicing demands. The overall pattern is one of substitution: proximity does not suppress citizen voice but rather redirects it from contentious to participatory channels.

Proposition 5 (Alignment amplifies participatory engagement). *The range of distances over which participatory engagement dominates is increasing in alignment: $\hat{d}'(\alpha) > 0$. In aligned districts, proximity produces stronger shifts toward participatory engagement.*

Proof. Differentiating $b_P(\alpha) - c_P(\hat{d}) = b_C - \bar{c}_C$ with respect to α yields $b'_P(\alpha) = c'_P(\hat{d}) \hat{d}'(\alpha)$. Since $b'_P(\alpha) > 0$ and $c'_P(\hat{d}) > 0$, it follows that $\hat{d}'(\alpha) > 0$. $\square$

When the state is responsive to citizen input (high α), the returns to participatory engagement are larger, making citizens willing to bear higher travel costs to engage formally. This predicts that the reduction in contentious engagement and the increase in participatory engagement from proximity should be more pronounced in politically aligned districts. In non-aligned districts, the lower responsiveness of the state limits the returns to formal engagement, and citizens may not shift away from contentious channels even when the headquarters is nearby.

A.4 Corruption and Monitoring

A further channel through which proximity affects public service delivery is accountability. After effort choices are made, each administrator can divert a fraction $\kappa_j \in [0, \bar{\kappa}]$ of resources for private gain, yielding private benefit $b \cdot \kappa_j$. Citizens monitor administrators with intensity

$$\mu(d_i), \quad \text{with } \mu'(d_i) < 0 \text{ and } \mu(0) = \bar{\mu} \leq 1, \quad (10)$$

so that closer households are better able to observe and report misconduct. Detection triggers a penalty $P > 0$. The monitoring-through-proximity mechanism builds on [Campante & Do \(2014\)](#), who show that isolated capital cities are associated with higher corruption, and with [Narasimhan & Weaver \(2024\)](#), who show that polity size can improve public service delivery under political accountability mechanisms. In the Ugandan context, proximity to the headquarters determines how effectively citizens can exercise this informational advantage.²²

²²Experimental evidence from Uganda supports the accountability channel modeled here. [Björkman & Svensson \(2009\)](#) show that community-based monitoring significantly improves the quality of health service

The RDC's net payoff from corruption is $b\kappa_R - \mu(d_i)P\kappa_R - \alpha\lambda\kappa_R$, where $\lambda > 0$ is the marginal welfare cost of diversion, internalized at rate α . Since the payoff is linear in κ_R , the RDC diverts resources if and only if $b - \alpha\lambda > \mu(d_i)P$. Two forces discipline the administrator: external monitoring by citizens ($\mu(d_i)P$) and internal alignment with local welfare ($\alpha\lambda$). Both are stronger when the household is closer and when the district is politically aligned.

Proposition 6 (Proximity deters corruption). *There exists a distance threshold $\hat{d}_c$ defined by $b - \alpha\lambda = \mu(\hat{d}_c)P$ such that corruption is deterred for $d_i < \hat{d}_c$ and occurs for $d_i > \hat{d}_c$. Since $\mu'(d_i) < 0$, households closer to the headquarters experience less corruption.*

Proposition 7 (Alignment amplifies the accountability effect of proximity). *The corruption threshold $\hat{d}_c(\alpha)$ is increasing in alignment: $\partial\hat{d}_c/\partial\alpha > 0$. In aligned districts, corruption is deterred over a wider range of distances.*

Proof. Implicit differentiation of $b - \alpha\lambda = \mu(\hat{d}_c)P$ with respect to α gives $-\lambda = \mu'(\hat{d}_c)P \frac{\partial\hat{d}_c}{\partial\alpha}$. Since $\lambda > 0$, $\mu'(\hat{d}_c) < 0$, and $P > 0$, it follows that $\frac{\partial\hat{d}_c}{\partial\alpha} = \frac{\lambda}{-\mu'(\hat{d}_c)P} > 0$. $\square$

Alignment introduces an additional deterrent beyond citizen monitoring: an aligned RDC internalizes the welfare cost of corruption, making diversion less attractive even at moderate distances from the headquarters. Taken together, these two propositions predict that corruption should be lower in areas closer to the administrative center, and that this reduction should be more pronounced in politically aligned districts.

Discussion. The model predicts that proximity to district headquarters improves public service delivery, shifts citizen engagement from contentious to participatory channels, and deters corruption. Political alignment between the elected LCV and the appointed RDC amplifies each of these effects. The main results in Section 6 test these predictions.

B Proofs of Propositions

B.1 Derivation of the Equilibrium

The administrators choose effort simultaneously in a Nash equilibrium. Each administrator takes the other's effort as given and maximizes their own payoff.

LCV's problem. The LCV chooses e_L to maximize (4). The first-order condition is:

$$\frac{\partial U_L}{\partial e_L} = \bar{\phi} - e_L = 0 \implies e_L^* = \bar{\phi}. \quad (11)$$

delivery in Ugandan health facilities, suggesting that citizen oversight can meaningfully discipline local officials.

RDC's problem. The RDC chooses e_R to maximize (5). The first-order condition is:

$$\frac{\partial U_R}{\partial e_R} = \alpha^2 \bar{\phi} - e_R = 0 \implies e_R^* = \alpha^2 \bar{\phi}. \quad (12)$$

Equilibrium public good. Substituting $e_L^* = \bar{\phi}$ and $e_R^* = \alpha^2 \bar{\phi}$ into (3):

$$\begin{aligned} g_i^* &= (e_L^* + \alpha e_R^*) \phi(d_i) \\ &= (\bar{\phi} + \alpha \cdot \alpha^2 \bar{\phi}) \phi(d_i) \\ &= \bar{\phi} (1 + \alpha^3) \phi(d_i). \end{aligned} \quad (13)$$

B.2 Proof of Proposition 1: Provision Decreasing in Distance

Statement. Equilibrium public good provision for household i is strictly decreasing in its distance to the headquarters: $\frac{\partial g_i^*}{\partial d_i} < 0$.

Proof. Differentiating (13) with respect to d_i :

$$\frac{\partial g_i^*}{\partial d_i} = (1 + \alpha^3) \bar{\phi} \phi'(d_i).$$

Since $\alpha \in (0, 1]$, the term $(1 + \alpha^3) > 0$. Since $\bar{\phi} > 0$ by construction. Since $\phi'(d_i) < 0$ by assumption. Therefore:

$$\frac{\partial g_i^*}{\partial d_i} = \underbrace{(1 + \alpha^3)}_{>0} \underbrace{\bar{\phi}}_{>0} \underbrace{\phi'(d_i)}_{<0} < 0.$$

B.3 Proof of Proposition 2: Alignment Premium

Statement. Equilibrium provision is strictly increasing in political alignment: $\frac{\partial g_i^*}{\partial \alpha} > 0$.

Proof. Differentiating (13) with respect to α :

$$\frac{\partial g_i^*}{\partial \alpha} = \bar{\phi} \phi(d_i) \frac{\partial}{\partial \alpha} (1 + \alpha^3) = 3\alpha^2 \bar{\phi} \phi(d_i).$$

Since $\alpha \in (0, 1]$, $\bar{\phi} > 0$, and $\phi(d_i) > 0$ for finite d_i :

$$\frac{\partial g_i^*}{\partial \alpha} = \underbrace{3\alpha^2}_{>0} \underbrace{\bar{\phi}}_{>0} \underbrace{\phi(d_i)}_{>0} > 0.$$

B.4 Proof of Proposition 3: Complementarity Between Proximity and Alignment

Statement. The marginal benefit of proximity is strictly increasing in alignment: $\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) > 0$.

Proof. The marginal benefit of proximity is the negative of the distance derivative (since lower d_i means higher proximity):

$$-\frac{\partial g_i^*}{\partial d_i} = -(1 + \alpha^3) \bar{\phi} \phi'(d_i).$$

This expression is positive since $\phi'(d_i) < 0$. Differentiating with respect to α :

$$\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) = -3\alpha^2 \bar{\phi} \phi'(d_i).$$

Since $\alpha^2 > 0$, $\bar{\phi} > 0$, and $\phi'(d_i) < 0$:

$$\frac{\partial}{\partial \alpha} \left(-\frac{\partial g_i^*}{\partial d_i} \right) = - \underbrace{3\alpha^2}_{>0} \underbrace{\bar{\phi}}_{>0} \underbrace{\phi'(d_i)}_{<0} > 0.$$

The positive cross-derivative establishes that distance and alignment are complements: proximity amplifies the returns to any effort directed at local services, and alignment ensures that a larger share of total effort is so directed.

B.5 Proof of Proposition 4: Proximity Shifts Engagement from Contentious to Participatory

Statement. A reduction in distance d_i increases participatory engagement and decreases contentious engagement.

Setup. The citizen at distance d_i chooses among three options: participatory engagement (net benefit $b_P(\alpha) - c_P(d_i)$), contentious engagement (net benefit $b_C - \bar{c}_C$), or no engagement (net benefit 0). The citizen chooses the option with the highest net payoff.

Participatory engagement dominates contentious engagement if and only if:

$$b_P(\alpha) - c_P(d_i) > b_C - \bar{c}_C. \tag{14}$$

Rearranging:

$$c_P(d_i) < b_P(\alpha) - b_C + \bar{c}_C.$$

Define the threshold $\hat{d}(\alpha)$ as the value of d_i at which the citizen is indifferent between the two forms of engagement:

$$c_P(\hat{d}(\alpha)) = b_P(\alpha) - b_C + \bar{c}_C. \quad (15)$$

Proof. Since $c'_P(d_i) > 0$ (the cost of participatory engagement is increasing in distance), the function c_P is strictly increasing. Therefore, condition (14) holds if and only if $d_i < \hat{d}(\alpha)$. This means:

- For $d_i < \hat{d}(\alpha)$: the citizen chooses participatory engagement (c_P is low enough).
- For $d_i > \hat{d}(\alpha)$: the citizen chooses contentious engagement (if $b_C - \bar{c}_C > 0$) or does not engage.

A reduction in d_i moves the citizen from the contentious to the participatory region whenever d_i crosses $\hat{d}(\alpha)$ from above. For a population of citizens with a continuous distribution of distances $F(d)$:

$$\Pr(\text{participatory}) = F(\hat{d}(\alpha)), \quad \Pr(\text{contentious}) = 1 - F(\hat{d}(\alpha)) - \Pr(\text{no engagement}).$$

Since F is increasing, a reduction in the distance of an individual citizen (which shifts probability mass below $\hat{d}$) increases the probability of participatory engagement and decreases the probability of contentious engagement.

B.6 Proof of Proposition 5: Alignment Amplifies Participatory Engagement

Statement. The range of distances over which participatory engagement dominates is increasing in alignment: $\hat{d}'(\alpha) > 0$.

Proof. Differentiate the indifference condition (15) with respect to α :

$$c'_P(\hat{d}(\alpha)) \hat{d}'(\alpha) = b'_P(\alpha).$$

Solving for $\hat{d}'(\alpha)$:

$$\hat{d}'(\alpha) = \frac{b'_P(\alpha)}{c'_P(\hat{d}(\alpha))}.$$

Since $b'_P(\alpha) > 0$ (the benefit of participatory engagement is increasing in alignment, as a more responsive state makes formal engagement more rewarding) and $c'_P(\hat{d}) > 0$ (the cost of

participatory engagement is increasing in distance):

$$\hat{d}'(\alpha) = \frac{\overbrace{b'_P(\alpha)}^{>0}}{\underbrace{c'_P(\hat{d}(\alpha))}_{>0}} > 0.$$

The positive sign means that higher alignment expands the set of distances for which participatory engagement is preferred. In aligned districts, citizens are willing to bear higher travel costs to engage formally because they expect the state to be more responsive to their input.

B.7 Proof of Proposition 6: Proximity Deters Corruption

Statement. There exists a distance threshold $\hat{d}_c$ such that corruption is deterred for $d_i < \hat{d}_c$ and occurs for $d_i > \hat{d}_c$.

Setup. After effort choices are made, the RDC can divert a fraction $\kappa_R \in [0, \bar{\kappa}]$ of resources for private gain. The RDC's net payoff from corruption is:

$$\Pi(\kappa_R) = b \kappa_R - \mu(d_i) P \kappa_R - \alpha \lambda \kappa_R = [b - \mu(d_i) P - \alpha \lambda] \kappa_R,$$

where $b > 0$ is the private benefit per unit diverted, $\mu(d_i)$ is the monitoring intensity with $\mu'(d_i) < 0$ and $\mu(0) = \bar{\mu}$, $P > 0$ is the penalty upon detection, and $\lambda > 0$ is the marginal welfare cost of diversion internalized at rate α .

Proof. Since $\Pi(\kappa_R)$ is linear in κ_R , the RDC's optimal strategy is a corner solution:

$$\kappa_R^* = \begin{cases} \bar{\kappa} & \text{if } b - \alpha \lambda > \mu(d_i) P, \\ 0 & \text{if } b - \alpha \lambda \leq \mu(d_i) P. \end{cases}$$

Corruption is deterred whenever $\mu(d_i) P \geq b - \alpha \lambda$, i.e., when monitoring is strong enough relative to the net temptation to divert. Assuming $b - \alpha \lambda > 0$ (corruption is tempting absent monitoring) and $\mu(0) P > b - \alpha \lambda$ (monitoring deters corruption at zero distance), there exists a unique threshold $\hat{d}_c$ solving:

$$\mu(\hat{d}_c) P = b - \alpha \lambda. \quad (16)$$

The threshold exists and is unique because $\mu(\cdot)$ is continuous and strictly decreasing. Since $\mu'(d_i) < 0$, monitoring weakens with distance, and therefore:

- For $d_i < \hat{d}_c$: $\mu(d_i) > \mu(\hat{d}_c)$, so $\mu(d_i) P > b - \alpha \lambda$, and corruption is deterred ($\kappa_R^* = 0$).
- For $d_i > \hat{d}_c$: $\mu(d_i) < \mu(\hat{d}_c)$, so $\mu(d_i) P < b - \alpha \lambda$, and corruption occurs ($\kappa_R^* = \bar{\kappa}$).

B.8 Proof of Proposition 7: Alignment Amplifies the Accountability Effect of Proximity

Statement. The corruption threshold $\hat{d}_c(\alpha)$ is increasing in alignment: $\partial \hat{d}_c / \partial \alpha > 0$.

Proof. Differentiate the threshold condition (16) with respect to α :

$$\mu'(\hat{d}_c) P \frac{\partial \hat{d}_c}{\partial \alpha} = -\lambda.$$

Solving for $\frac{\partial \hat{d}_c}{\partial \alpha}$:

$$\frac{\partial \hat{d}_c}{\partial \alpha} = \frac{-\lambda}{\mu'(\hat{d}_c) P} = \frac{\lambda}{-\mu'(\hat{d}_c) P}.$$

Since $\lambda > 0$, $-\mu'(\hat{d}_c) > 0$ (because $\mu'(\hat{d}_c) < 0$), and $P > 0$:

$$\frac{\partial \hat{d}_c}{\partial \alpha} = \frac{\overbrace{\lambda}^{>0}}{\underbrace{(-\mu'(\hat{d}_c))}_{>0} \underbrace{P}_{>0}} > 0.$$

A higher α increases the range of distances over which corruption is deterred. The mechanism is that an aligned RDC internalizes a larger share of the welfare cost of diversion ($\alpha\lambda$ is larger), which reduces the net temptation to engage in corruption ($b - \alpha\lambda$ falls). This means that a lower level of external monitoring is sufficient to deter corruption, extending the deterrence region to more distant households. In aligned districts, both the internal alignment cost and the external monitoring force work together to discipline administrators, generating stronger accountability effects of proximity.

C Variables construction

Education. The first set of outcomes relates to education. Like many other public services, education in Uganda is provided at both the central and district levels. While districts oversee primary education, secondary education falls under the jurisdiction of the central government. Therefore, the analysis includes indicators for both education levels.

Although the NSDS collects school attendance data, it does not specify the grade level of children. To address this limitation, I construct two school attendance variables based on schooling age brackets. *School attendance (prim. age)* measures attendance for 4,726 households with children aged 6–12, while *School attendance (second. age)* captures attendance for 3,663 households with children aged 13–18.

Following the same criteria, I construct variables for *Distance to school*, *School travel times*,

and measures of perceived quality including *High quality of teachers*, *High quality of facilities*, and *School has an auditor*. The variable *Public school* is an indicator variable that takes the value of one if the household has any school-age children attending a public institution. *School travel time* measures the amount of time, in minutes, that families spend traveling to school.

Health. Healthcare provision in Uganda is divided between the central government and district governments, each responsible for different levels of service. Districts manage health centers, whereas the central government operates hospitals.

At the district level, I analyze several indicators, including *Government health center*, which measures access to public health centers for 2,361 households that vaccinated their children in the past year. *HH member sick/injured* captures the proportion of household members who reported illness or injury within the last 30 days. *Health center travel time* records the number of minutes taken to reach a health center, considering only households that visited a health facility in the last 30 days. At the community level, I assess access to Government health centers, focusing on their proximity *Government health center is within walking distance*, staffing (availability of *Full-time staff*), resources (*Ambulance presence*), management challenges (*Absenteeism concerns*), and oversight (*District-level supervision on a monthly basis*).

For central government provision, the analysis focuses on three indicators: *Government hospital*, *Government hospital within walking distance*, and *Insufficient PHC funds*. The latter captures whether central government funds for primary health care are insufficient in the community.

Water. The third set of outcomes examines water services, which are also provided by both district and central governments. District governments are responsible for supplying water for household consumption, while the central government provides water for productive activities.

Key district-level water indicators include *Improved water source*, which identifies households with access to safe drinking water in both dry and wet seasons, following World Health Organization guidelines (WHO, 2017). *Water distance* records the self-reported distance (in kilometers) from households to their water source. *Water committee* captures whether a functioning water committee exists within the household's sub-county. Additionally, community-level indicators include access to *Piped water*, *Boreholes*, and *Protected wells*.

For central government-provided water, the dataset includes indicators for *Improved water* and *Water distance* for sources used for production (e.g., agriculture, livestock, and fisheries) in both wet and dry seasons. Additional community-level indicators assess access to *Water for crop production*, *Water for cattle production*, and *Water for fish production* within walking distance.

Electricity. Electricity provision falls under the jurisdiction of the central government. The analysis includes *HH has electricity*, which measures household access to electricity, as well

as *HH connected to grid* and *HH interested in grid*, which indicate household access to and demand for grid electricity. Community-level indicators include *Community has electricity* and *Community connected to the grid*, measuring electrification at the community level.

Roads. The fifth set of outcomes examines road infrastructure, which is provided at both the central and district levels. The central government focuses on inter-district connectivity, while district governments are responsible for roads linking sub-counties and villages.

Although the NSDS does not explicitly classify roads by provider, the provider type can be inferred from road quality. Central government roads are typically paved (*Tarmac*), whereas district roads are predominantly unpaved (*Murram*). Additionally, district roads are categorized as *Feeder* or *Communal* roads. These household-level information is complemented by community-level information on community leaders' perceptions of *Road quality* and the extent of *Road maintenance* by different levels of government.

Agricultural Inputs. The final set of outcomes assesses the availability and use of agricultural inputs, including *Hybrid seeds*, *Planting materials*, *Pesticides*, and *Fertilizer*. Although the main national input distribution programmes, the National Agricultural Advisory Services (NAADS) and Operation Wealth Creation (OWC), are coordinated by the Ministry of Agriculture, Animal Industry and Fisheries, agricultural extension is a district mandate under the Local Governments Act, and inputs are distributed through district production departments and sub-county extension officers. Household-level variables measure whether these inputs were used in the past year and their availability in the household's sub-county. Additional community-level indicators capture access to *Agricultural markets*, *Livestock markets*, *General markets*, *Agricultural extension services*, *Maize/rice and coffee mills*, and *Milk cooling systems*.

Institutional Knowledge and Traditional Institutions. To measure households' awareness of district-level state institutions, I construct indicator variables from the NSDS capturing whether the household knows the location of the *Magistrate court*, the *Police*, the *District land office*, the *Office of the Auditor General*, and the *Judiciary*. *Family civil servant* indicates whether a household member is employed as a civil servant. I complement these with measures from the Afrobarometer to capture the balance between formal and traditional institutions. *Resolved by government* and *Resolved by traditional* indicate whether households resolve disputes through government or traditional institutions, respectively. *Contacted MP* and *Contacted traditional leader* indicate whether the respondent contacted a member of parliament or a traditional leader at least once during the past year. Column (6) of each panel in Table 5 reports a summary index constructed as described under *Summary Indices* below. The *Institutional knowledge* index aggregates the five NSDS knowledge indicators. The *State reliance* index aggregates the four Afrobarometer indicators, with *Resolved by traditional* and *Contacted traditional leader* entering with reversed sign, so that higher values indicate greater reliance on formal state channels; *Family civil servant* is excluded because it comes from the NSDS.

Citizen Engagement. Participatory engagement with the district government is measured with three NSDS indicators. *Involved in LCV meetings* captures whether the household is involved in LCV decision-making and the management of district resources. *Met RDC* indicates whether the household has ever attended an activity organized by the Resident District Commissioner, and *RDC Baraza* whether the RDC has held a baraza (a public accountability meeting) in the community. Contentious and collective forms of engagement come from the Afrobarometer: *Collective civic action* indicates whether the respondent joined others to request government action, while *Protest* and *Contacted media* indicate participation in a demonstration or protest march and contacting the media about politics or community affairs, each measured as any occurrence during the past year. Column (4) of each panel in Table 6 reports a summary index of the three indicators in that panel (*Participatory engagement* for the NSDS and *Contentious engagement* for the Afrobarometer), constructed as described under *Summary Indices* below; higher values indicate more engagement of the corresponding type.

Satisfaction with Local Administrators. Three NSDS indicators capture households' evaluations of district leaders: *LCV performance*, which indicates a positive evaluation of the elected LCV chairperson's performance; *RDC satisfaction*, which indicates that the household is satisfied or very satisfied with the RDC's work; and *Civil servants performance*, which captures a positive evaluation of district civil servants.

Corruption. Corruption outcomes distinguish awareness from direct experience. From the NSDS, *Nepotism*, *Embezzlement*, and *Absenteeism* indicate whether the household has heard of each form of corruption, while *Victim nepotism*, *Victim embezzlement*, and *Victim absenteeism* indicate whether the household has been a victim of the corresponding practice. Bribery experiences come from the Afrobarometer: *Bribe health*, *Bribe police*, *Bribe avoid police*, and *Bribe ID* indicate whether the respondent paid a bribe at least once during the past year to obtain medical care, to obtain police assistance, to avoid problems with the police, or to obtain an identity document, in each case conditional on having sought the corresponding service. Table 7 also reports three summary indices constructed as described under *Summary Indices* below: a *Corruption awareness* index over the three NSDS awareness indicators, a *Corruption victimisation* index over the three NSDS victimisation indicators, and a *Bribery* index over the four Afrobarometer bribery indicators. Because each bribery indicator is defined only for respondents who sought the corresponding service, the bribery index is computed for respondents with at least one non-missing component. Higher values of all three indices indicate more corruption.

Democratic Perceptions and Trust. All variables in this group come from the Afrobarometer. *Is democracy* indicates that the respondent considers Uganda a democracy (whether full or with problems), and *Satisf democracy* that the respondent is fairly or very satisfied with the way democracy works in Uganda. *Free vote choice* and *Free join orgs* indicate that the respondent feels somewhat or completely free to choose whom to vote for and to join any political orga-

nization. The rule of law perceptions (*President ignores law*, *Unequal law*, *Judges influenced*, and *Officials impunity*) take the value of one when the respondent believes the corresponding situation occurs often or always. The trust variables (*Trust localgov*, *Trust president*, *Trust parliament*, *Trust ruling party*, *Trust opposition*, and *Trust courts*) indicate that the respondent trusts the corresponding institution somewhat or a lot. The social capital variables in Appendix Table A22 follow the same construction for trust in relatives, neighbours, other citizens, people of other religions, and people of other ethnicities. The last column of each panel in Table 8 reports a summary index of the outcomes in that panel, constructed as described under *Summary Indices* below: a *Democratic perceptions* index (Panel A), a *Rule of law perceptions* index (Panel B), and a *Trust in government* index (Panel C). The indices follow the orientation of their components, so that higher values indicate more favourable democratic perceptions, more negative rule-of-law perceptions, and more trust in government, respectively.

Political Alignment. A district is classified as aligned when the elected LCV chairperson and the centrally appointed RDC share political affiliation. Since RDCs are appointed by the president, alignment is measured with an indicator taking the value of one when the district's LCV chairperson elected in the 2016 local elections belongs to the ruling party (NRM). Districts created after 2016 are assigned the affiliation of the LCV chairperson recorded at their creation.

Summary Indices. Table A1 lists the components entering each of the ten service indices used in the main analysis and the ten mechanism and perception indices in Tables 5, 6, 7, and 8. Components are standardized to mean zero and standard deviation one over the full survey sample and averaged with equal weights; variables for which higher values indicate worse access (or reliance on traditional rather than state institutions) enter with reversed sign, so that higher values of every service index correspond to better service access, and higher values of the mechanism and perception indices to more institutional knowledge, greater reliance on the state, more participatory or contentious engagement, more corruption awareness, victimisation, and bribery, more favourable democratic perceptions, more negative rule-of-law perceptions, or more trust in government.

D Additional Identification and Robustness Checks

This appendix collects the additional identification and robustness exercises referred to in the main text. Table A2 reports the headquarters placement regressions discussed in Section 5.2.1, and Figure A1 shows the reduced-form binned scatters for the main outcome indices referred to in Section 5.1. The remaining subsections show that the instrument does not proxy for proximity to large cities, that proximity does not predict placebo outcomes unrelated to administrative access, and that the main estimates survive the exclusion of households around the headquarters town, a GIS-based measure of proximity, the comparison with OLS, an alternative construction of the instrument, and a sample restricted to newly created districts. The last subsection reports

TABLE A1: Components of the Summary Indices

Index	Components
Education (district)	School attendance (prim. age); distance to school (rev.); school travel time (rev.); high quality of teachers; high quality of facilities; school has an auditor — all for primary-age children
Education (central gov.)	Same six components, for secondary-age children
Health (district)	Gov. health facility for immunization; HH member sick/injured in last 30 days (rev.); health center travel time (rev.); full-time staff at health center; absenteeism in health center (rev.); motorized ambulance in health center; monthly district supervision of health center
Health (central gov.)	Government hospital available; gov. hospital within walking distance; PHC funds not enough (rev.)
Water (district)	Improved water source (dry and wet seasons); water distance (dry and wet seasons, rev.); water committee; piped water available; borehole available; protected well available
Water (central gov.)	Improved water source for productive use (dry and wet seasons); water distance for productive use (dry and wet seasons, rev.); water for crop, cattle, and fish production within walking distance
Electricity (central gov.)	HH has electricity; HH connected to grid; HH interested in grid; community has electricity
Roads (district)	Sub-county has murram road; sub-county has feeder road; sub-county has communal road (rev.); high quality murram, feeder, and community roads; murram, feeder, and community roads mainly repaired by the district
Roads (central gov.)	Sub-county has tarmac road; high quality tarmac road; tarmac roads mainly repaired by the central government
Agricultural inputs (district)	Use and sub-county availability of hybrid seeds, planting materials, pesticides, and fertilizer
Institutional knowledge (NSDS)	Knows the magistrate court; the police; the district land office; the Office of the Auditor General; the judiciary
State reliance (Afrobarometer)	Dispute resolved by government; contacted an MP; dispute resolved by traditional institutions (rev.); contacted a traditional leader (rev.) — contacts in the past year
Corruption awareness (NSDS)	Has heard of nepotism; of embezzlement; of absenteeism
Corruption victimisation (NSDS)	Household was a victim of nepotism; of embezzlement; of absenteeism
Bribery (Afrobarometer)	Paid a bribe in the past year for medical care; for police assistance; to avoid problems with the police; for an identity document — each conditional on having sought the service
Democratic perceptions (Afrobarometer)	Uganda is a democracy; satisfied with democracy; free to choose whom to vote for; free to join any political organization
Rule of law perceptions (Afrobarometer)	President often ignores the law; people are treated unequally under the law; judges are influenced; officials go unpunished — higher values indicate more negative perceptions
Trust in government (Afrobarometer)	Trust in local government; the president; parliament; the ruling party; the opposition; the courts
Participatory engagement (NSDS)	Involved in LCV decision-making; ever attended an RDC activity; RDC held a baraza in the community
Contentious engagement (Afrobarometer)	Joined others to request government action; participated in a protest or demonstration; contacted the media — all in the past year

Notes: Components of each Anderson (2008) summary index. Each component is standardized to mean zero and standard deviation one before averaging; components marked (rev.) enter with reversed sign so that higher values of every service index indicate better service access; higher values of the mechanism indices indicate more institutional knowledge, greater reliance on formal state channels, more corruption awareness, victimisation, or bribery, more favourable democratic perceptions, more negative rule-of-law perceptions, more trust in government, or more engagement of the corresponding type. Household-level components come from the NSDS household questionnaire and community-level components (e.g., walking-distance and availability measures) from the community questionnaire.

weak-instrument-robust confidence sets for the alignment subsamples of Section 9.

TABLE A2: FUTURE HEADQUARTERS' PLACEMENT: POPULATION DENSITY AND NIGHT LIGHTS

	New headquarter in 2014-2020			
	(1)	(2)	(3)	(4)
Population Density in 2010	-0.000231 (0.000873)	0.000458 (0.000933)		
Night Lights in 2013			-0.142 (0.346)	-0.0745 (0.345)
Mean of Dependent Variable	0.135	0.135	0.135	0.135
Observations	178	178	178	178
District FE	No	Yes	No	Yes

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Logistic regression of future district headquarters' placement (2014-2020) on population density as of 2010 and night lights as of 2013. Each observation corresponds to one sub-county of the 24 newly created districts between 2014 and 2020. Robust standard errors clustered at the district level in parenthesis.

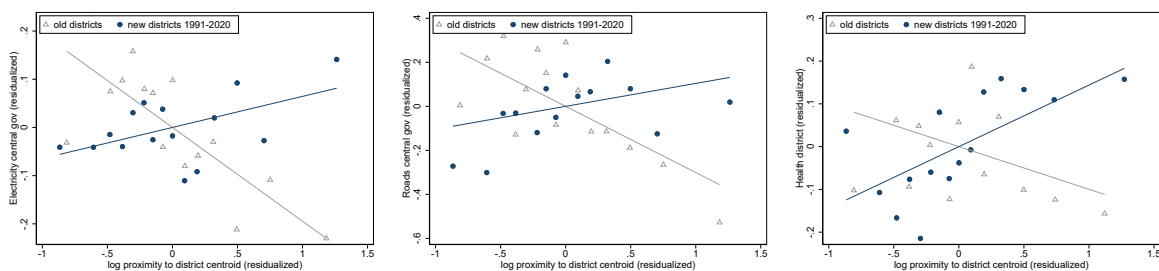


FIGURE A1: Reduced forms: binned scatters of the residualized electricity (left), roads — central government (centre), and health — district (right) indices against residualized proximity to the district centroid, separately for households in new districts (1991–2020) and old districts. Construction as in Figure 2.

D.1 Distance to large cities

One potential concern is that the instrument captures households' proximity to large urban centers rather than to administrative headquarters. If distance to the district centroid simply reflects distance to large cities, it would not provide independent variation in proximity to administrative centers. To address this, I include three alternative measures of urban proximity as mediators in the first stage: the household's distance to the closest large city (defined as the top 20 most populous cities in Uganda), the household's distance to the closest large city within its region, and the household's distance to Kampala. Table A3 shows that the instrument remains strong and stable across all specifications. The coefficient on the interaction term ranges from 0.373 to 0.437 and remains highly significant throughout, while the F-statistic stays above 19 in all columns.

TABLE A3: FIRST STAGE: DISTANCE HOUSEHOLD - LARGE TOWN

	Log SR Proximity Household - District Headquarters				
	Improved water source				
	(1)	(2)	(3)	(4)	(5)
Log Proximity HH-Centroid $\times$ New District 1991-2020	0.429*** (0.080)	0.383*** (0.081)	0.437*** (0.082)	0.425*** (0.082)	0.373*** (0.085)
Log Proximity HH - District's centroid	-0.051 (0.063)	-0.002 (0.065)	-0.057 (0.064)	-0.047 (0.065)	0.007 (0.069)
New district between 1991-2020	1.696*** (0.220)	1.565*** (0.221)	1.712*** (0.222)	1.677*** (0.227)	1.516*** (0.236)
Log Distance Household - Closest Large City		-0.090*** (0.027)			-0.099*** (0.029)
Log Distance Household - Closest Large City within Region			-0.063** (0.030)		-0.057* (0.032)
Log Distance Household - Kampala				0.065 (0.073)	0.174** (0.073)
Constant	-0.857*** (0.270)	-0.622** (0.266)	-0.760*** (0.264)	-1.086*** (0.321)	-1.126*** (0.307)
Mean of Dependent Variable	-2.686	-2.686	-2.686	-2.686	-2.686
Observations	8,417	8,417	8,417	8,417	8,417
Number of Sub-counties	967	967	967	967	967
IV F-stat Kleibergen-Paap rk Wald	28.745	22.546	28.509	27.138	19.266
Controls	Yes	Yes	Yes	Yes	Yes

Notes: * p<0.1, ** p<0.05, *** p<0.01. Controls include: district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Lake Victoria, sub-county mean elevation (in m.a.s.l.), four major regions. Standard errors are clustered at the Parish level.

D.2 Placebo tests

I perform placebo regressions to test whether the instrument captures unobserved geographical features rather than administrative proximity. If instrumented proximity to the district headquarters predicted outcomes that are plausibly unrelated to it, the instrument would likely be picking up unobservable characteristics, indicating omitted variable bias. Suitable placebo outcomes are therefore ones that would respond to such characteristics but are not coordinated by district administrations. The three outcomes available in the NSDS have this property and cover different channels: the use of improved cookstoves depends on household wealth and local fuel markets, the use of air transportation, and involvement in mining on the location of mineral deposits, which is fixed by geology and unrelated to district borders. If the instrument captured local development, remoteness, or natural geography rather than administrative proximity, effects should appear on at least one of these outcomes. Table A4 reports the 2SLS estimates. All three are close to zero and statistically insignificant. Table A5 reports the corresponding test for lower-tier local government institutions (LC1, LC2, and LC3), discussed in Section 5.3.

D.3 Distance to the Headquarters Town

Another potential concern is that the estimated effects reflect the growth of the headquarters town itself rather than improved administrative access for the surrounding population. If the results were driven entirely by households located in the immediate vicinity of the new headquarters, they would speak to urban agglomeration around the administrative center rather than to the reach of the state. To address this, Table A6 re-estimates the main specification excluding households progressively closer to their district headquarters: those in the same parish as the headquarters point, and those within 2 and 5 kilometres. The electricity results are

TABLE A4: Placebo Outcomes and Proximity to District Headquarters (2SLS)

	Cookstove (1)	Air transport (2)	Minerals (3)
Log Proximity to District HQ	-0.016 (0.033)	-0.002 (0.007)	-0.022 (0.018)
Observations	8415	8417	8415
Number of clusters	967	967	967
Mean of DV	0.929	0.006	0.019
IV F-stat (KP rk Wald)	28.749	28.745	28.683

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of placebo variables on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A5: Contact with Local Government Institutions and Proximity to District Headquarters (2SLS)

	Knows LCI (1)	Distance LCI (2)	Knows LCII (3)	Distance LCII (4)	Knows LCIII (5)	Distance LCIII (6)
<i>Panel A: State presence</i>						
Log Proximity to District HQ	0.022 (0.036)	0.112 (0.160)	0.107 (0.104)	-0.851** (0.404)	0.065 (0.090)	-0.065 (0.090)
p-val	[0.528]	[0.482]	[0.304]	[0.035]	[0.471]	[0.471]
q-val	{0.785}	{0.785}	{0.785}	{0.268}	{0.785}	{0.785}
Observations	8414	8044	8413	4286	8413	8413
Number of clusters	967	966	967	901	967	967
Mean of DV	0.956	0.651	0.509	1.945	0.485	1.515
IV F-stat (KP rk Wald)	28.597	25.897	28.592	27.234	28.592	28.592
	HH decision involvement (1)	Ever in LC1 (2)	LC1 meeting freq (3)	Attends LC1 (4)	LC1 performance (5)	LC1 quality change (6)
<i>Panel B: Citizen participation</i>						
Log Proximity to District HQ	0.136* (0.070)	0.009 (0.045)	-0.129* (0.075)	-0.013 (0.076)	0.064 (0.069)	0.011 (0.025)
p-val	[0.050]	[0.843]	[0.085]	[0.865]	[0.357]	[0.658]
q-val	{0.344}	{1.000}	{0.344}	{1.000}	{0.909}	{1.000}
Observations	8414	8417	8417	8417	8217	8143
Number of clusters	967	967	967	967	967	967
Mean of DV	0.471	0.097	0.190	0.306	0.835	0.962
IV F-stat (KP rk Wald)	28.655	28.745	28.745	28.745	29.514	29.509

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' contact with local government institutions on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Unadjusted p-values are in brackets. Anderson's sharpened False Discovery Rate (FDR) q-values, computed within each panel, are in curly brackets.

essentially unchanged across all three donuts, and the roads coefficients become larger the more of the headquarters' surroundings are excluded, indicating that these effects are not an artifact of headquarters-town growth. The health index attenuates by roughly a fifth when households within 5 kilometres are excluded but remains significant at the 10 percent level, suggesting

that part (though not most) of the health effect is concentrated near the headquarters town, consistent with the location of district health facilities.

TABLE A6: Donut Specifications: Excluding Households Near District Headquarters

	Baseline (1)	Excl. HQ parish (2)	Excl. <2 km (3)	Excl. <5 km (4)
<i>Panel A: Electricity (central government)</i>				
Log Proximity to District HQ	0.428*** (0.137)	0.413*** (0.136)	0.453*** (0.164)	0.464** (0.188)
IV F-stat (KP rk Wald)	28.75	29.88	22.32	21.77
Observations	8,417	8,152	7,502	6,123
<i>Panel B: Roads (central government)</i>				
Log Proximity to District HQ	0.706*** (0.261)	0.689*** (0.262)	0.984*** (0.305)	1.115*** (0.357)
IV F-stat (KP rk Wald)	28.75	29.88	22.32	21.77
Observations	8,417	8,152	7,502	6,123
<i>Panel C: Health (district)</i>				
Log Proximity to District HQ	0.387*** (0.121)	0.376*** (0.120)	0.377*** (0.136)	0.301* (0.156)
IV F-stat (KP rk Wald)	33.35	34.48	27.14	22.79
Observations	7,595	7,378	6,838	5,700

Notes: *p<0.1, **p<0.05, ***p<0.01. Each column re-estimates the main 2SLS specification excluding households near their district headquarters: Column 2 drops enumeration areas located in the same parish as the district headquarters point (35 enumeration areas); Columns 3 and 4 drop enumeration areas within 2 and 5 kilometres of the headquarters (125 and 304 enumeration areas, respectively). All regressions use the main specification's controls, weights, and clustering; Kampala and Wakiso are excluded as in the main tables. Standard errors clustered at the enumeration area level are in parenthesis.

D.4 Alternative Distance Variable: GIS Distance

While the household's self-reported distance to the district headquarters arguably captures effective proximity (it reflects not only straight-line distance but also the available transportation network) it may suffer from over- or under-reporting. To address this, Table A7 replaces the self-reported measure with the GIS straight-line proximity from the household's enumeration area to the headquarters point, keeping the instrument and all controls unchanged. The two measures are highly but imperfectly correlated (0.80; Appendix Figure A2), consistent with noisy but informative self-reports. The 2SLS estimates with the GIS measure are statistically significant for all three headline outcomes and, if anything, larger in magnitude, while the first-stage F-statistics remain above 20 for both measures. OLS estimates are three to four times smaller than the 2SLS estimates with the self-reported measure and five to six times smaller with the GIS measure, a gap discussed in the next subsection.

D.5 OLS Estimates

Appendix Table A8 compares the 2SLS estimates with their OLS counterparts for the ten service indices. The OLS coefficients are positive and statistically significant for most outcomes,

TABLE A7: Alternative Endogenous Variable: Self-Reported vs. GIS Proximity to District Headquarters

	2SLS Self-reported (1)	2SLS GIS (2)	OLS Self-reported (3)	OLS GIS (4)
<i>Panel A: Electricity (central government)</i>				
Log Proximity to District HQ	0.428*** (0.137)	0.557** (0.240)	0.102*** (0.020)	0.121*** (0.028)
IV F-stat (KP rk Wald)	28.75	21.19	—	—
Observations	8,417	9,313	8,417	9,313
<i>Panel B: Roads (central government)</i>				
Log Proximity to District HQ	0.706*** (0.261)	0.924** (0.401)	0.222*** (0.035)	0.165*** (0.051)
IV F-stat (KP rk Wald)	28.75	21.18	—	—
Observations	8,417	9,303	8,417	9,303
<i>Panel C: Health (district)</i>				
Log Proximity to District HQ	0.387*** (0.121)	0.593*** (0.205)	0.111*** (0.020)	0.128*** (0.033)
IV F-stat (KP rk Wald)	33.35	20.10	—	—
Observations	7,595	8,320	7,595	8,320

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Columns 1 and 3 use the household's self-reported proximity to the district headquarters as the variable of interest; Columns 2 and 4 replace it with the GIS straight-line proximity from the household's enumeration area to the headquarters point. The instrument, controls, weights, and clustering follow the main specification; Kampala and Wakiso are excluded as in the main tables. The GIS measure removes self-reporting error but provides less variation, as it does not reflect transportation networks; first-stage F-statistics nevertheless remain above 20 for both measures. Standard errors clustered at the enumeration area level are in parenthesis.

confirming the descriptive association between proximity to headquarters and service access. However, for the outcomes where the 2SLS estimates are significant, the OLS coefficients are three to five times smaller. This pattern is informative as it goes against the most direct selection concern. If headquarters had been systematically placed in already better served locations, the naive OLS estimates would likely be larger than the 2SLS estimates. Instead, OLS is substantially smaller, being consistent with two explanations. First, both the self reported and GIS based measures are noisy proxies for effective proximity, so OLS is likely affected by attenuation bias, which the instrument helps correct. Second, the 2SLS estimates identify local average treatment effects for compliers, that is, households whose proximity to headquarters was shifted by redistricting. These households were previously more remote, and the marginal returns to administrative access are plausibly larger for them. The recentering exercise points in the same direction: correcting for predictable exposure to the redistricting process moves the estimates away from zero rather than toward it (Appendix F). This suggests that naive comparisons understate, rather than overstate, the effect of proximity.

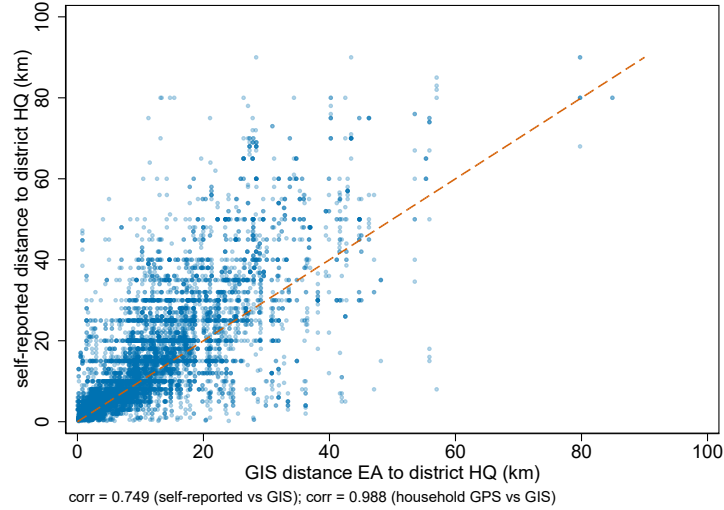


FIGURE A2: Calibration of the self-reported distance measure: household self-reported distance to the district headquarters against the GIS straight-line distance from the household's enumeration area to the headquarters point (in kilometres). The dashed line is the 45-degree line. The correlation between the two measures is 0.75; the correlation between household GPS-based and enumeration area GIS distances is 0.99, indicating that the gap reflects self-reporting rather than geocoding.

D.6 Alternative Instrument: Parish-Level Distance

As an additional robustness check, I replace the enumeration area geocoordinates with parish centroid distances to construct an alternative version of the instrument. Table A9 shows that the results are similar to the ones obtained in Table 3. If anything, several coefficients are larger in magnitude and more precisely estimated under this alternative, while the first stage remains strong with F-statistics exceeding 23 across all specifications. These results show that the findings are robust to the choice of geocoordinates used to compute household distances.

D.7 Alternative Sample: Only New Districts

The main specification identifies the effect of proximity from the differential relationship between centroid proximity and headquarters proximity in new versus old districts, interacting centroid proximity with the new-district indicator and using old districts as a comparison group. Because the government placement rule applies only to newly created districts, an alternative is to restrict the sample to these districts, where the location of the centroid directly determines the location of the headquarters. In this subsample the interaction is no longer necessary: the household's proximity to the district centroid can be used on its own as an instrument for its proximity to the district capital. Because the placement rule ties the centroid to the headquarters mechanically within new districts, the instrument is considerably stronger here, with first-stage F-statistics ranging from 29 to 49, well above conventional thresholds.

Table A10 reports the results. All ten point estimates are positive, and the two most robust

TABLE A8: OLS and 2SLS Estimates: Effects on Services

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
OLS	0.060** (0.028)	0.111*** (0.020)	0.054*** (0.021)	0.003 (0.016)	-0.005 (0.019)
2SLS	0.282 (0.214)	0.387*** (0.121)	0.279* (0.157)	0.089 (0.110)	-0.085 (0.145)
IV F-stat (KP rk Wald)	22.24	33.35	28.75	28.75	28.75
Observations	7,137	7,595	8,417	8,417	8,417
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
OLS	0.090*** (0.026)	0.215*** (0.033)	0.073*** (0.019)	0.222*** (0.035)	0.102*** (0.020)
2SLS	0.151 (0.192)	0.233 (0.201)	0.372** (0.146)	0.706*** (0.261)	0.428*** (0.137)
IV F-stat (KP rk Wald)	22.62	28.75	28.75	28.75	28.75
Observations	5,553	8,417	8,417	8,417	8,417

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Each column reports the coefficient on the household's (log) self-reported proximity to the district headquarters from two estimators: OLS, and the paper's main 2SLS specification instrumenting proximity with the interaction between the enumeration area's log proximity to the district centroid and the new-district indicator. The 2SLS rows reproduce the main results tables. Both estimators use the same controls, sampling weights, and clustering; Kampala and Wakiso are excluded as in the main tables. Standard errors clustered at the enumeration area level are in parenthesis.

findings of the main specification survive: proximity significantly improves access to district-provided health services (0.317, significant at the 1 percent level) and to centrally provided electricity (0.149, significant at 5 percent). District roads (0.143) and secondary education (0.255) become marginally significant, while water and roads under central government retain the expected positive sign but lose statistical significance. The loss of precision reflects the fact that restricting the sample to new districts drops almost half of the observations and removes the identifying variation coming from the comparison with old districts. The magnitudes are also not directly comparable to the main estimates, since the within new district design identifies effects from a different mix of compliers. Overall, this exercise indicates that the main results are not an artifact of the new-versus-old district contrast, as the central findings are recovered

TABLE A9: Effects on Services — PA Distance Instrument (2SLS)

	Schooling (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: District services</i>					
Log Proximity to District HQ	0.222 (0.156)	0.204* (0.115)	0.329** (0.137)	0.019 (0.102)	-0.042 (0.122)
Observations	7170	7627	8529	8529	8529
Number of clusters	974	966	984	984	984
Mean of DV	-0.002	-0.020	-0.005	0.006	0.027
IV F-stat (KP rk Wald)	25.481	26.367	24.385	24.385	24.385
	Schooling (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: Central govt services</i>					
Log Proximity to District HQ	0.004 (0.188)	0.264 (0.216)	0.307*** (0.119)	0.886*** (0.229)	0.353*** (0.113)
Observations	5582	8363	8529	8529	8529
Number of clusters	967	963	984	984	984
Mean of DV	-0.001	0.016	-0.015	-0.061	-0.038
IV F-stat (KP rk Wald)	16.993	23.989	24.385	24.385	24.385

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km^2 as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's parish centroid to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

using only the variation generated within newly created districts, where the placement rule directly applies.

D.8 Weak-Instrument-Robust Inference in the Alignment Subsamples

Tables A11 and A12 report, for every outcome in Section 9, the 2SLS coefficient, the Anderson–Rubin 95 percent confidence set and the first-stage F-statistic in the aligned and non-aligned subsamples separately. The body tables integrate these sets only for the Afrobarometer outcomes, where the subsample first stages are weakest; the full set is reported here for completeness.

TABLE A10: Effects on Services — New Districts Only (2SLS)

	Schooling (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: District services</i>					
Log Proximity to District HQ	0.155 (0.113)	0.317*** (0.093)	0.049 (0.080)	0.143* (0.081)	0.094 (0.079)
Observations	3692	3973	4242	4242	4242
Number of clusters	472	470	474	474	474
Mean of DV	-0.045	-0.073	-0.067	0.063	0.126
IV F-stat (KP rk Wald)	49.487	47.274	45.888	45.888	45.888
	Schooling (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: Central govt services</i>					
Log Proximity to District HQ	0.255* (0.132)	0.156 (0.127)	0.080 (0.073)	0.246 (0.165)	0.149** (0.066)
Observations	2857	4242	4242	4242	4242
Number of clusters	470	474	474	474	474
Mean of DV	-0.015	-0.234	-0.080	-0.288	-0.171
IV F-stat (KP rk Wald)	28.925	45.888	45.888	45.888	45.888

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of outcome index variables on self-reported household's distance to the district's capital. The indices are constructed to have a mean of 0 and a standard deviation of 1. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid. Standard errors clustered at the Parish level are in parenthesis.

TABLE A11: Weak-Instrument-Robust Inference in the Alignment Subsamples (Part A: Services, Engagement, Institutions)

	Aligned districts			Non-aligned districts		
	2SLS (SE) (1)	AR 95% set (2)	KP F (3)	2SLS (SE) (4)	AR 95% set (5)	KP F (6)
<i>Panel A: Services (NSDS 2021)</i>						
School (district)	0.413 (0.307)	[−0.29, 1.30]	9.9	−0.018 (0.275)	[−0.68, 0.55]	16.7
Health (district)	0.324* (0.167)	[0.00, 0.75]	17.0	0.383** (0.157)	[0.11, 0.91]	11.8
Water (district)	0.187 (0.218)	[−0.34, 0.69]	14.4	0.308 (0.203)	[−0.04, 1.01]	12.5
Roads (district)	0.101 (0.157)	[−0.20, 0.53]	14.4	0.007 (0.150)	[−0.32, 0.40]	12.5
Agricultural inputs (district)	0.105 (0.195)	[−0.32, 0.58]	14.4	−0.277* (0.165)	[−0.75, 0.06]	12.5
School (central)	−0.026 (0.309)	[−1.03, 0.65]	9.4	−0.026 (0.244)	[−0.82, 0.48]	9.7
Health (central)	0.157 (0.268)	[−0.59, 0.68]	14.4	0.070 (0.344)	[−0.84, 0.82]	12.5
Water (central)	0.241 (0.184)	[−0.14, 0.73]	14.4	0.337 (0.213)	[−0.05, 1.03]	12.5
Roads (central)	1.013*** (0.384)	[0.31, 2.12]	14.4	0.202 (0.274)	[−0.39, 0.93]	12.5
Electricity (central)	0.424** (0.193)	[0.05, 0.96]	14.4	0.254 (0.167)	[−0.07, 0.78]	12.5
<i>Panel B: Citizen engagement</i>						
Involved in LCV meetings	0.179* (0.097)	[0.01, 0.47]	14.4	−0.011 (0.083)	[−0.22, 0.18]	12.5
Met the RDC	0.072 (0.066)	[−0.06, 0.24]	14.4	0.010 (0.043)	[−0.09, 0.12]	12.5
Attended RDC baraza	0.065 (0.042)	[−0.02, 0.18]	14.4	0.000 (0.040)	[−0.10, 0.10]	12.5
Collective civic action	−0.498** (0.249)	[−3.60, −0.19]	6.4	0.058 (0.059)	[−0.11, 0.21]	19.8
Protest	−0.164** (0.076)	[−1.06, −0.07]	6.4	0.024 (0.026)	[−0.03, 0.11]	19.8
Contacted media	−0.195* (0.108)	[−1.50, −0.06]	6.4	−0.025 (0.049)	[−0.17, 0.09]	19.8
Participatory index	0.321* (0.172)	[0.01, 0.82]	14.4	0.004 (0.126)	[−0.32, 0.29]	12.5
Contentious index	−0.860** (0.383)	[−5.98, −0.43]	6.4	0.054 (0.116)	[−0.24, 0.36]	19.8
<i>Panel C: Institutional knowledge and state-traditional substitution</i>						
Knows magistrate court	0.385*** (0.123)	[0.17, 0.77]	14.3	0.010 (0.120)	[−0.27, 0.31]	12.4
Knows police	0.120 (0.077)	[−0.03, 0.33]	14.3	−0.054 (0.041)	[−0.16, 0.04]	12.5
Knows land office	0.119 (0.086)	[−0.07, 0.33]	14.3	0.032 (0.062)	[−0.09, 0.23]	12.4
Knows Auditor General	0.052 (0.050)	[−0.06, 0.17]	14.4	0.125** (0.061)	[0.01, 0.31]	12.5
Knows judiciary	0.149 (0.102)	[−0.09, 0.38]	14.4	−0.039 (0.078)	[−0.21, 0.18]	12.5
Family member civil servant	0.086** (0.040)	[0.01, 0.21]	14.4	0.044 (0.059)	[−0.08, 0.21]	12.4
Conflict resolved by government	0.007 (0.224)	[−0.64, 1.29]	6.3	0.152* (0.085)	[−0.01, 0.46]	20.7
Conflict resolved by trad. leader	−0.158 (0.195)	[−1.78, 0.22]	6.3	−0.011 (0.050)	[−0.14, 0.12]	20.7
Contacted MP	0.168 (0.125)	[−0.07, 1.16]	6.4	0.043 (0.073)	[−0.19, 0.19]	19.8
Contacted trad. leader	−0.275 (0.199)	[−2.38, 0.04]	6.4	−0.011 (0.073)	[−0.18, 0.21]	19.8
Knowledge index	0.413** (0.161)	[0.10, 0.86]	14.4	0.049 (0.123)	[−0.21, 0.40]	12.5
State reliance index	0.362 (0.320)	[−0.19, 3.29]	6.4	0.113 (0.085)	[−0.12, 0.33]	19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Each row reports, for the subsamples of politically aligned districts (LCV chairperson and RDC of the same party after the 2016 elections) and non-aligned districts, the 2SLS coefficient on log proximity to the district headquarters with its standard error clustered at the parish level, the Anderson–Rubin 95 percent confidence set, which is robust to weak instruments, and the Kleibergen–Paap rk Wald F-statistic of the first stage. The subsample regressions are those underlying Tables 9 to 14; specification, weights and instrument are as in Table 3. Because aligned and non-aligned districts share no clusters, the interaction between proximity and alignment reported in those tables is the difference between the two subsample coefficients; where the non-aligned first stage is strong and its coefficient is close to zero, an aligned confidence set that excludes zero implies that the interaction also differs from zero. Anderson–Rubin sets are obtained by test inversion over a grid of 500 points spanning 15 times the 2SLS confidence interval (weakiv, Finlay, Magnusson and Schaffer); an unbounded set indicates that the instrument is too weak to bound the coefficient at the 95 percent level.

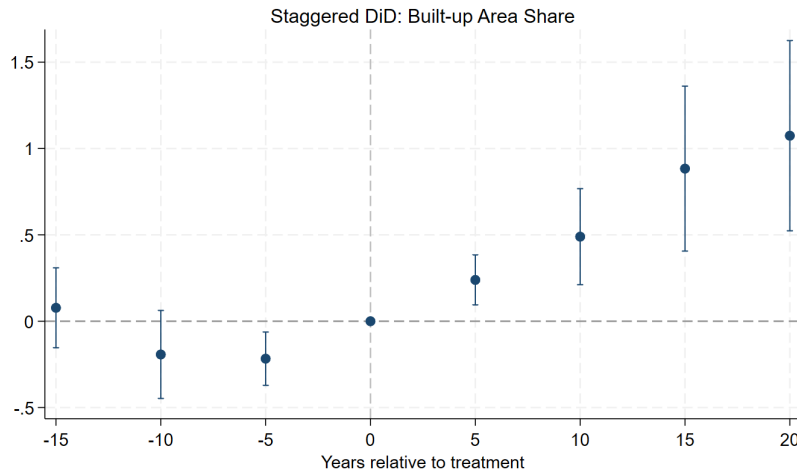
TABLE A12: Weak-Instrument-Robust Inference in the Alignment Subsamples (Part B: Satisfaction, Corruption, Democratic Perceptions)

	Aligned districts			Non-aligned districts		
	2SLS (SE) (1)	AR 95% set (2)	KP F (3)	2SLS (SE) (4)	AR 95% set (5)	KP F (6)
<i>Panel D: Satisfaction with local leaders (NSDS 2021)</i>						
LCV performance	0.055 (0.101)	[−0.18, 0.29]	14.4	0.237* (0.130)	[0.03, 0.72]	12.5
RDC satisfaction	0.208** (0.086)	[0.05, 0.46]	14.4	−0.056 (0.084)	[−0.31, 0.11]	12.5
Civil servants performance	0.108 (0.120)	[−0.12, 0.43]	14.7	0.062 (0.107)	[−0.12, 0.41]	12.6
<i>Panel E: Corruption</i>						
Aware of nepotism	0.221* (0.117)	[−0.01, 0.53]	14.3	−0.225** (0.104)	[−0.56, −0.04]	12.5
Aware of embezzlement	0.170 (0.106)	[−0.05, 0.45]	14.3	−0.138 (0.113)	[−0.46, 0.09]	12.5
Aware of absenteeism	0.063 (0.112)	[−0.21, 0.31]	14.3	−0.106 (0.094)	[−0.39, 0.08]	12.5
Victim of nepotism	−0.047 (0.030)	[−0.14, 0.00]	14.4	0.002 (0.022)	[−0.05, 0.06]	12.5
Victim of embezzlement	−0.011 (0.036)	[−0.10, 0.07]	14.4	−0.080* (0.043)	[−0.21, 0.00]	12.5
Victim of absenteeism	−0.061 (0.055)	[−0.21, 0.05]	14.4	−0.108 (0.083)	[−0.33, 0.07]	12.5
Bribe for medical care	−0.483* (0.248)	[−3.42, −0.15]	6.5	−0.019 (0.066)	[−0.18, 0.16]	26.1
Bribe for police help	−0.398 (0.385)	[−3.78, 0.26]	6.8	−0.299** (0.121)	[−0.81, −0.09]	16.8
Bribe to avoid police	−0.115 (0.309)	(−∞, ∞)	2.9	0.034 (0.112)	[−0.25, 0.42]	16.0
Bribe for ID document	0.284 (0.243)	[−0.42, 2.13]	6.6	−0.034 (0.141)	[−0.47, 0.32]	12.9
Awareness index	0.308 (0.190)	[−0.08, 0.79]	14.3	−0.321* (0.188)	[−0.91, 0.02]	12.5
Victimisation index	−0.184 (0.132)	[−0.56, 0.06]	14.4	−0.277* (0.159)	[−0.74, 0.05]	12.5
Bribery index	−0.823* (0.500)	(−∞, −0.20]	4.8	−0.008 (0.112)	[−0.28, 0.29]	24.2
<i>Panel F: Democratic perceptions (Afrobarometer R10)</i>						
Country is a democracy	−0.015 (0.089)	[−0.39, 0.30]	6.4	−0.002 (0.038)	[−0.11, 0.09]	21.1
Satisfied with democracy	0.138 (0.161)	[−0.15, 1.43]	6.4	0.031 (0.063)	[−0.15, 0.18]	21.5
Free to vote	0.047 (0.075)	[−0.09, 0.54]	7.1	−0.113*** (0.032)	[−0.21, −0.04]	18.9
Free to join organisations	−0.099 (0.102)	[−0.55, 0.15]	7.8	−0.033 (0.050)	[−0.19, 0.08]	19.6
President ignores the law	−0.176 (0.169)	[−1.00, 0.13]	8.8	0.151** (0.067)	[0.02, 0.41]	16.7
Unequal under the law	0.108 (0.150)	[−0.34, 0.70]	7.4	−0.113 (0.089)	[−0.43, 0.06]	18.5
Judges influenced	−0.208 (0.201)	[−1.95, 0.11]	6.7	−0.014 (0.086)	[−0.30, 0.16]	17.7
Officials go unpunished	−0.087 (0.184)	[−1.18, 0.34]	6.8	−0.014 (0.088)	[−0.31, 0.17]	19.5
Trust local government	0.121 (0.123)	[−0.18, 0.78]	7.0	−0.129* (0.077)	[−0.45, −0.01]	20.8
Trust president	−0.050 (0.146)	[−0.64, 0.65]	6.0	−0.083 (0.067)	[−0.28, 0.06]	20.0
Trust parliament	0.023 (0.155)	[−0.62, 0.60]	6.4	0.012 (0.079)	[−0.21, 0.20]	19.8
Trust ruling party	0.146 (0.137)	[−0.15, 1.11]	6.4	−0.126* (0.074)	[−0.40, 0.01]	19.8
Trust opposition	0.002 (0.146)	[−0.75, 0.44]	6.4	0.079 (0.059)	[−0.04, 0.28]	19.8
Trust courts	0.032 (0.144)	[−0.37, 0.70]	7.0	0.091 (0.056)	[−0.10, 0.20]	20.4
Democracy index	0.029 (0.198)	[−0.45, 1.09]	6.9	−0.109 (0.078)	[−0.35, 0.06]	20.1
Rule of law index	−0.156 (0.303)	[−2.21, 0.47]	6.6	−0.011 (0.131)	[−0.44, 0.26]	19.0
Trust index	0.094 (0.181)	[−0.43, 1.07]	6.4	−0.048 (0.092)	[−0.37, 0.13]	19.8

Notes: *p<0.1, **p<0.05, ***p<0.01, two-sided tests. Each row reports, for the subsamples of politically aligned districts (LCV chairperson and RDC of the same party after the 2016 elections) and non-aligned districts, the 2SLS coefficient on log proximity to the district headquarters with its standard error clustered at the parish level, the Anderson–Rubin 95 percent confidence set, which is robust to weak instruments, and the Kleibergen–Paap rk Wald F-statistic of the first stage. The subsample regressions are those underlying Tables 9 to 14; specification, weights and instrument are as in Table 3. Because aligned and non-aligned districts share no clusters, the interaction between proximity and alignment reported in those tables is the difference between the two subsample coefficients; where the non-aligned first stage is strong and its coefficient is close to zero, an aligned confidence set that excludes zero implies that the interaction also differs from zero. Anderson–Rubin sets are obtained by test inversion over a grid of 500 points spanning 15 times the 2SLS confidence interval (weakiv, Finlay, Magnusson and Schaffer); an unbounded set indicates that the instrument is too weak to bound the coefficient at the 95 percent level.

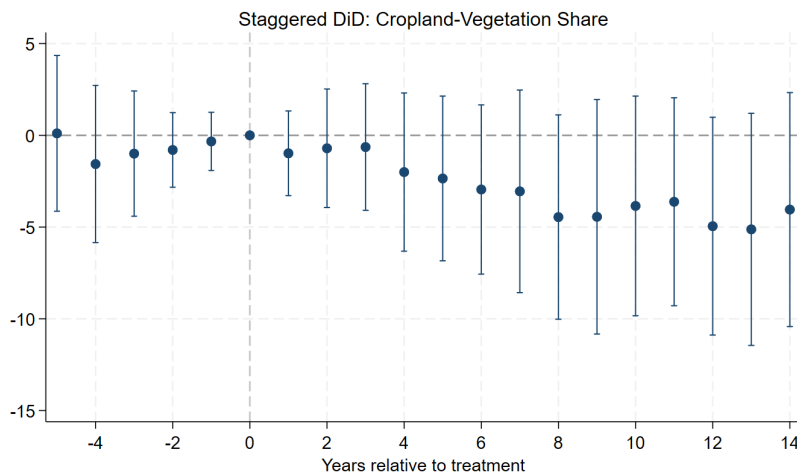
E Figures

FIGURE A3: EVENT STUDY: DISTRICT HEADQUARTERS CREATION ON BUILT-UP AREA SHARE



Notes: Event study of the [de Chaisemartin & D'Haultfœuille \(2020\)](#) staggered difference-in-differences estimator of the parish built-up area share on district headquarters creation. The outcome is the share of a parish that is built-up, measured from the JRC Global Human Settlement Layer built-up surface product (GHS-BUILT-S; 100 m resolution), available in five-year epochs from 1990 to 2025. Event time is measured in five-year epochs and shown in years on the horizontal axis: the three points to the left of zero are placebo (pre-trend) estimates and the four points to the right are the post-creation effects. Vertical bars are 95 percent confidence intervals from standard errors clustered at the parish level.

FIGURE A4: EVENT STUDY: DISTRICT HEADQUARTER CREATION ON PARISH CROPLAND-VEGETATION SHARE



Notes: Event study graph of the estimated coefficients of the [de Chaisemartin & D'Haultfœuille \(2020\)](#) staggered difference in difference estimator of cropland-vegetation share on headquarters creation.

F Recentered Instrument and Randomization Inference

The instrument interacts an arguably exogenous shock, the placement of district headquarters following district creation, with non random exposure: a household's fixed location relative to the geometry of its mother district. Even if the realized partition is as good as random among the feasible partitions, households located near the geometric interior of any feasible fragment mechanically receive high instrument values under every counterfactual configuration (Borusyak & Hull, 2023). This appendix implements the correction proposed by Borusyak & Hull (2023). I compute the expected instrument μ_i across counterfactual redistricting configurations, control for it directly, construct the recentered instrument $z_i - \mu_i$, and conduct randomization inference over placebo redistrictings.

Counterfactual redistricting design

The building blocks are the 1991 county borders,²³ which were fixed before the entire redistricting wave and therefore cannot respond to it. Holding fixed which mothers split and into how many terminal 2020 districts, I compute, for every splitting mother, all feasible partitions of its counties into the realized number of contiguous fragments. The feasible sets range from a single partition (eleven mothers whose splits had no spatial alternative at the county level) to 5,822 partitions for Mbarara. The computation of the counterfactuals is exhaustive everywhere, so μ_i is an exact expectation rather than a simulation average. In each counterfactual, the fragment containing the 1991 district headquarters remains the mother district, all other fragments are new districts, and the instrument is recomputed exactly as in the actual data: the logarithm of the household's proximity to its fragment's centroid, interacted with the new-district indicator.²⁴ Figure A5 illustrates the design for Mbarara, the mother district with the richest counterfactual set.

The counterfactual redistrictings substantially reshuffle households. Among enumeration areas in mother districts that split, 47 percent are assigned to a different fragment under at least one feasible alternative partition. As a result, the recentered instrument retains about two thirds of the standard deviation of the raw instrument, leaving meaningful variation to identify the effect. At the same time, a large share of the raw instrument is predictable from a household's fixed position in the geometry of the mother district, with the correlation between z_i and the expected instrument μ_i reaching 0.74.

²³IPUMS shapefile for the 1991 Uganda census (156 counties).

²⁴As in the main analysis, centroids falling outside half-moon shaped fragments are forced to lie within the polygon. Randomization inference draws one partition per mother per configuration, using independent random streams from a single seed, so that each of the 1,000 draws constitutes one national counterfactual configuration.

Redrawing Mbarara: actual vs counterfactual county partitions (5,822 feasible partitions)

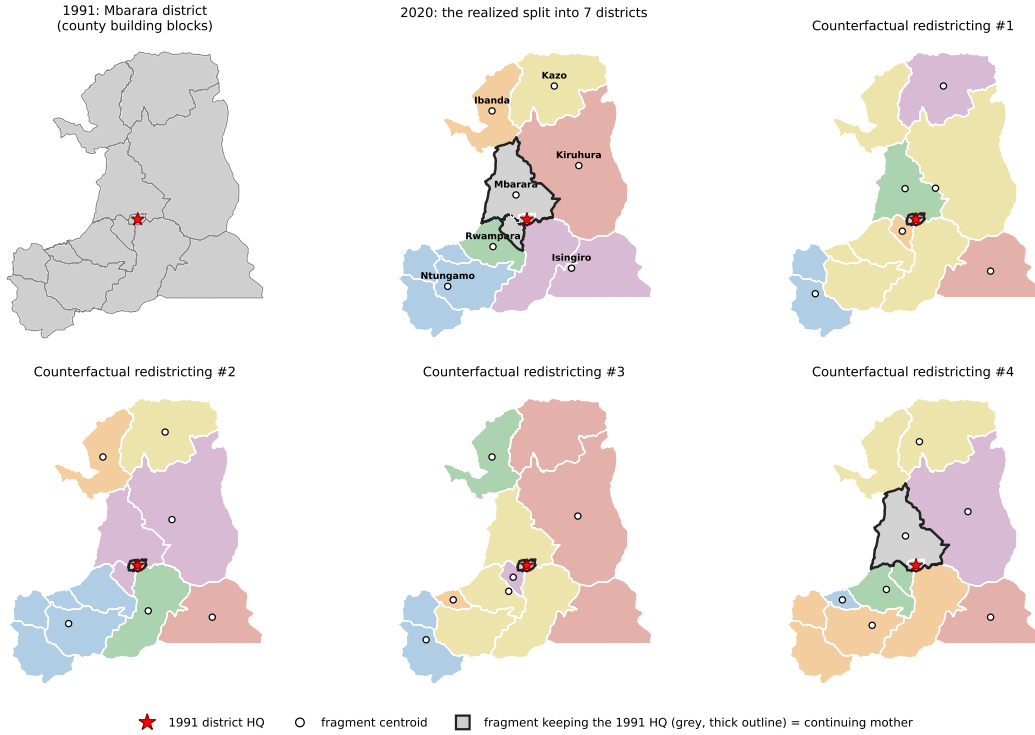


FIGURE A5: Redrawing Mbarara: actual and counterfactual county partitions. The 1991 district (top left) is composed of its county building blocks; the realized 2020 configuration (top centre) splits it into seven districts. The remaining panels show four of the 5,822 feasible contiguous partitions into seven fragments. The fragment retaining the 1991 headquarters (red star) remains the mother district (grey, thick outline); white dots mark fragment centroids, from which the instrument is computed.

Results

Table A13 compares three specifications for each service index: the baseline 2SLS estimate, an expected-instrument-adjusted specification that adds μ_z , μ_{prox} , and μ_{new} as controls,²⁵ and a specification that uses the pure recentered instrument $z - \mu_z$. Adding the expected-instrument controls leaves the results qualitatively unchanged and generally increases the estimated effects. For example, the coefficient rises from 0.43 to 0.80 for electricity, from 0.39 to 0.52 for district health, and from 0.28 to 0.50 for district water. The corresponding first-stage F -statistics range from approximately 10 to 25. These results indicate that the main findings are not driven only by households' predictable geographical exposure to the redistricting process.

The pure recentered instrument, by contrast, has a weak first stage, with Kleibergen–Paap F -statistics of around 4 or below, so I do not interpret its 2SLS magnitudes. A weak first stage

²⁵Three regressors respond to how the mother is partitioned, so each enters with its counterfactual expectation. For a given household these are the average, over all feasible partitions of its mother, of the excluded instrument (μ_z), of proximity to the fragment centroid (μ_{prox}), and of the new-district indicator (μ_{new} , the share of partitions placing the household in a new district rather than the mother rump). Controlling for all three, rather than μ_z alone, prevents predictable geometry from re-entering through the proximity and new-district controls.

also renders Anderson–Rubin confidence sets constructed from the recentered instrument itself uninformative, as they widen toward the entire parameter space. I therefore base inference on two complementary procedures, each robust to weak identification but in a different way. First, I report Anderson–Rubin 95 percent confidence sets for the μ -control specification, whose first stage is strong enough (Kleibergen–Paap F between roughly 10 and 25) for these sets to be bounded and informative while retaining correct size irrespective of instrument strength. They exclude zero for health (district) [0.26, 0.90], health (central government) [0.08, 1.17], water (district) [0.17, 0.97], water (central government) [0.18, 0.88], roads (central government) [0.37, 2.15], and electricity [0.47, 1.45]. Second, for the recentered instrument I conduct randomization inference, whose validity is design-based and therefore does not depend on the strength of the first stage at all. For each of 1,000 counterfactual national redistricting configurations I recompute the reduced-form coefficient of each outcome on the corresponding recentered placebo instrument and compare the realized estimate with this placebo distribution (Figure A6). The realized first-stage configuration is itself an extreme draw, with an omnibus randomization p -value of 0.001. Reading the reduced forms with attention to sign, seven of the ten service indices are both significant and positively signed (randomization p -values at or below 0.02, most at 0.001), matching the main tables, and these seven include all six headline outcomes, namely health and water at both tiers, roads (central government), and electricity. District roads is indistinguishable from the placebo distribution. The remaining two indices are extreme but negatively signed. Agricultural inputs (district) is also negative in the main 2SLS and plausibly reflects that input delivery is not organized around the headquarters town, while schools (central government) is a precise zero in the main specification whose recentered reduced form is unstable given its weak first stage.

Randomization inference: reduced form across 1,000 counterfactual redistricting configurations
(recentered instrument, μ -control specification; orange line = actual estimate)

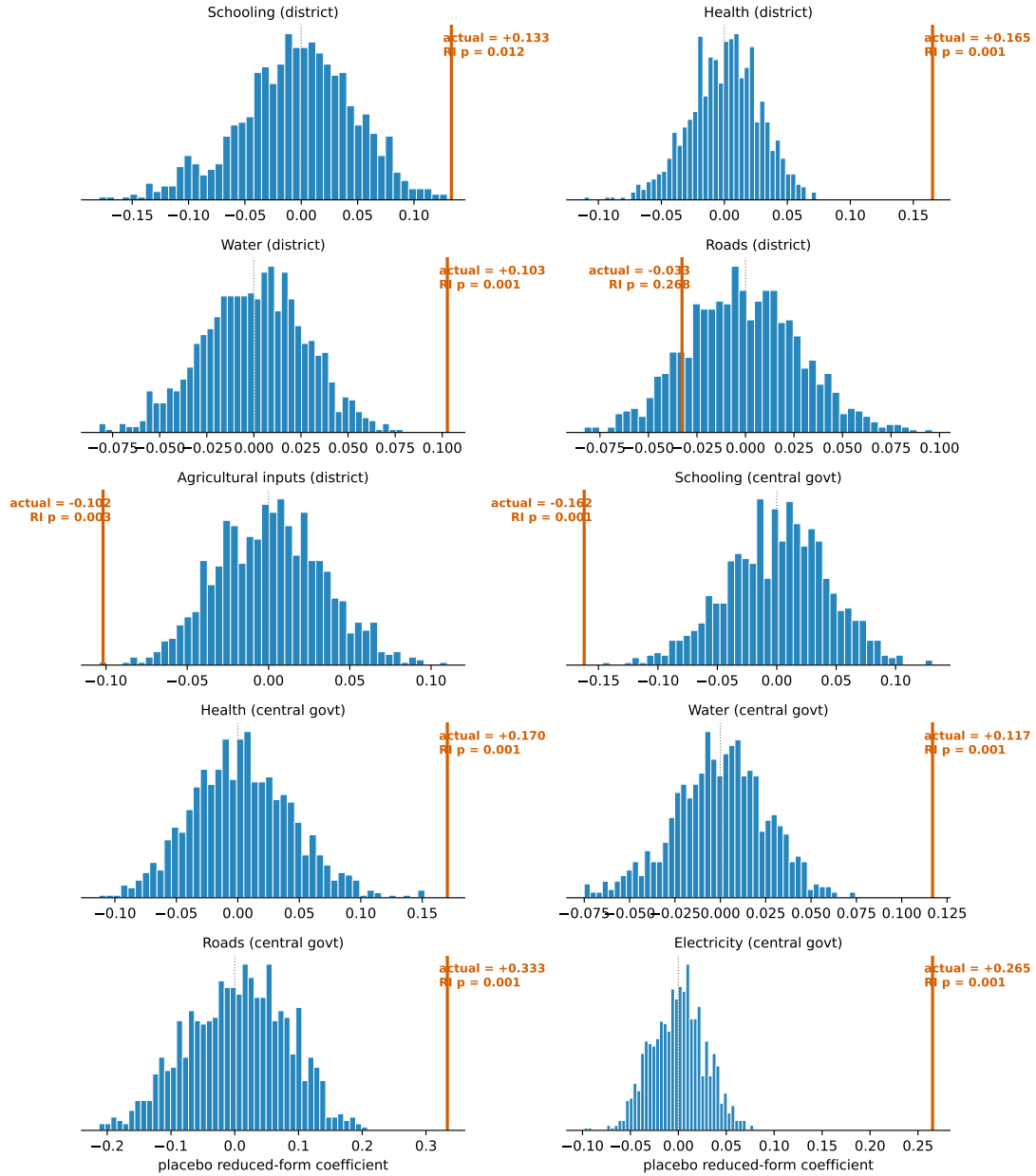


FIGURE A6: Randomization inference. Each panel shows the distribution of reduced-form coefficients on the recentered instrument across 1,000 counterfactual national redistricting configurations (μ -control specification); the orange line marks the estimate under the realized configuration, with its two-sided randomization p-value.

TABLE A13: Recentered Instrument: Effects on Services (2SLS)

	School (1)	Health (2)	Water (3)	Roads (4)	Agr inputs (5)
<i>Panel A: NSDS 2021 District services</i>					
Baseline 2SLS	0.282 (0.214)	0.387*** (0.121)	0.279* (0.157)	0.089 (0.110)	-0.085 (0.145)
2SLS, μ controls	0.464* (0.249)	0.516*** (0.148)	0.496*** (0.176)	-0.012 (0.129)	-0.291 (0.180)
2SLS, recentered IV ($z - \mu$)	1.420 (0.935)	-0.155 (0.349)	1.079** (0.533)	-0.858* (0.498)	-0.048 (0.378)
KP F (baseline)	22.24	33.35	28.75	28.75	28.75
KP F (μ controls)	17.58	24.51	17.46	17.46	17.46
KP F (recentered)	2.79	4.44	4.17	4.17	4.17
RI p (μ controls, $S=1,000$)	[0.002]	[0.001]	[0.001]	[0.081]	[0.001]
AR 95% set (μ controls)	[-0.00, 1.11]	[0.26, 0.90]	[0.17, 0.97]	[-0.31, 0.26]	[-0.75, 0.06]
Observations	7,137	7,595	8,417	8,417	8,417
	School (1)	Health (2)	Water (3)	Roads (4)	Electricity (5)
<i>Panel B: NSDS 2021 Central govt services</i>					
Baseline 2SLS	0.151 (0.192)	0.233 (0.201)	0.372** (0.146)	0.706*** (0.261)	0.428*** (0.137)
2SLS, μ controls	0.058 (0.261)	0.573** (0.244)	0.449*** (0.154)	0.970** (0.377)	0.795*** (0.205)
2SLS, recentered IV ($z - \mu$)	3.454 (4.219)	0.048 (0.479)	0.620* (0.345)	2.146* (1.222)	0.572 (0.387)
KP F (baseline)	22.62	28.75	28.75	28.75	28.75
KP F (μ controls)	9.86	17.46	17.46	17.46	17.46
KP F (recentered)	0.67	4.17	4.17	4.17	4.17
RI p (μ controls, $S=1,000$)	[0.020]	[0.001]	[0.001]	[0.001]	[0.001]
AR 95% set (μ controls)	[-0.60, 0.68]	[0.08, 1.17]	[0.18, 0.88]	[0.37, 2.15]	[0.47, 1.45]
Observations	5,553	8,417	8,417	8,417	8,417

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (cluster-robust). Borusyak–Hull (2023) recentering exercise. The expected instrument μ_i is the exact mean of the instrument over all feasible contiguous re-partitions of each splitting 1991 mother district into its realized number of 2020 districts, holding fixed which mothers split and into how many units; building blocks are 1991-vintage counties (IPUMS), and every mother’s feasible set is enumerated exhaustively. Column blocks: the paper’s 2SLS specification; adding ($\mu_z, \mu_{prox}, \mu_{new}$) as controls; and the recentered excluded instrument $z - \mu$. All regressions use the paper’s controls, weights, and clustering (NSDS: EA; AB: town/village); Kampala and Wakiso excluded as in the main tables. RI p : two-sided randomization-inference p-value of the reduced form on $z - \mu$ across $S=1,000$ national placebo redistricting configurations (one independent draw per mother per configuration, seed 20260706); first-stage omnibus RI $p = 0.001$ (NSDS) and 0.003 (AB). AR 95% set: weak-instrument-robust Anderson–Rubin confidence set (cluster-robust). Recentered-column magnitudes are estimated off within-mother reassignment variation with a different complier mix and should not be compared to baseline magnitudes.

G Additional results

TABLE A14: Service Availability and Proximity to District HQ (2SLS)

	School (1)	Health clinic (2)	Piped water (3)	Electricity (4)	Paved road (5)	Police station (6)
<i>Panel A: Afrobarometer R10</i>						
Log Proximity to District HQ	0.006 (0.165)	0.166 (0.201)	0.495** (0.224)	0.034 (0.203)	0.358** (0.162)	0.504** (0.209)
Observations	2184	2168	2168	2176	2168	2176
Number of clusters	271	269	269	271	269	270
Mean of DV	0.788	0.613	0.306	0.540	0.214	0.312
IV F-stat (KP rk Wald)	19.446	19.883	17.858	19.591	19.418	19.464

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to public goods and services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A15: Schooling and Proximity to District Headquarters (2SLS)

	School attendance (1)	Distance to school (2)	Travel time (3)	Teacher quality (4)	Facility quality (5)	Audited school (6)
<i>Panel A: District services</i>						
Log Proximity to District HQ	-0.128* (0.078)	-1.165** (0.536)	-19.130*** (6.807)	0.021 (0.067)	0.046 (0.065)	0.343 (0.267)
Observations	4566	3329	3329	4563	4564	5581
Number of clusters	951	916	916	951	951	638
Mean of DV	0.714	1.710	31.421	0.804	0.809	0.559
IV F-stat (KP rk Wald)	26.298	22.676	22.676	26.298	26.244	9.399
	School attendance (1)	Distance to school (2)	Travel time (3)	Teacher quality (4)	Facility quality (5)	Audited school (6)
<i>Panel B: Central govt services</i>						
Log Proximity to District HQ	-0.169 (0.105)	-0.606 (0.689)	-10.457 (7.829)	0.104 (0.087)	0.067 (0.082)	0.127 (0.225)
Observations	3546	2132	2132	3539	3541	3659
Number of clusters	943	827	827	943	943	420
Mean of DV	0.700	2.007	34.856	0.769	0.780	0.695
IV F-stat (KP rk Wald)	20.727	16.442	16.442	20.948	20.697	15.054

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to schooling services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Column 6 of each panel is based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A16: Health Services and Proximity to District Headquarters (2SLS)

	Public hospital (1)	Injury (2)	Injury travel time (3)	Health facility score (4)	HC absenteeism (5)	HC ambulance (6)	District HC supervision (7)
<i>Panel A: District services</i>							
Log Proximity to District HQ	-0.013 (0.090)	-0.040 (0.034)	-14.329* (8.670)	29.451*** (9.847)	-0.265* (0.138)	0.265* (0.160)	0.162 (0.161)
Observations	2291	8415	3252	6948	6821	6909	6619
Number of clusters	826	967	869	791	778	786	750
Mean of DV	0.865	0.146	36.098	26.347	0.215	0.303	0.787
IV F-stat (KP rk Wald)	26.663	28.803	23.237	33.461	30.568	30.178	29.761
	HC type 4 (1)	Distance gov hospital (2)	PHC insufficient (3)				
<i>Panel B: Central govt services</i>							
Log Proximity to District HQ	0.146** (0.066)	0.049 (0.099)	-0.029 (0.157)				
Observations	6979	8417	6819				
Number of clusters	795	967	776				
Mean of DV	0.075	0.240	0.233				
IV F-stat (KP rk Wald)	33.260	28.745	32.624				

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to health services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–7 of Panel A and all columns of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

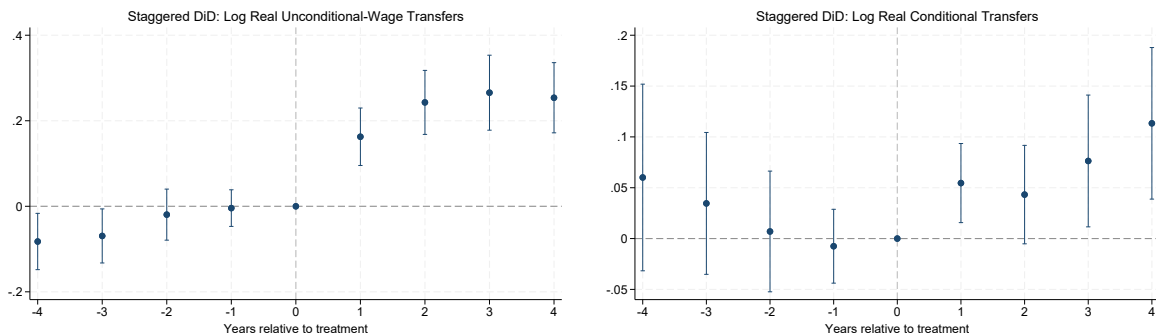


FIGURE A7: Event study: composition of the territory transfer response. Same specification and sample as Figure 5, for the wage component of the unconditional block grant (left, the duplicated administrative payroll) and conditional service-delivery grants (right), in log real FY2017/18 shillings. The joint placebo test has a p-value of 0.02 for the unconditional-wage component (with individually small placebo estimates) and 0.34 for conditional grants.

TABLE A17: Water Services and Proximity to District Headquarters (2SLS)

	Water source dry (1)	Water distance dry (2)	Water source wet (3)	Piped water (4)	Borehole (5)	Protected well (6)	
<i>Panel A: District services</i>							
Log Proximity to District HQ	0.128 (0.111)	-0.225 (0.194)	-0.029 (0.089)	0.212 (0.136)	0.184 (0.158)	0.293 (0.188)	
Observations	8417	7229	8417	8371	8371	8374	
Number of clusters	967	952	967	961	961	961	
Mean of DV	0.834	0.786	0.889	0.331	0.636	0.302	
IV F-stat (KP rk Wald)	28.745	30.837	28.745	28.094	28.094	27.880	
	Water src dry outside (1)	Water dist dry outside (2)	Water src wet outside (3)	Water dist wet outside (4)	WAP crops (5)	WAP cattle (6)	WAP fish (7)
<i>Panel B: Central govt services</i>							
Log Proximity to District HQ	0.145 (0.110)	-0.141 (0.228)	0.039 (0.075)	-0.065 (0.166)	0.042 (0.030)	0.084** (0.042)	0.123* (0.063)
Observations	8417	7177	8416	5603	7815	8356	7762
Number of clusters	967	949	967	894	901	959	895
Mean of DV	0.784	0.814	0.855	0.746	0.993	0.969	0.961
IV F-stat (KP rk Wald)	28.745	28.853	28.682	28.733	25.727	27.845	25.966

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to water services on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–6 of Panel A and Columns 5–7 of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A18: Road Infrastructure and Proximity to District Headquarters (2SLS)

	Trunk road (1)	Feeder road (2)	Communal road (3)	Murram road quality (4)	Feeder road quality (5)	Communal road quality (6)	Murram maintenance (7)	Feeder maintenance (8)	Communal maintenance (9)
Panel A: District services									
Log Proximity to District HQ	-0.119 (0.110)	0.058 (0.062)	-0.134** (0.058)	-0.005 (0.152)	0.149 (0.143)	0.268** (0.117)	-0.314** (0.138)	0.075 (0.143)	-0.034 (0.052)
Observations	8416	8416	8415	6220	7814	7758	8336	8364	8372
Number of clusters	967	967	967	708	894	888	957	960	961
Mean of DV	0.634	0.883	0.914	0.288	0.205	0.099	0.157	0.751	0.026
IV F-stat (KP rk Wald)	28.702	28.702	28.674	21.068	27.638	27.202	27.662	27.655	27.642
	Trunk road (1)	Tarmac road quality (2)	Tarmac maintenance (3)						
Panel B: Central govt services									
Log Proximity to District HQ	0.262** (0.118)	0.672** (0.317)	0.117 (0.146)						
Observations	8416	6236	8362						
Number of clusters	967	721	960						
Mean of DV	0.436	0.775	0.866						
IV F-stat (KP rk Wald)	28.702	13.092	28.125						

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to roads on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Columns 4–9 of Panel A and Columns 2–3 of Panel B are based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A19: Agricultural Inputs (District) and Proximity to District Headquarters (2SLS)

	Hybrid seeds (1)	Planting material (2)	Pesticides (3)	Fertilizer (4)
<i>Panel A: Inputs Availability</i>				
Log Proximity to District HQ	-0.080 (0.073)	0.005 (0.096)	-0.125 (0.094)	-0.070 (0.072)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.205	0.325	0.238	0.106
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745
	Hybrid seeds (1)	Planting material (2)	Pesticides (3)	Fertilizer (4)
<i>Panel B: Inputs Use</i>				
Log Proximity to District HQ	-0.115** (0.059)	0.108 (0.101)	0.043 (0.062)	0.006 (0.044)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.137	0.351	0.151	0.053
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of the reported availability (Panel A) and household use (Panel B) of agricultural inputs on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A20: Electricity (Central Government) and Proximity to District Headquarters (2SLS)

	HH electricity (1)	Connected grid (2)	Interested grid (3)	Electricity availability (4)
<i>Panel A: Central govt services</i>				
Log Proximity to District HQ	0.276*** (0.101)	0.197** (0.091)	0.009 (0.037)	0.237* (0.127)
Observations	8417	8417	8417	8417
Number of clusters	967	967	967	967
Mean of DV	0.463	0.175	0.055	0.824
IV F-stat (KP rk Wald)	28.745	28.745	28.745	28.745

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' access to electricity on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis. Column 4 is based on information from the community questionnaire, collected once per enumeration area and assigned to all households in the enumeration area; observation counts refer to households.

TABLE A21: Priority Projects Implemented and Proximity to District HQ (2SLS, community level)

	New school (1)	Classroom (2)	Health facility (3)	Water (4)	Electricity (5)	Roads bridges (6)	Share top-3 implemented (7)	Any top-3 implemented (8)
Log Proximity to District HQ (EA)	-0.017 (0.030)	0.308*** (0.105)	0.190** (0.079)	0.084 (0.126)	0.017 (0.138)	0.312** (0.152)	0.319*** (0.099)	0.232* (0.136)
Observations	595	515	651	743	617	436	933	933
Mean of DV	0.022	0.064	0.055	0.229	0.209	0.135	0.304	0.501
IV F-stat (KP rk Wald)	36.911	24.773	35.653	37.159	24.678	18.671	54.747	54.747

Notes: *p<0.1, **p<0.05, ***p<0.01. Community-level regressions (one observation per enumeration area) of outcomes from the NSDS community questionnaire on the log GIS distance from the enumeration area to the district's capital. In Columns (1)–(6) the outcome equals one if the community ranked the project among its three most important and the project was implemented in the village or parish in the past three years; each column is estimated on the communities that ranked that project. Column (7) is the share of the community's top-three priorities that were implemented in the past three years and Column (8) equals one if any of them was implemented. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument is the interaction of distance from the enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Heteroskedasticity-robust standard errors are in parenthesis.

TABLE A22: Trust in Others and Social Capital and Proximity to District HQ (2SLS)

	Trust relatives (1)	Trust neighbours (2)	Trust citizens (3)	Trust other religion (4)	Trust other ethnicity (5)
<i>Panel A: Afrobarometer R10</i>					
Log Proximity to District HQ	-0.013 (0.077)	0.058 (0.089)	0.131 (0.108)	0.192* (0.109)	0.257** (0.126)
Observations	2176	2179	2177	2179	2167
Number of clusters	271	271	271	271	271
Mean of DV	0.800	0.652	0.537	0.587	0.503
IV F-stat (KP rk Wald)	19.564	19.299	19.038	19.340	19.179
	Empowerment (1)	Collective action (2)	Bridging groups (3)	General trust (4)	
<i>Panel B: NSDS 2021</i>					
Log Proximity to District HQ	0.084 (0.074)	0.179** (0.080)	0.055 (0.047)	-0.050 (0.075)	
Observations	8416	8416	4334	8416	
Number of clusters	967	967	926	967	
Mean of DV	0.637	0.591	0.080	0.441	
IV F-stat (KP rk Wald)	28.702	28.702	26.874	28.702	

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of citizens' trust in others and social capital on self-reported household's distance to the district's capital. All regressions control for district ever split between 1991-2020, lake Victoria district, districts' area in Km² as of 1991, household's enumeration area distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of distance from the household's enumeration area to the district's centroid and an indicator for new districts between 1991-2020. Standard errors clustered at the Parish level are in parenthesis.

TABLE A23: Trust in Government and Proximity to District HQ: 2017 Afrobarometer (2SLS)

	Trust localgov (1)	Trust president (2)	Trust parliament (3)	Trust ruling party (4)	Trust opposition (5)	Trust courts (6)
<i>Panel A: Afrobarometer R7 (2017)</i>						
Log Proximity to District HQ	-0.085 (0.100)	0.018 (0.099)	-0.142 (0.106)	0.047 (0.083)	-0.092 (0.130)	-0.025 (0.080)
Observations	995	1014	1009	1025	995	987
Number of clusters	129	129	129	129	129	129
Mean of DV	0.587	0.711	0.554	0.642	0.361	0.626
IV F-stat (KP rk Wald)	10.770	10.062	9.688	10.029	10.004	10.106

Notes: *p<0.1, **p<0.05, ***p<0.01. Coefficients from regressions of trust in government variables from the 2017 Afrobarometer (Round 7) on self-reported household's distance to the district's capital. Each variable indicates that the respondent trusts the corresponding institution somewhat or a lot. All regressions control for district ever split, lake Victoria district, districts' area in Km² as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of the household's distance to the district's centroid and an indicator for new districts between 1991-2016, using district boundaries and headquarters as of 2017. Standard errors clustered at the enumeration area (town/village) level are in parenthesis. The first-stage Kleibergen-Paap F statistic is close to the conventional threshold of 10, so magnitudes should be read with caution; the estimates are reported to show that the absence of trust effects documented in the 2024 round already holds in 2017.

TABLE A24: Service Availability and Proximity to District HQ Across Afrobarometer Rounds (2SLS)

	School (1)	Health clinic (2)	Piped water (3)	Electricity grid (4)	Paved road (5)	Police station (6)
<i>Panel A: Afrobarometer R7 (2017)</i>						
Log Proximity to District HQ	-0.070 (0.174)	-0.293 (0.242)	0.268* (0.153)	0.155 (0.228)	-0.105 (0.285)	-0.059 (0.260)
Observations	1048	1040	1048	1048	1048	1048
Number of clusters	129	128	129	129	129	129
Mean of DV	0.809	0.692	0.191	0.275	0.489	0.366
IV F-stat (KP rk Wald)	9.621	9.806	9.621	9.621	9.621	9.621
	School (1)	Health clinic (2)	Piped water (3)	Electricity grid (4)	Paved road (5)	Police station (6)
<i>Panel B: Afrobarometer R10 (2024)</i>						
Log Proximity to District HQ	0.006 (0.165)	0.166 (0.201)	0.495** (0.224)	0.034 (0.203)	0.358** (0.162)	0.504** (0.209)
Observations	2184	2168	2168	2176	2168	2176
Number of clusters	271	269	269	271	269	270
Mean of DV	0.788	0.613	0.306	0.540	0.214	0.312
IV F-stat (KP rk Wald)	19.446	19.883	17.858	19.591	19.418	19.464

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Coefficients from regressions of enumeration-area service availability indicators, recorded by Afrobarometer enumerators (not self-reported), on self-reported household's distance to the district's capital. Outcomes are identical across the two rounds and indicate whether the enumeration area has a school, a health clinic, a piped water system, an electricity grid, a paved road surface over the last five kilometers, and a police station. All regressions control for district ever split, lake Victoria district, districts' area in Km^2 as of 1991, household's distance to district's capital as of 1991 (in log.), district contains a lake other than Victoria, sub-county mean elevation (in m.a.s.l.), and four major regions. The instrument for the household's distance to the district's capital is the interaction of the household's distance to the district's centroid and an indicator for new districts, defined over 1991-2016 with 2017 boundaries in Panel A and over 1991-2020 with 2020 boundaries in Panel B. Standard errors clustered at the enumeration area (town/village) level are in parenthesis. The first-stage Kleibergen-Paap F statistic in the 2017 round is close to the conventional threshold of 10, so Panel A magnitudes should be read with caution.

TABLE A25: Non-zero-sum fiscal accounting around district splits

	Own pre-split counterfactual	Trend-adjusted counterfactual	Matched-trend counterfactual
<i>Panel A. Net extra spending across the territory, $(M_{post} + C_{post}) - \widehat{M}_{cf}$</i>			
Mean (billions UGX)	10.11	5.37	5.60
Median (billions UGX)	7.39	2.83	3.00
Splits with positive excess	20/24	19/24	19/24
<i>Panel B. As percent of the mother's pre-split budget</i>			
Mean (%)	35.26	19.21	20.42
Median (%)	28.34	11.84	9.99
Standard deviation (%)	37.98	30.59	32.12
<i>Panel C. Underlying real budget levels (billions of FY2017/18 UGX)</i>			
Mother, pre-split (M_{pre}), median		31.05	
Mother, post-split (M_{post}), median		23.01	
Child, first two years (C_{post}), median		14.38	

Notes: The table asks whether the spending a mother district loses when it splits is recovered elsewhere, or whether the split raises total spending across the original territory. The sample is the 24 districts created within the panel window (calendar 2015–2020; approved local-government transfers FY2012/13–FY2023/24), and all amounts are in real FY2017/18 shillings. For each split at creation year T , M_{pre} is the mother's average budget over the two fiscal years ending at T , M_{post} its average over $(T+1, T+2)$, and C_{post} the child's average over its first two years. The excess spending generated by the split is $(M_{post} + C_{post}) - \widehat{M}_{cf}$, the post-split total across mother and child minus the budget the undivided territory would have had, $\widehat{M}_{cf}$. The three columns differ only in that counterfactual. Column (1) sets $\widehat{M}_{cf} = M_{pre}$, asking whether post-split spending exceeds the mother's own pre-split level. Because all local budgets grew over the period, columns (2) and (3) instead let the counterfactual grow at the pace of comparable districts that never split: $\widehat{M}_{cf} = M_{pre} \times g$, where g is the median real growth factor over the same windows of the 19 territories with no split during the panel window (column 2), or of those same territories in the mother's own quartile of pre-split budget (column 3). Kampala, Wakiso, and municipal and city councils are excluded throughout. Both trend-adjusted counterfactuals leave a positive excess in 19 of the 24 splits and an average net cost near 20 percent of the mother's pre-split budget, indicating that district creation raises total spending rather than redistributing a fixed envelope.

H Administrative Unit Proliferation in Comparative Perspective

Table A26 places Uganda’s district creation in comparative perspective. It reports, for sub-Saharan African countries with verifiable counts and for five countries outside the region, the number of subnational units of a given tier at two dates. Uganda’s increase is the largest in the region, but the direction is shared by most countries listed, and the constitutional or statutory criteria for creating units typically refer to population and to bringing services closer to citizens.

TABLE A26: Administrative Unit Proliferation in Sub-Saharan Africa and Elsewhere

Country	Unit	Initial (year)	Later (year)	Change (%)	Source
<i>Panel A: Sub-Saharan Africa</i>					
Uganda	Districts	38 (1991)	136 (2020)	+258	This paper
Kenya	Districts	41 (1992)	158 (2009)	+285	Hassan (2016); Kenya National Bureau of Statistics (2010)
Ghana	Districts	110 (1988)	216 (2012)	+96	Resnick (2017)
Nigeria	Local government areas	301 (1976)	774 (1996)	+157	Adeyemi (2013)
DR Congo	Provinces	11 (2014)	26 (2015)	+136	Law (2026)
Benin	Départements	6 (1990)	12 (2000)	+100	Grossman & Lewis (2014)
Angola	Municípios	164 (2024)	326 (2025)	+99	Law (2026)
Togo	Préfectures	21 (1990)	35 (2010)	+67	Grossman & Lewis (2014)
Zambia	Districts	72 (2011)	116 (2018)	+61	Law (2026)
Burkina Faso	Provinces	30 (1996)	45 (1997)	+50	Billing (2019)
Senegal	Départements	30 (1990)	45 (2010)	+50	Grossman & Lewis (2014)
Ethiopia	Woredas	556 (1990)	736 (2010)	+32	Grossman & Lewis (2014)
Tanzania	Districts	119 (1990)	149 (2012)	+25	Grossman & Lewis (2014)
Mali	Cercles	40 (1990)	49 (2005)	+23	Grossman & Lewis (2014)
Mozambique	Districts	128 (2012)	151 (2013)	+18	Law (2026)
Malawi	Districts	24 (1990)	28 (2003)	+17	Grossman et al. (2017)
Sierra Leone	Districts	14 (2016)	16 (2017)	+14	Law (2026)
<i>Consolidations</i>					
Rwanda	Districts	106 (2005)	30 (2006)	−72	Law (2026)
South Sudan	States	10 (2011)	32 (2017); 10 (2020)	0	Law (2026)
<i>Panel B: Other regions</i>					
United States	Counties	292 (1790)	2,713 (1900)	+829	U.S. Bureau of the Census (1994)
Indonesia	Districts and cities	302 (1999)	514 (2014)	+70	Bazzi & Gudgeon (2021)
Vietnam	Provinces	40 (1979)	64 (2004)	+60	Law (2026)
India	Districts	466 (1991)	640 (2011)	+37	Rajan & Malghan (2022)
Brazil	Municipalities	4,124 (1989)	5,507 (1997)	+34	Dahis & Szerman (2024)

Notes: The table reports the number of subnational units of the tier indicated at two dates, and the percentage change between them. Panel A lists sub-Saharan African countries for which counts could be verified against an academic source, a census, or the change history compiled by Law (2026); Grossman & Lewis (2014) document that almost half of sub-Saharan African countries increased their number of administrative units by at least 20 percent between 1990 and 2010. Tiers are not comparable across countries: some countries subdivided the top tier (Nigeria, DR Congo, South Sudan) and others the local government tier (Ghana, Kenya, Uganda, Zambia, Tanzania). Ghana had 65 districts before the 1988 reform that created 110, and further waves raised the count to 260 by 2019. Kenya’s districts were replaced by 47 counties under the 2010 Constitution. Ethiopian woreda counts are sensitive to definitional changes across censuses. Rwanda consolidated its districts in 2006 on fiscal-sustainability grounds; South Sudan’s 2015 and 2017 state creations were reversed in 2020. Uganda’s counts correspond to the analysis sample.